



The BIM principle and philosophy



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BIM digest

By noreply@blogger.com (Taewook Kang)

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Description

안녕하세요. Engineering을 쉽게 풀어서 다하다가 podcast에 대해 이야기를 나눌 때 Engineering digest는 현재 공학에서 화학, Software, Graphics 그리고 BIM입니다. 앞으로는 방송을 편하고 유연하게 부담없이... 기대주시길 바랍니다. 방송내용은 iPod에서 검색하실 수 있습니다. - Tae Wook, Kang. 감

View in iTunes

Free

Category: Educational Technology

Name

- 1 No.122 - BIM 최신 솔루션 ...
- 2 No.121 - 건설분야 IoT 기...
- 3 No. 120 - 최근 BIM 소프트...

need to use a ground station to view the s
info to listen to Pihawk's status tones.



Status LED

Initializing, hold the XB still and level

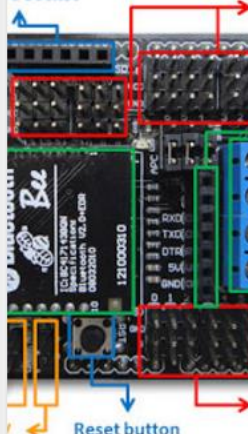
Acquiring GPS, please wait

Autopilot ready, GPS locked



```
Blink.begin(auth);
}
void loop()
{
  Blink.run();
}
```

Socket



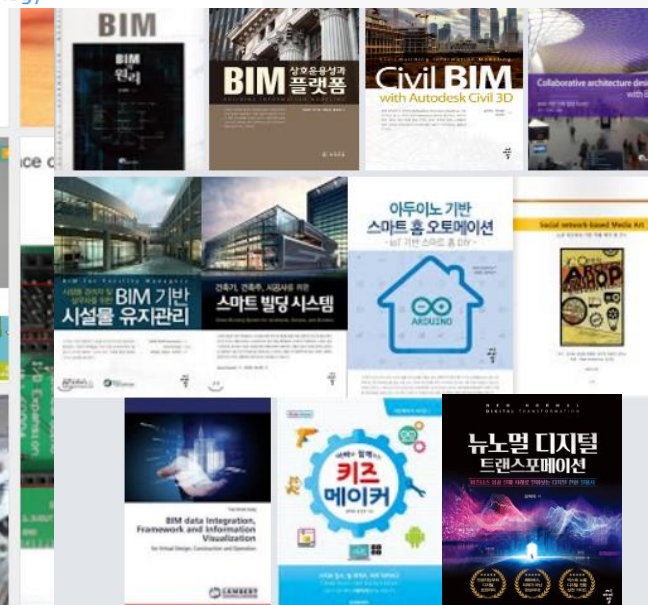
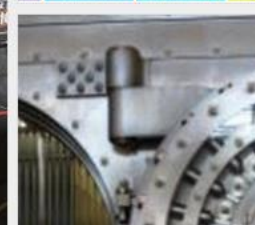
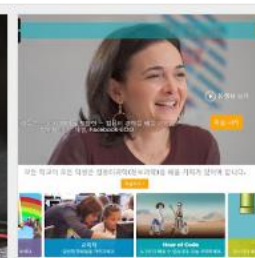
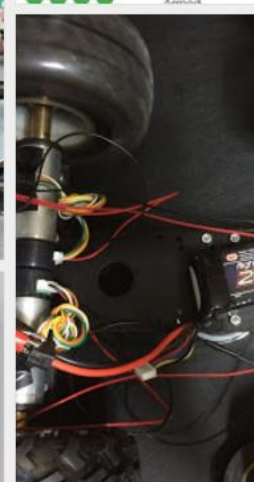
Reset button

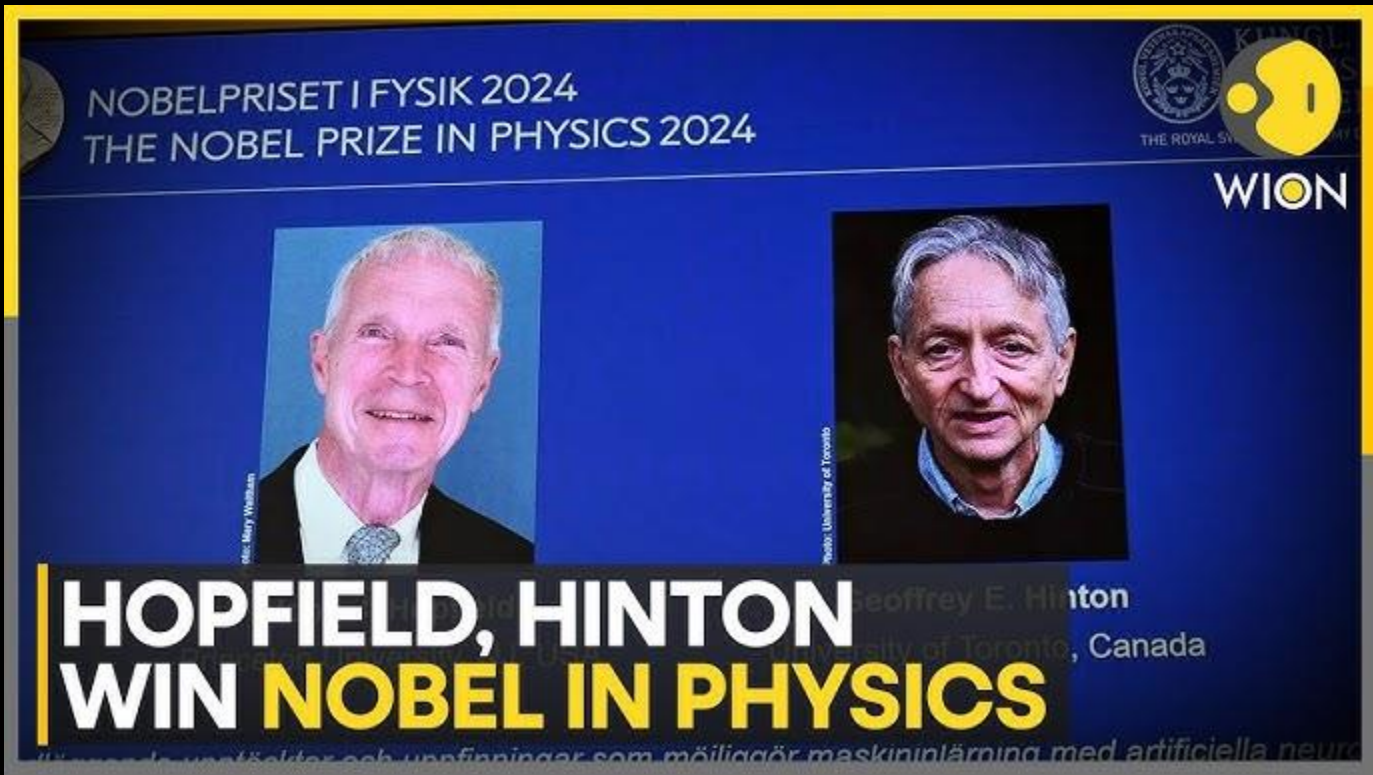


FORCE FEEDBACK
Most manufacturing robots
are too dangerous to work
without. Reaction network controls.



Initializing, hold the XB still and level
Acquiring GPS, please wait
Autopilot ready, GPS locked
Autopilot





Members of the Nobel Committee for Chemistry at the Royal Swedish Academy of Sciences explain the work of 2024 Nobel Prize in Chemistry winners David Baker, Demis Hassabis and John M. Jumper. JONATHAN NACKSTRAND/AFP via Getty Images

AI Pioneers Geoffrey Hinton And John Hopfield Win Nobel Prize For Physics | Latest News | WION

 머니투데이

"추격조'에 GPU 몰아주면 韓 딥시크 10개 나와"

"글로벌 AI 추격조를 만들어 '3년간 한국 데이터를 다 가져다 써라' 해야 합니다. 정부가 연내 GPU를 1만개를 확보해 상하반기 각 5개 기업에 2000...

Feb 6, 2025

 한국일보

[현장] "GPU·데이터 몰아주면 한국에서도 딥시크 10개 나온다"

6일 과학기술정보통신부가 개최한 중국 인공지능(AI) 모델 '딥시크'의 등장 이후 국내 AI 산업의 경쟁력을 점검하는 회의에서 거대언어모델(LLM)을...

Feb 7, 2025

 중앙일보

韓 AI 회사 "지금이 오히려 기회...딥시크 10개 만들 방법 있다" [팩플]

중국 생성 인공지능(AI) 모델 딥시크의 등장은 한국 AI 산업에 위기일까 기회일까. 국내 AI 기업들은 'AI 추격조'를 만드는 등 정부의 전폭적 지원이...

Feb 6, 2025

 노컷뉴스

[Q&A]AI 강국되려면 GPU 잡아라..."품귀현상에도 구할 수 있다"

"더 많은 인프라를 가지고 양질의 데이터를 확보해서 AI 모델을 만드는 것이 중요하다" (배경훈 LG AI연구원장) "GPU 1만 개 확보해서 10개 회사에..."

1 month ago



ChatGPT 회원 가입

Claude 회원 가입

Microsoft Copilot: Your AI companion 회원 가입

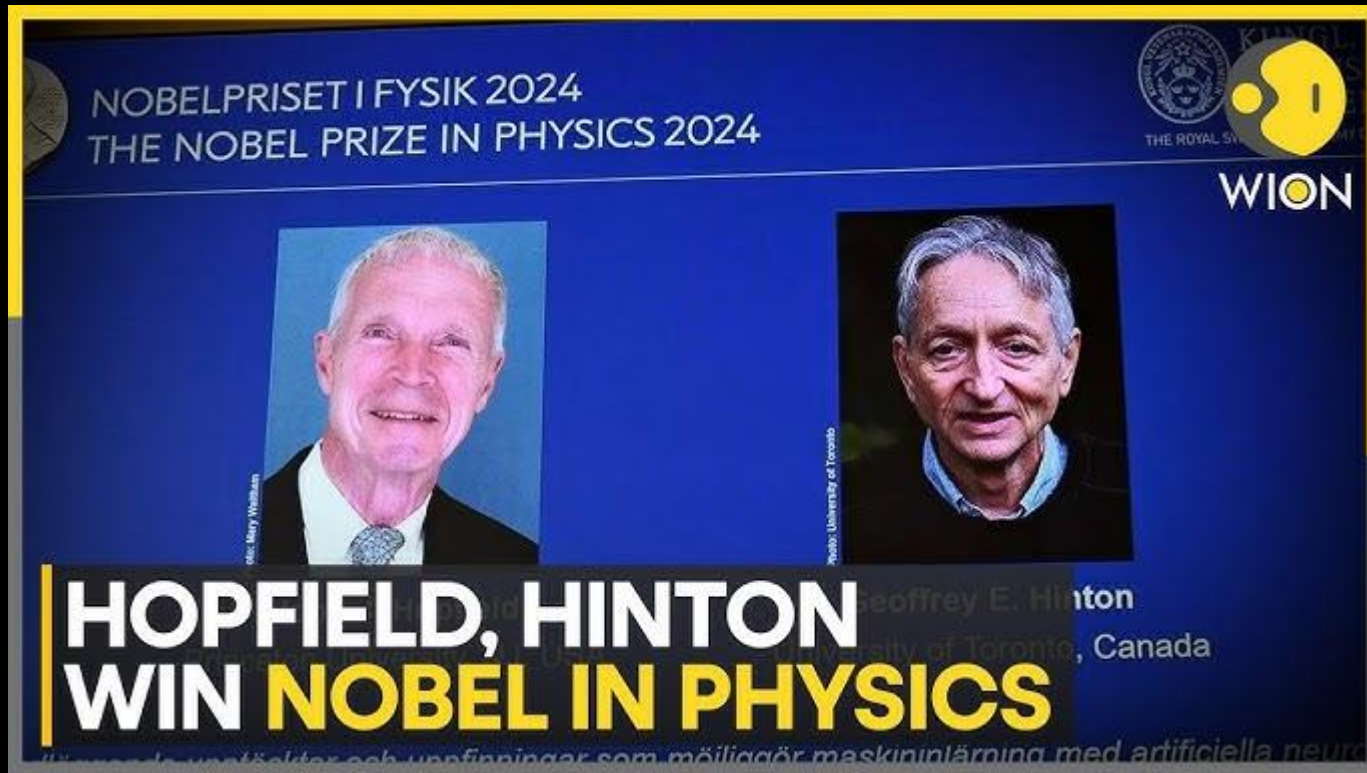
Hugging Face – The AI community building the future. 회원 가입 (무료 옵션)

GitHub 회원 가입 (무료 옵션)

LLM & AI Agent Trend

mac999/LLM-RAG-Agent-Tutorial: LLM-RAG-
Agent-Tutorial

AX Era



Members of the Nobel Committee for Chemistry at the Royal Swedish Academy of Sciences explain the work of 2024 Nobel Prize in Chemistry winners David Baker, Demis Hassabis and John M. Jumper. JONATHAN NACKSTRAND/AFP via Getty Images

AI Pioneers Geoffrey Hinton And John Hopfield Win Nobel Prize For Physics | Latest News | WION

AX Era

● Live



AX Era



evening, Sam



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일시중지(k)

S

claude@anthropic.com



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Large Language Models as General Pattern Machines

Suvir Mirchandani Fei Xia Pete Florence Brian Ichter
Danny Driess Montserrat Gonzalez Arenas Kanishka Rao
Dorsa Sadigh Andy Zeng

general-pattern-machines.github.io



Google DeepMind

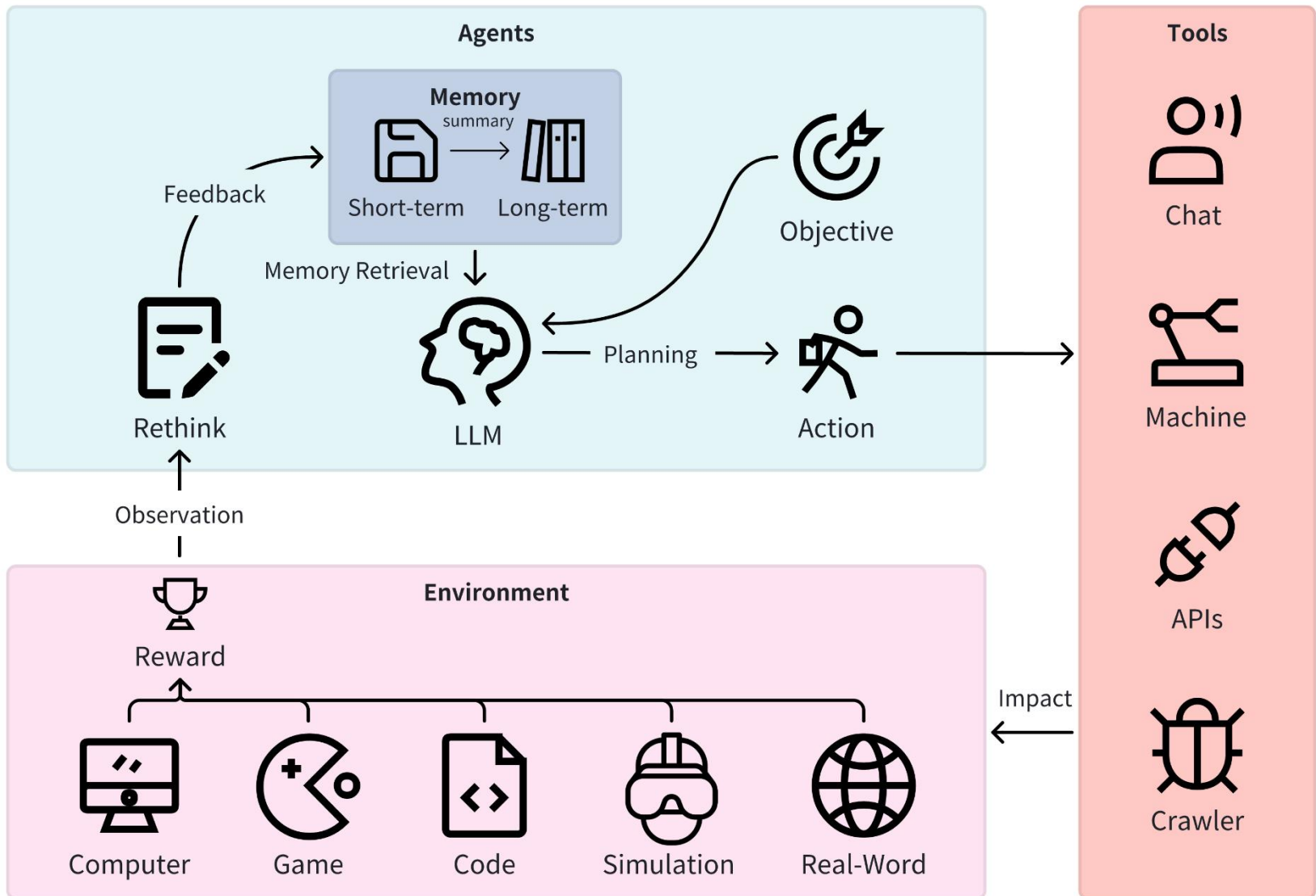
Stanford



TECHNISCHE
UNIVERSITÄT
BERLIN

TITLE	Google Scholar	CITED BY	YEAR
Large language models as general pattern machines S Mirchandani, F Xia, P Florence, B Ichter, D Driess, MG Arenas, K Rao, ... arXiv preprint arXiv:2307.04721		103	2023

Multi AI Agent



Vibe coding. No coding

바이브
코딩



TechCrunch Latest Startups Venture Apple Security AI Apps | Events Podcasts Newsletters

Vibe Coding: The Future of Software Development or Just a Trend? - Lovable Blog

Lovable becomes a unicorn with \$200M Series A just 8 months after launch

Anna Heim · 5:11 AM PDT · July 17, 2025

This Game Created by AI 'Vibe Coding' Makes \$50,000 a Month. Yours Probably Won't

flypieter.com was initially made in just 30 minutes with AI tools and is now generating thousands of dollars a month. The future of AI-assisted game development will not be that simple.

EMANUEL WALBERG · MAR 5, 2025 AT 12:03 PM



A game created with AI in just 30 minutes is generating more than \$50,000 a month, its creator claims—and could be a peek at how AI can, and can't, change game design in the future.

MORE LIKE THIS

WIX ACQUIRES SIX-MONTH-OLD AI "VIBE CODING" STARTUP BASE44 FOR \$80M CASH

AI Business Capital Markets News Slider By James Dargan · June 20, 2025

Published on July 28, 2025 · In IT Services

Cognizant's Vibe Coding Lesson for Indian IT

Cognizant aims to generate 50% of its code using AI within a year and states it has already achieved 20% of that target.



BASE44

Build Apps with AI in Minutes | Base44

This Game Created by AI 'Vibe Coding' Makes \$50,000 a Month. Yours Probably Won't, Wix Acquires Six-month-old AI "Vibe Coding" Startup Base44 for \$80M Cash, Cognizant's Vibe Coding Lesson for Indian IT, Vibe Coded a Website With My Daughter Using an AI Tool Called Bolt - Business Insider

Vibe coding. No coding

Cursor



Copilot



Claude CODE



```
@ mod.rs
src > listeners > @ mod.rs
46   async fn get_certificates(&self) -> Option<Vec<
47       None
48   }
49   /// Called to get SHI-specific certificates
50   /// Returns a tuple of (certificate_chain.pem,
51   async fn get_certificates_for_shi(&self, server:
52       self.get_certificates().await // Fall back
53   }
54   }
55   }
56   pub type TlsAcceptCallbacks = Box<dyn TlsAccept +
57   }
58   struct TransportStackBuilder {
59       l4: ServerAddress,
60       tls: Option<TlsSettings>,
61   }
62   }
63   impl TransportStackBuilder {
64       pub fn build(&mut self, #[cfg(unix)] upgrade_li
65       TransportStack {
66           l4: ListenerEndpoint::new(self.l4.clone(),
67           tls: self.tls.take().map(|tls| Arc::new
68           #[cfg(unix)]
69           upgrade_listeners,
70       }
71   }
72   }
73   }
74   }
```

CHAT COMPOSER

+ @ mod.rs Current File X
Ask anything (HLL), TL to select
Claude-3.5-sonnet @ Mention chat codebase



Vibe coding. No coding

OpenAI Codex: Transforming Software Development with AI Agents

OpenAI Codex: Transforming Software Development with AI Agents

[OpenAI Codex: Transforming Software Development with AI Agents - DevOps.com](#)

BY: TOM SMITH ON MAY 18, 2025

Code Generation



Smart IDEs



Frontend + Deployment

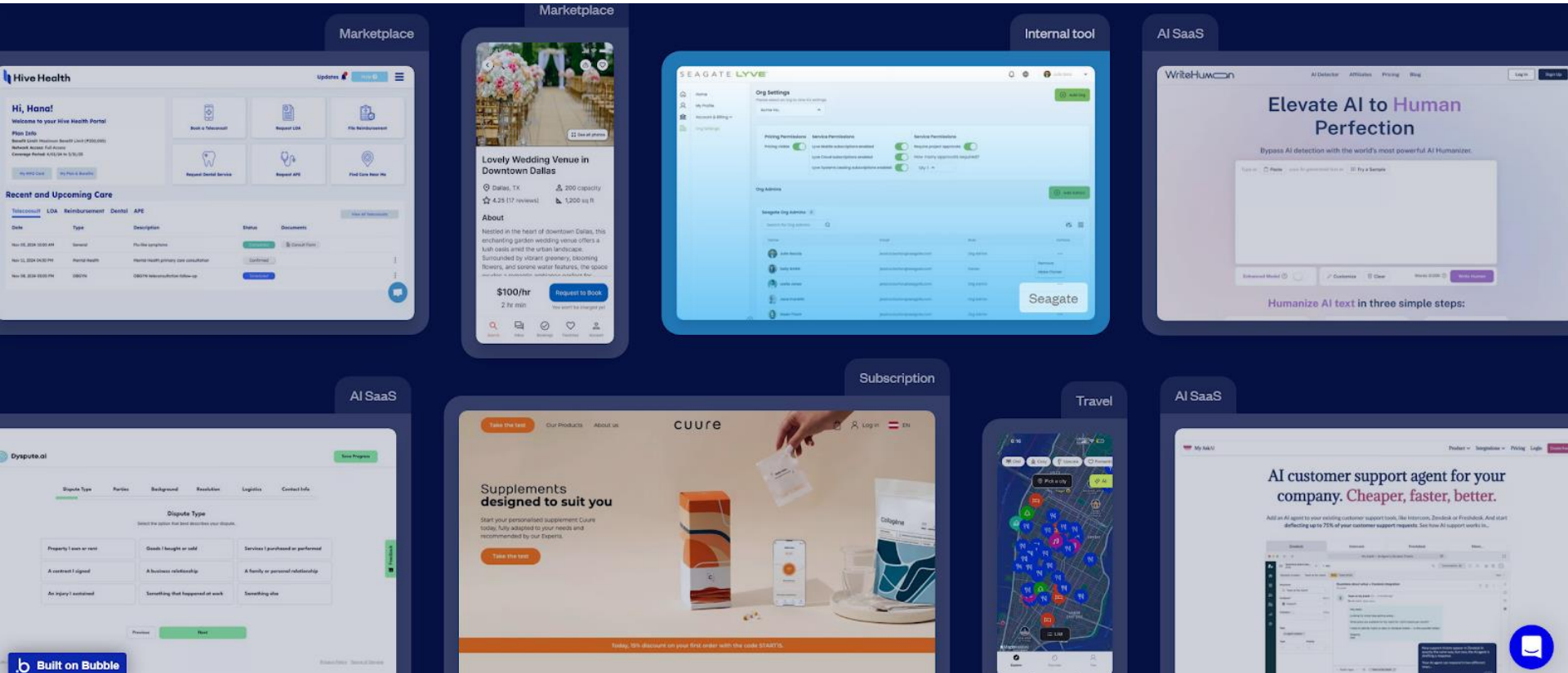


[Gemini CLI vs Claude Code vs Cursor – Which is the best option for coding? – Bind AI IDE](#)
[Is Vibe Coding the Future of Software Development? A Deep Dive into AI's Role](#)

Vibe coding. No coding

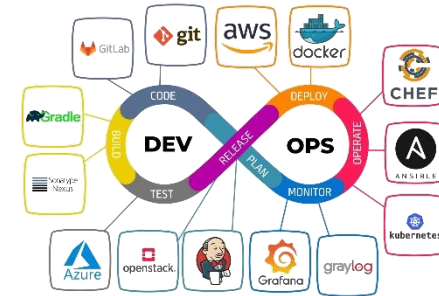
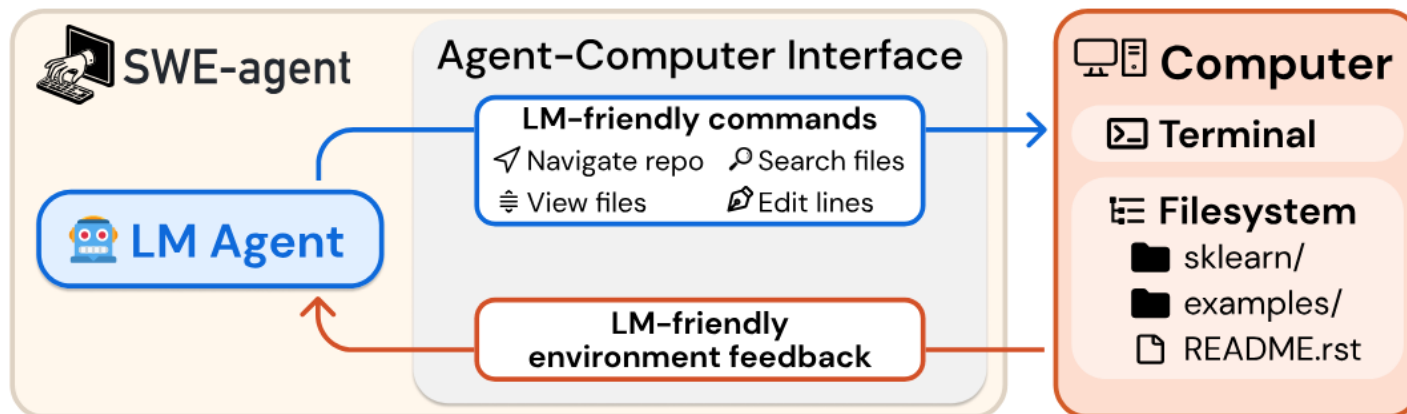
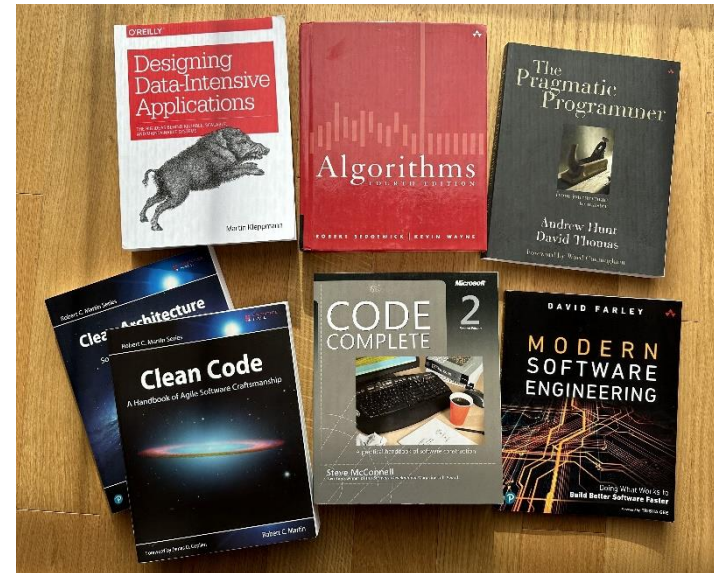
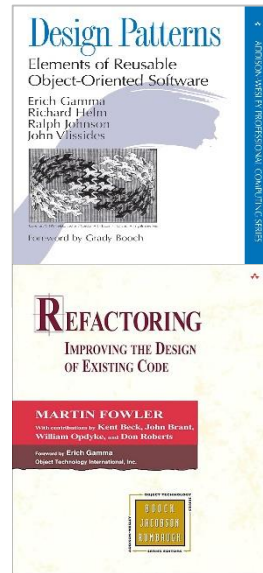
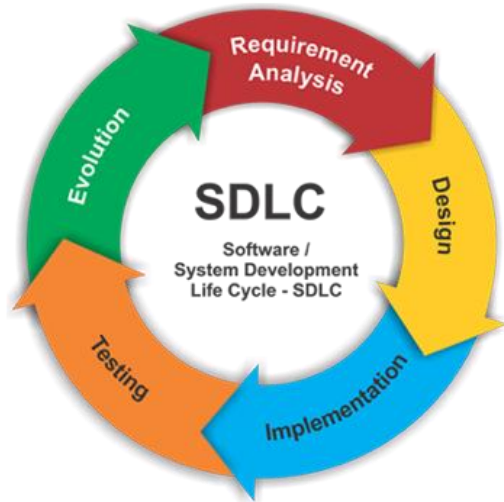
노코드(No-code) 확산

Daddy Makers: 노코드 서비스 비교 분석하기



Vibe coding. No coding

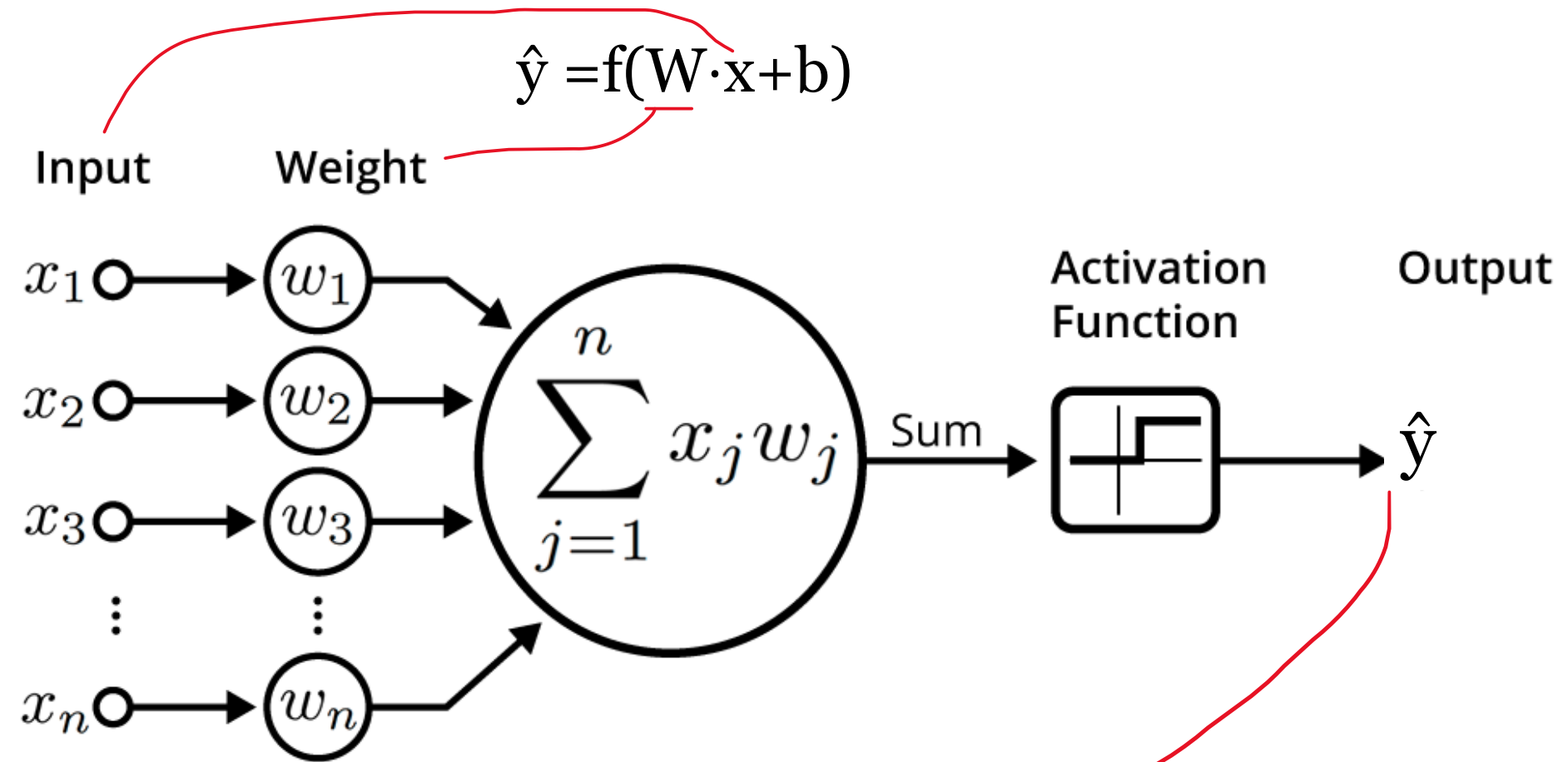
소프트웨어 엔지니어링 AI 에이전트 기술



AI foundation

mac999/AI foundation tutorial

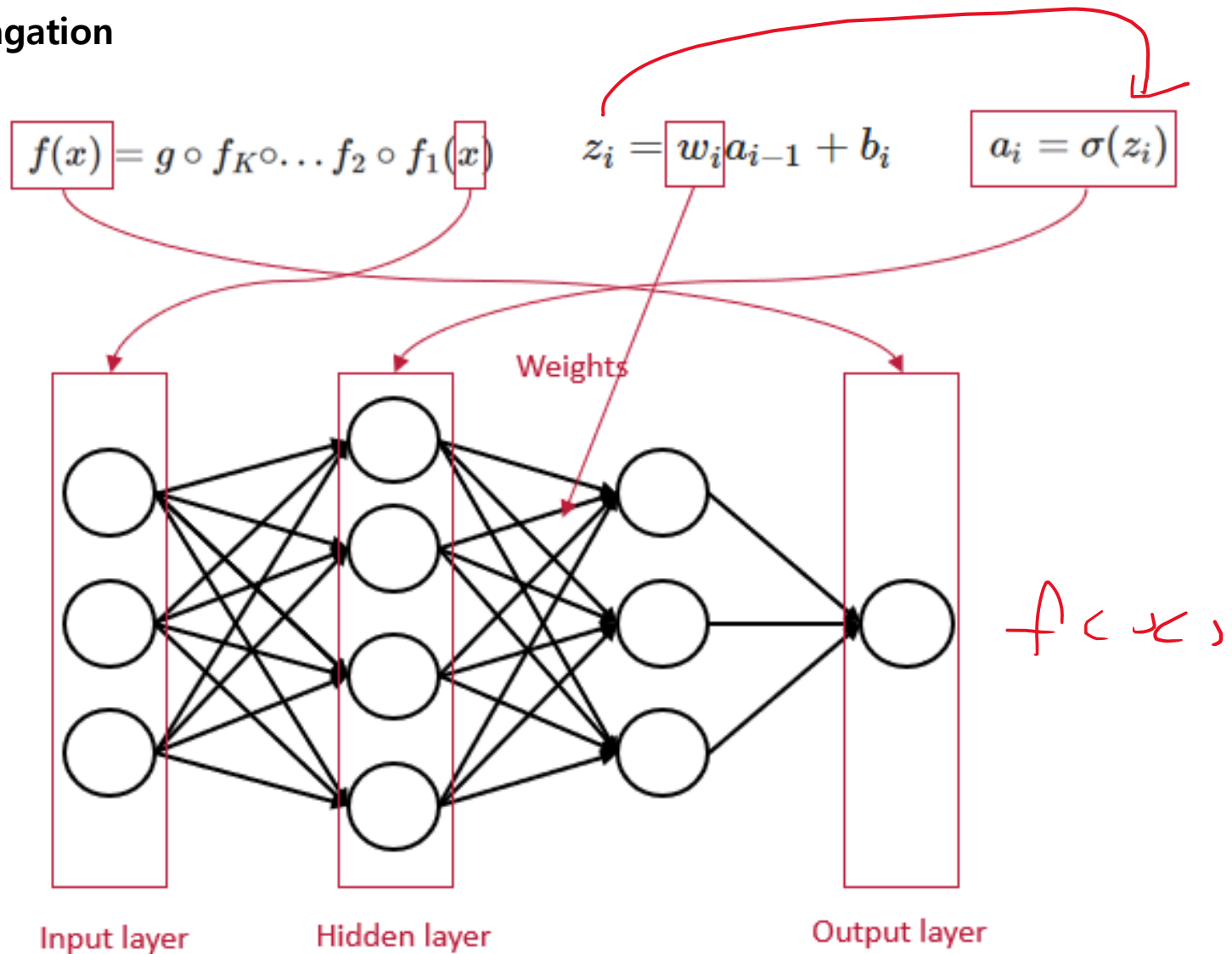
Deep learning



$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2.$$

Deep learning

Backpropagation



Deep learning

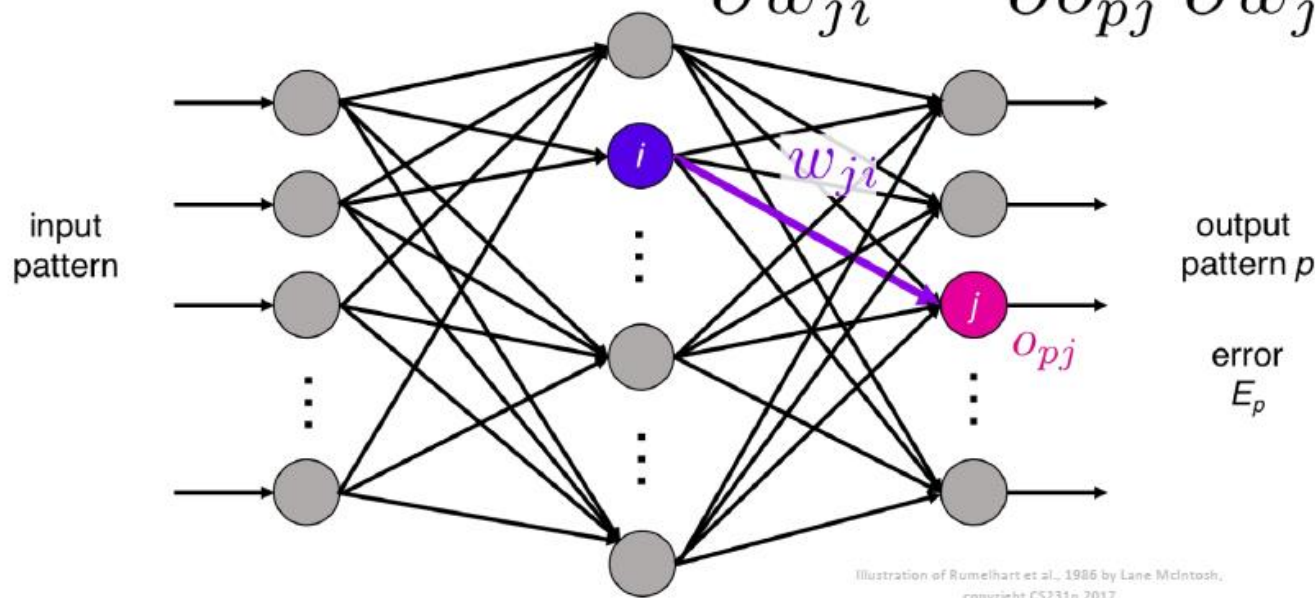
Backpropagation

$$\hat{y} = \underline{f(\underline{w \cdot x + b})}$$

$$\Delta w_i = -\alpha \frac{dE}{dw_i}$$

$$w_{next} = w + \Delta w$$

$$\frac{\partial E_p}{\partial w_{ji}} = \frac{\partial E_p}{\partial o_{pj}} \frac{\partial o_{pj}}{\partial w_{ji}}$$

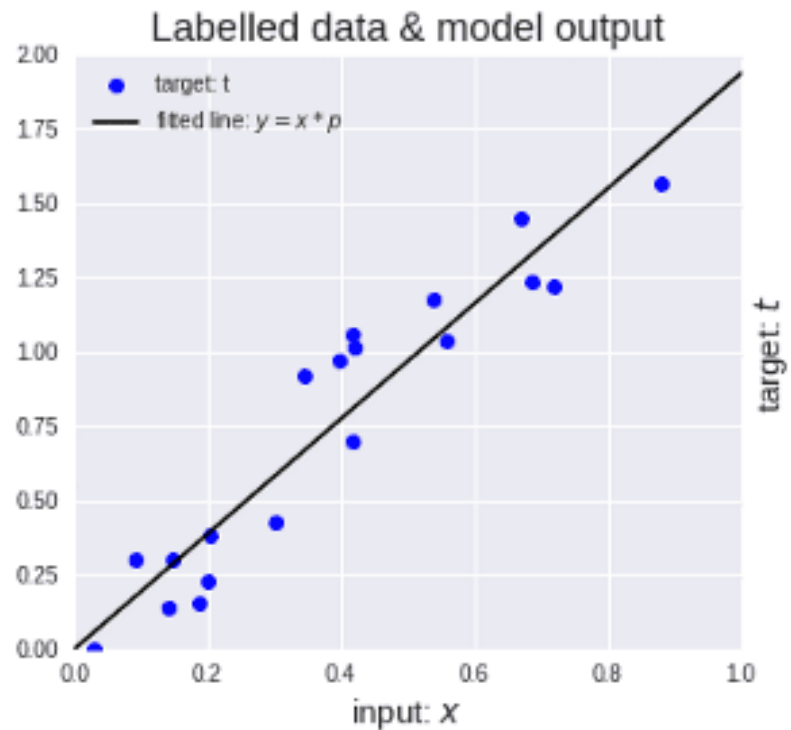
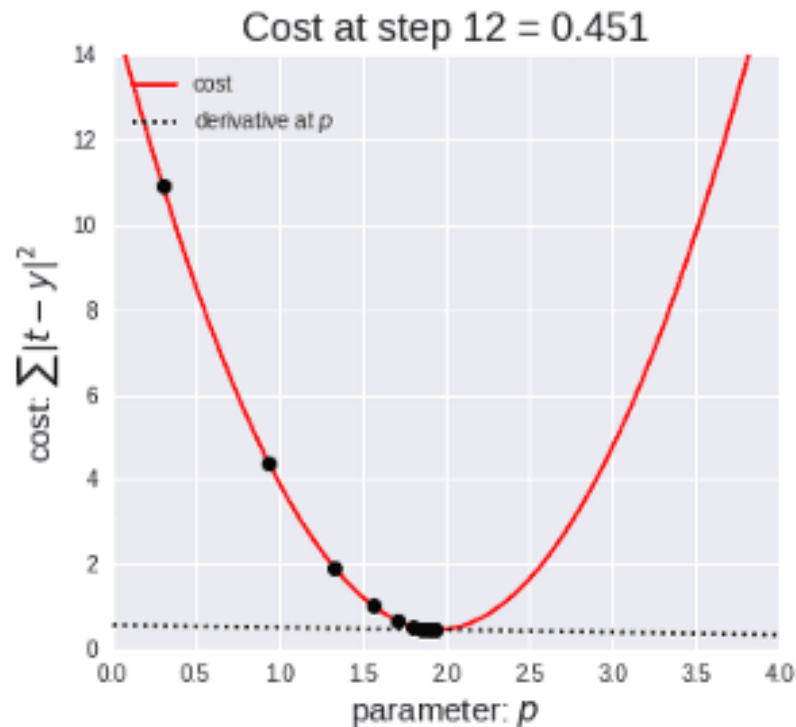


Backprop: Rumelhart, Hinton, and Williams, 1986

Deep learning

$$\hat{y} = f(W \cdot x + b)$$

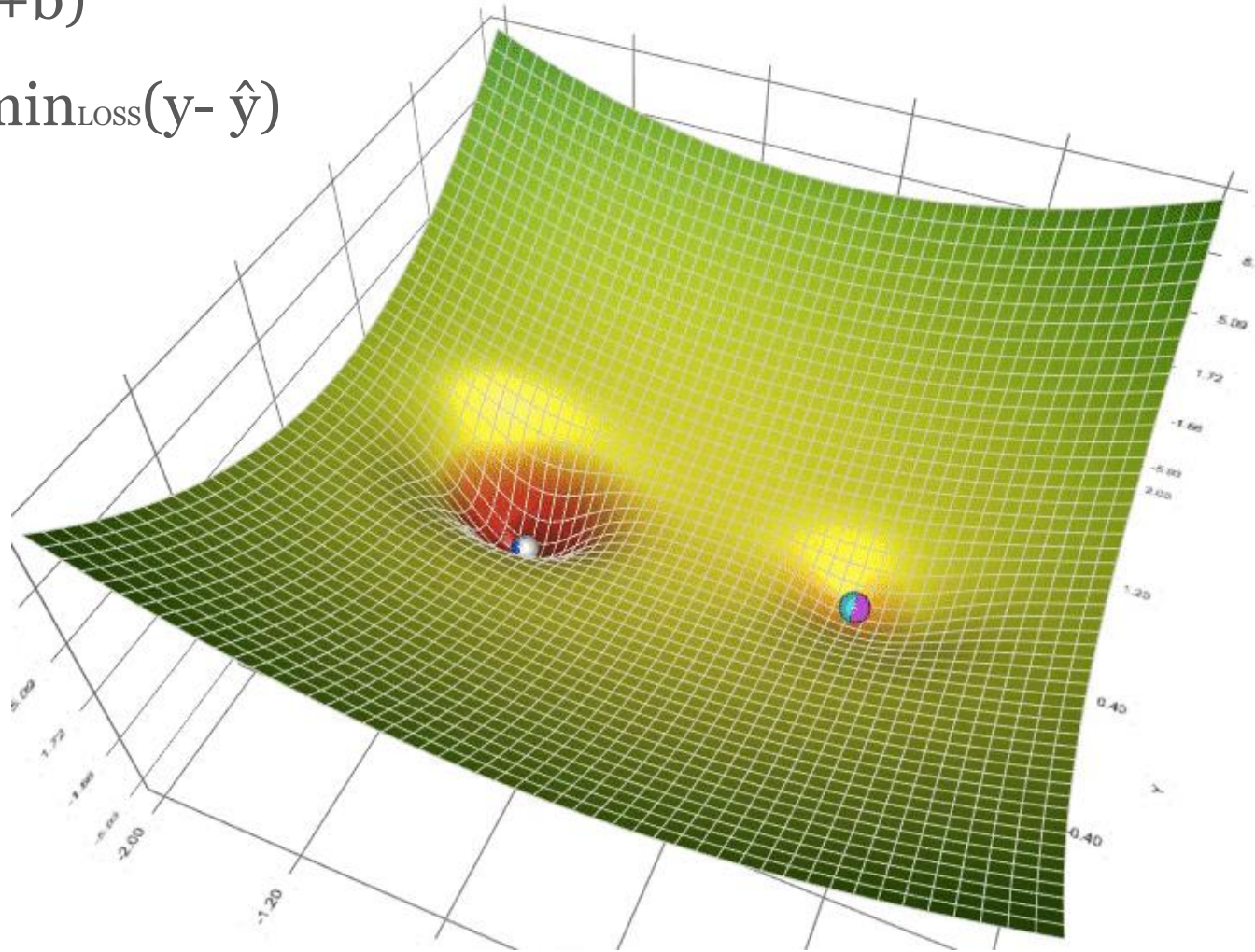
$$\text{target} = \min_{\text{LOSS}}(y - \hat{y}) \quad w_{\text{new}} = w_{\text{old}} + \eta \Delta w$$



Deep learning

$$\hat{y}=f(W \cdot x+b)$$

$$\text{target}=\min_{\text{LOSS}}(y-\hat{y})$$



Transformer

mac999/Al_foundation_tutorial

Token

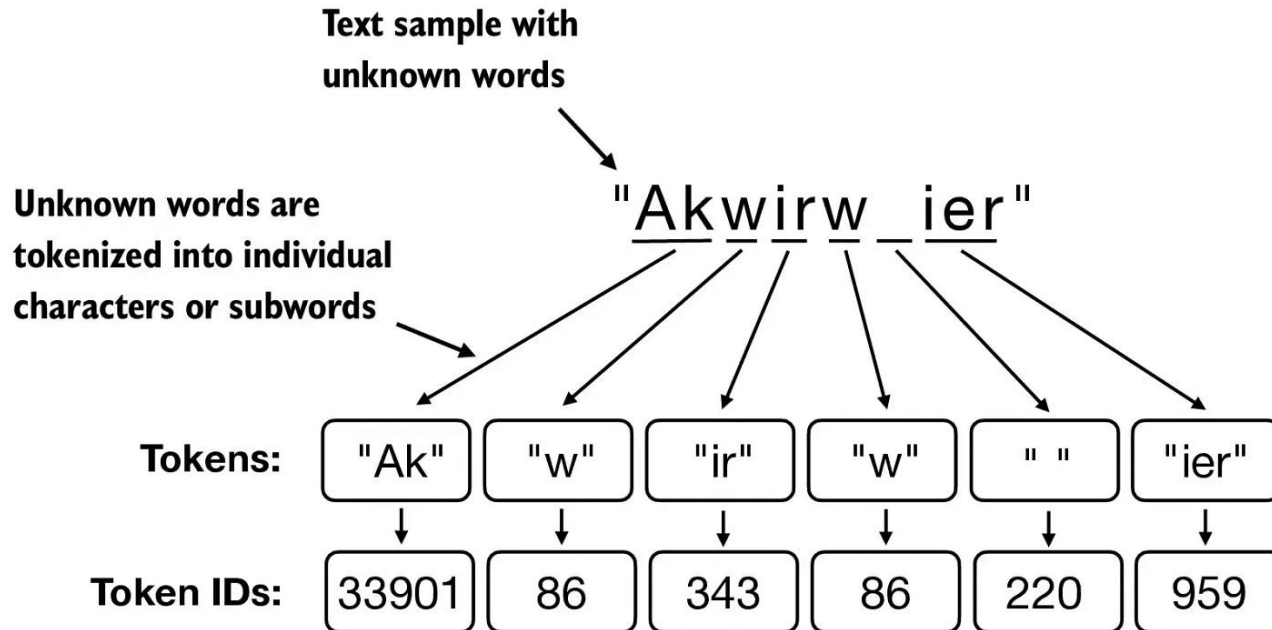
✓ a c

Token	ID
How	10
Are	4
You	13

	Tokens			Token IDs		
how are you	how	are	you	10	5	13
hi there	hi	there		4	6	
hi	hi			4		

Token

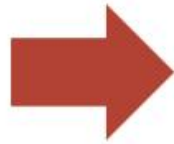
BPE(Byte Pair Encoding)



```
tokenizer.encode('123234234')
[10163, 24409, 24409]
tokenizer.encode('123')
[10163]
tokenizer.encode('234')
[24409]
tokenizer.encode('234')
[24409]
```

Embedding

Vocabulary:
Man, woman, boy,
girl, prince,
princess, queen,
king, monarch

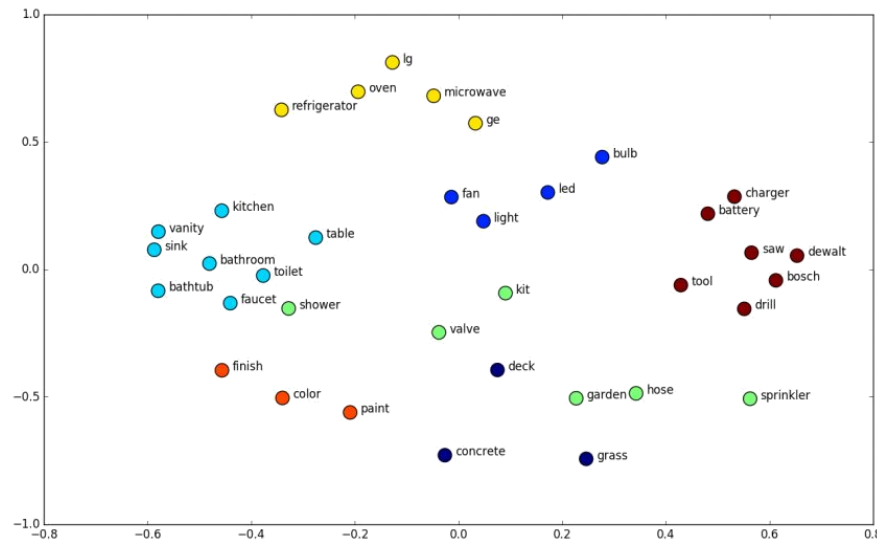


	Femininity	Youth	Royalty
Man	0	0	0
Woman	1	0	0
Boy	0	1	0
Girl	1	1	0
Prince	0	1	1
Princess	1	1	1
Queen	1	0	1
King	0	0	1
Monarch	0.5	0.5	1

Each word gets a
1x3 vector

Similar words...
similar vectors

[Intro to Word Embeddings and
Vectors for Text Analysis.](http://suriyadeepan.github.io/)



Embedding

Word2Vec (Google, 2013)

Skip-gram과 CBOW(Continuous Bag of Words) 두 가지 학습 방법 사용.

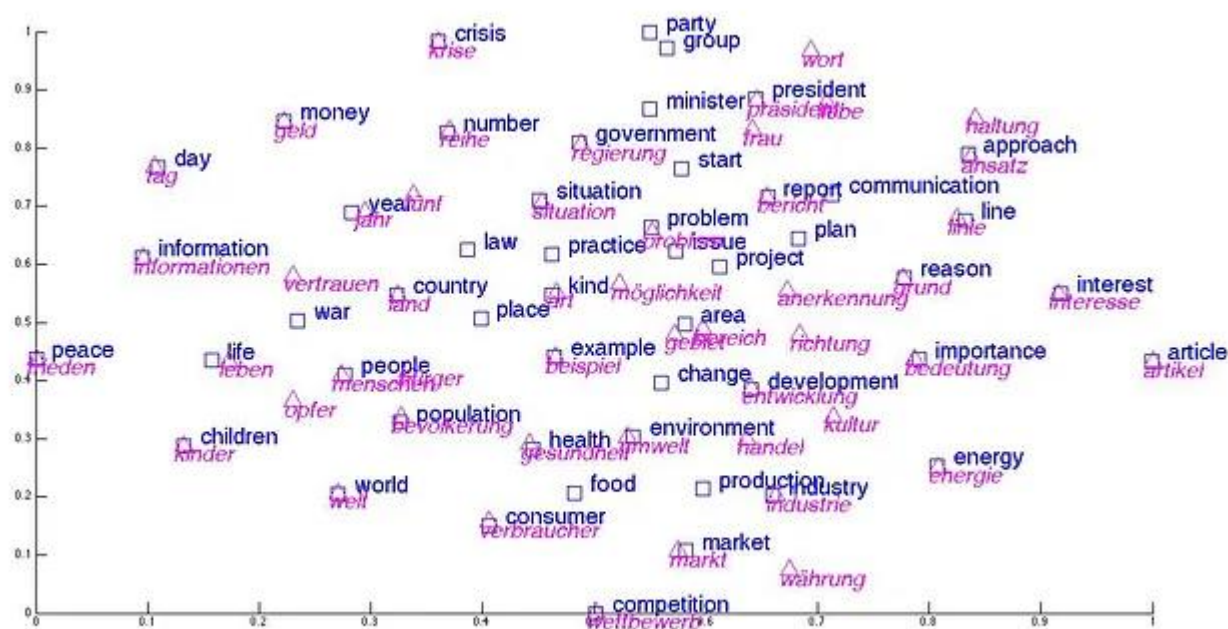
Skip-gram은 중심 단어를 입력으로 주고, 주변 단어를 예측. CBOW는 주변단어들로 중심단어를 예측

"The quick brown fox jumps" 문장 중심 단어 " fox"의 윈도우 크기가 2라면, 주변 단어는 " brown " 과 " jumps " 가 됨

Word2Vec은 주변 단어를 예측하는 loss를 최소화하며 학습. 임베딩 모델들은 주로 대규모 텍스트 데이터에서 비지도 학습 방식으로 학습

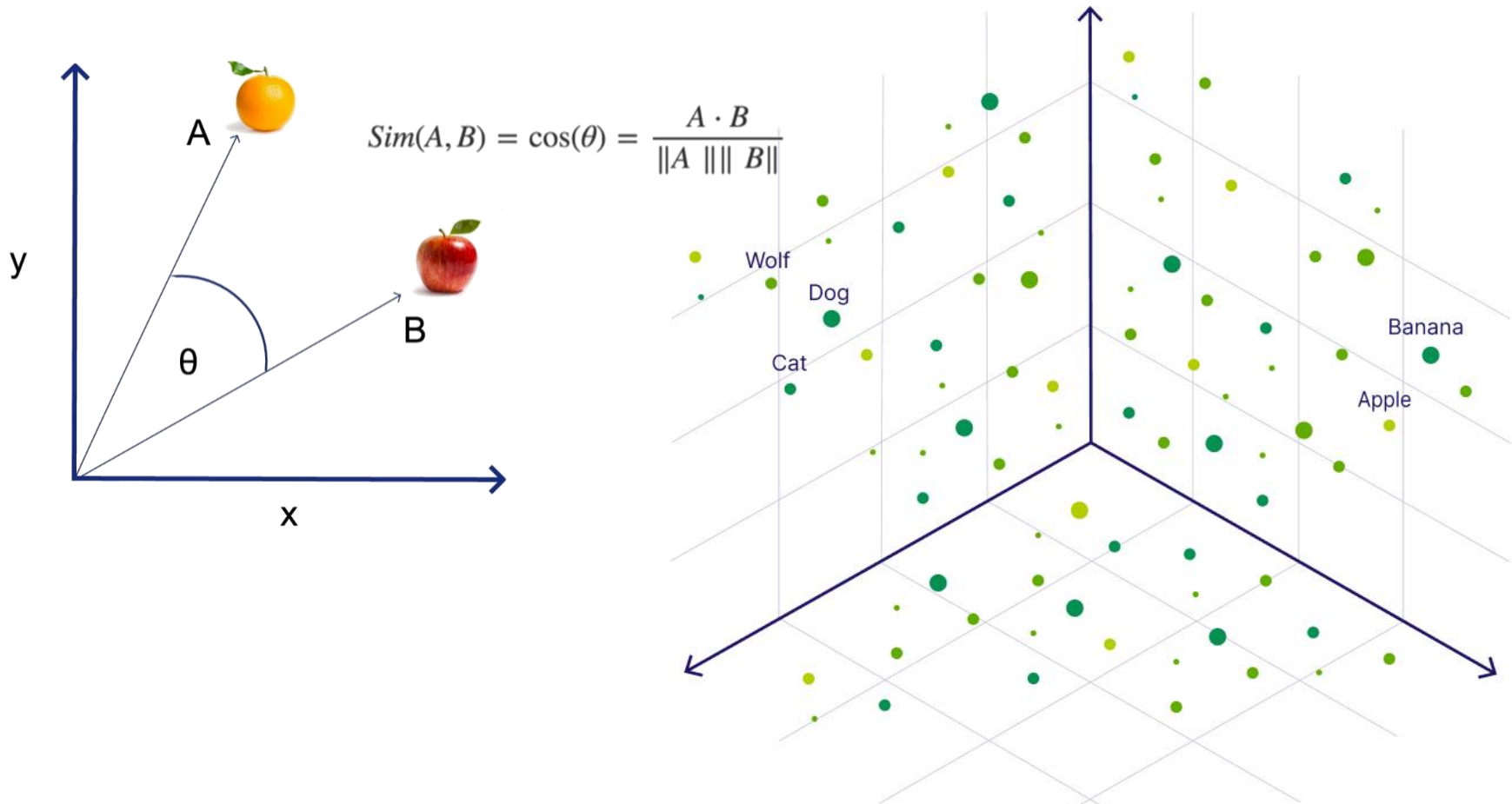
```
for center_word, context_words in dataset:
    center_vec = word_embedding(center_word)
    for context in context_words:
        context_vec = word_embedding(context)
        loss += negative_sampling_loss(center_vec, context_vec)
    loss.backward()
    optimizer.step()
```

Embedding



Embedding vector distance

Cosine similarity

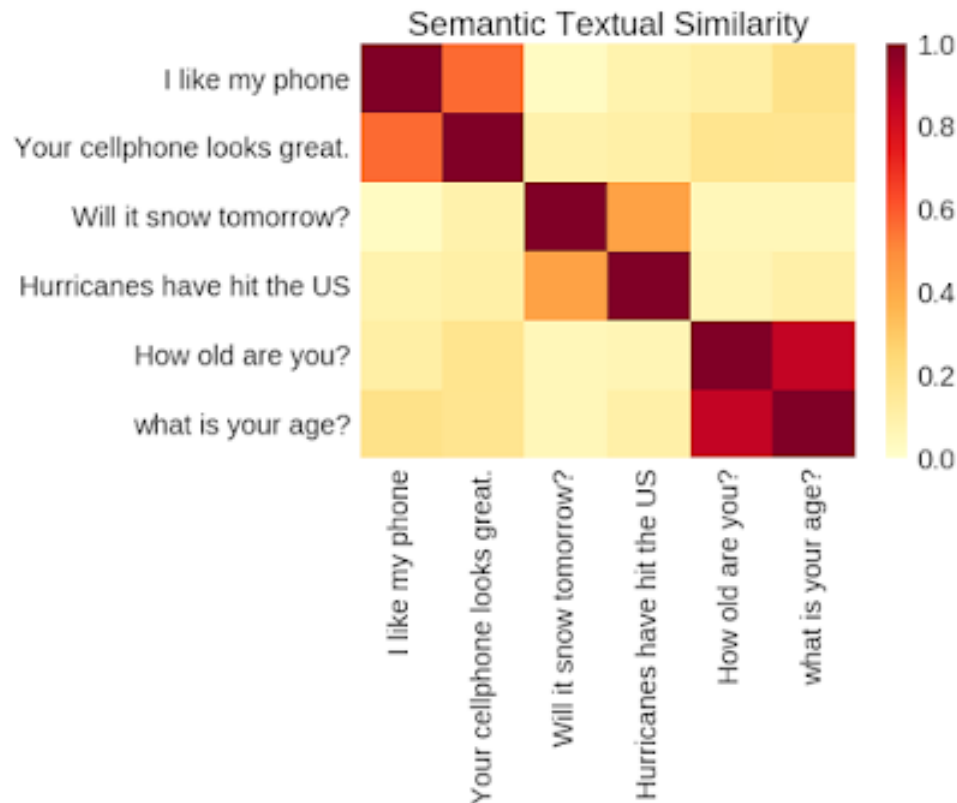


```
distance = F.cosine_similarity(embedding("apple"), embedding("banana"))
```

Sequence semantic similarity

If	the universe	has	an explanation of its existence	, that explanation is God.
	ARG0	VERB	ARG1	

If the universe has an explanation of its existence	,	that explanation	is	God	.
ARGM-ADV		ARG1	VERB	ARG2	



Computing semantic similarity of texts based on deep graph learning with ability to use semantic role label information | Scientific Reports

Advances in Semantic Textual Similarity

Attention

The FBI is chasing a criminal on the run .

The FBI is chasing a criminal on the run .

The FBI is chasing a criminal on the run .

The FBI is chasing a criminal on the run .

The FBI is chasing a criminal on the run .

The FBI is chasing a criminal on the run .

The FBI is chasing a criminal on the run .

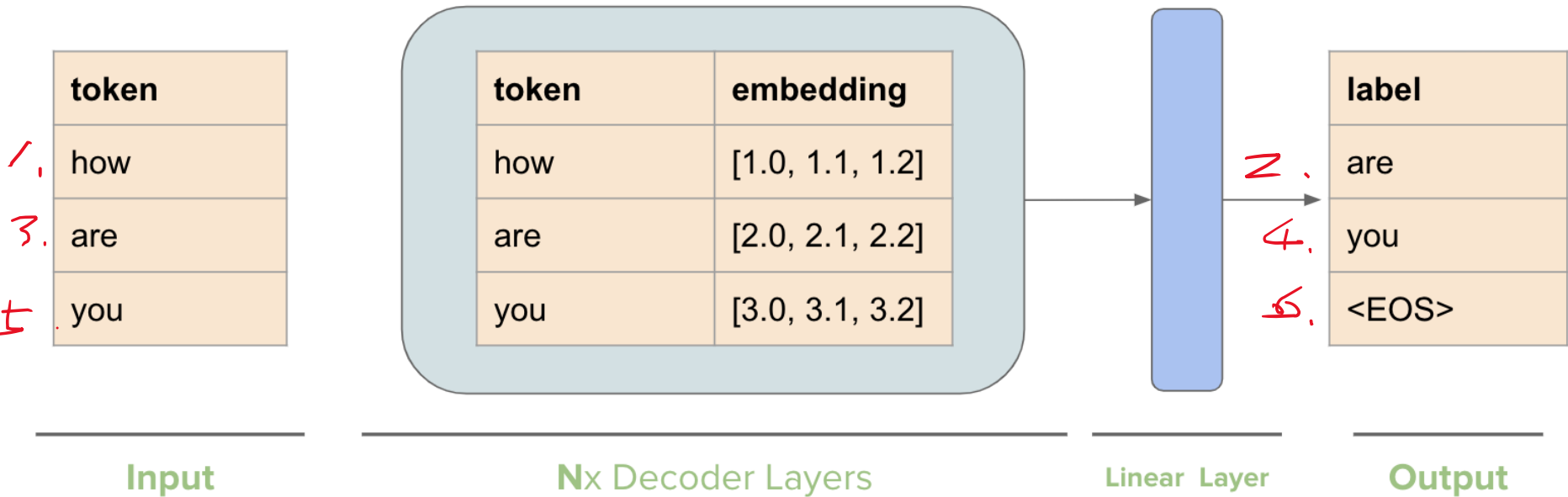
The FBI is chasing a criminal on the run .

The FBI is chasing a criminal on the run .

The FBI is chasing a criminal on the run .

Transform Problem

auto-regressive manner



Sequence context calculation

Query
Key

token	embedding
how	[1.0, 1.1, 1.2]
are	[2.0, 2.1, 2.2]
you	[3.0, 3.1, 3.2]

Query/Key Dot Product

	← Key →		
	how	are	you
↑ Query ↓	how	how are	how you
	are	are are	are you
	you	you are	you you

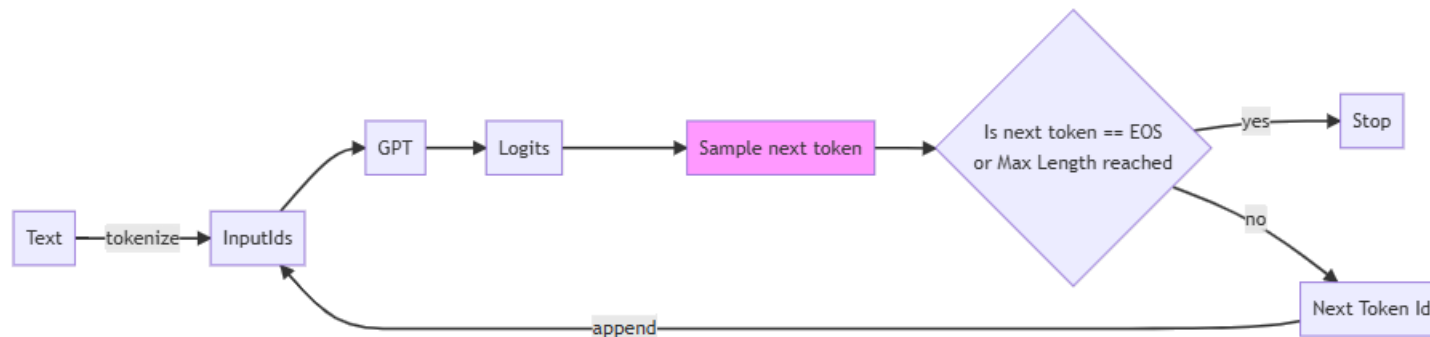
Sequence context calculation

Cosine similarity for sequence context calculation

Query/Key Dot Product

how → are
how are → you
how are you → <EOS>

		Key					label
		how	are	you			
Query	how	how	how	are	how		are
	are	are	are	are	are		you
	you	you	you	are	you		<EOS>



Decoding strategies in
Decoder models (LLMs)
- Sanjaya's Blog

Sequence context calculation

Attention Mask

Query/Key Dot Product					Attention Mask				
← Key →									
	how		are		you		how	are	you
how	how	are	are	you	you	how	0	-inf	-inf
are	are	are	are	you	you	are	0	0	-inf
you	you	are	are	you	you	you	0	0	0

Attention equation

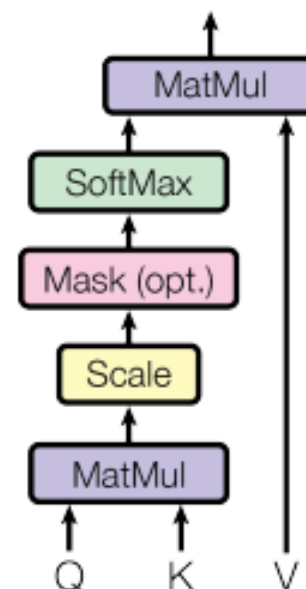
Query = $Q \times Q_w$

Key = $K \times K_w$

Value = $V \times V_w$

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V$$

Scaled Dot-Product Attention



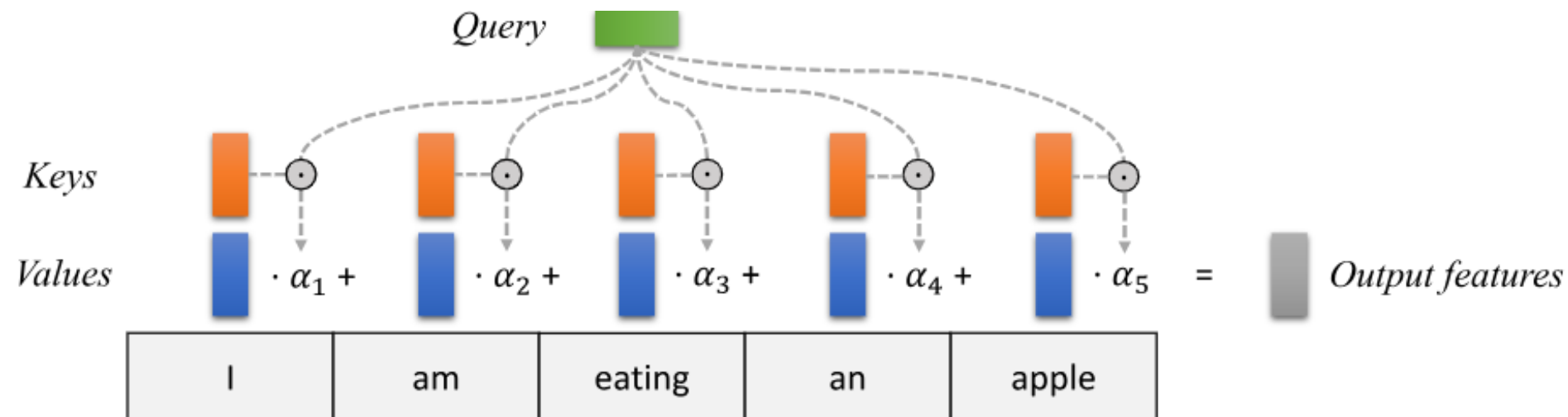
$$\begin{bmatrix} \textit{how} & \textit{are} & \textit{you} \\ \textit{how} & 0.5 & 0.1 & 0.4 \\ \textit{are} & 0.1 & 0.8 & 0.3 \\ \textit{you} & 0.4 & 0.3 & 0.1 \end{bmatrix}_{3 \times 3}$$

$$\begin{bmatrix} \textit{how} & \textit{are} & \textit{you} \\ \textit{how} & 0.38 & 0.26 & 0.35 \\ \textit{are} & 0.23 & 0.48 & 0.29 \\ \textit{you} & 0.38 & 0.34 & 0.28 \end{bmatrix}_{3 \times 3}$$

$$\begin{bmatrix} \textit{how} & \textit{are} & \textit{you} \\ \textit{how} & 0.38 & 0.26 & 0.35 \\ \textit{are} & 0.23 & 0.48 & 0.29 \\ \textit{you} & 0.38 & 0.34 & 0.28 \end{bmatrix}_{3 \times 3}$$

$$\begin{bmatrix} 1.1 & 1.2 \\ 2.1 & 2.2 \\ 3.1 & 3.2 \end{bmatrix}_{3 \times 2}$$

Attention equation

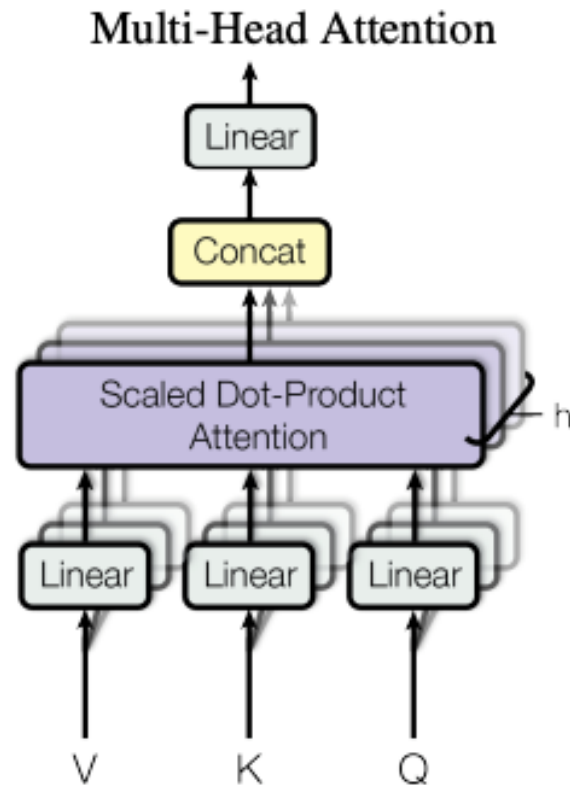


Token	Token ID
how	10
are	3
you	5

$$\begin{array}{c} \text{how} \\ \text{are} \\ \text{you} \end{array} \begin{array}{c} 10 \\ 3 \\ 5 \end{array} \longrightarrow \begin{bmatrix} 1.1 & 1.2 \\ 2.1 & 2.2 \\ 3.1 & 3.2 \end{bmatrix}_{3 \times 2} \longrightarrow \begin{bmatrix} \textit{how} & 2.0630 & 2.1630 \\ \textit{are} & 2.1523 & 2.2523 \\ \textit{you} & 2.0020 & 2.1020 \end{bmatrix}_{3 \times 2}$$

Context Feature

Multihead Attention



Position encoding

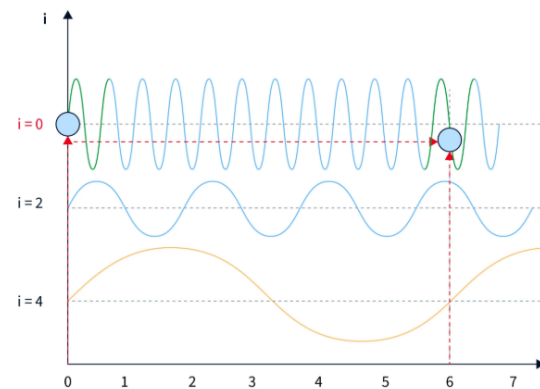
how	are	you	?	See	you	soon	<EOS>	<PAD>	<PAD>
1	2	3	4	5	6	7	8	9	10

$$PE_{(pos,i)} = \begin{cases} \sin\left(\frac{pos}{10000^{i/d_{model}}}\right) & \text{if } i \bmod 2 = 0 \\ \cos\left(\frac{pos}{10000^{(i-1)/d_{model}}}\right) & \text{otherwise} \end{cases}$$

pos = position of word in sequence

d = embedding dimension

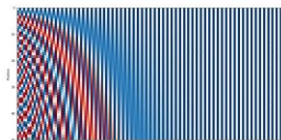
i = index in embedding vector. $0 \leq i < d/2$



Positional Encoding Matrix for d=4 n=100

k i=0 i=0 i=1 i=1

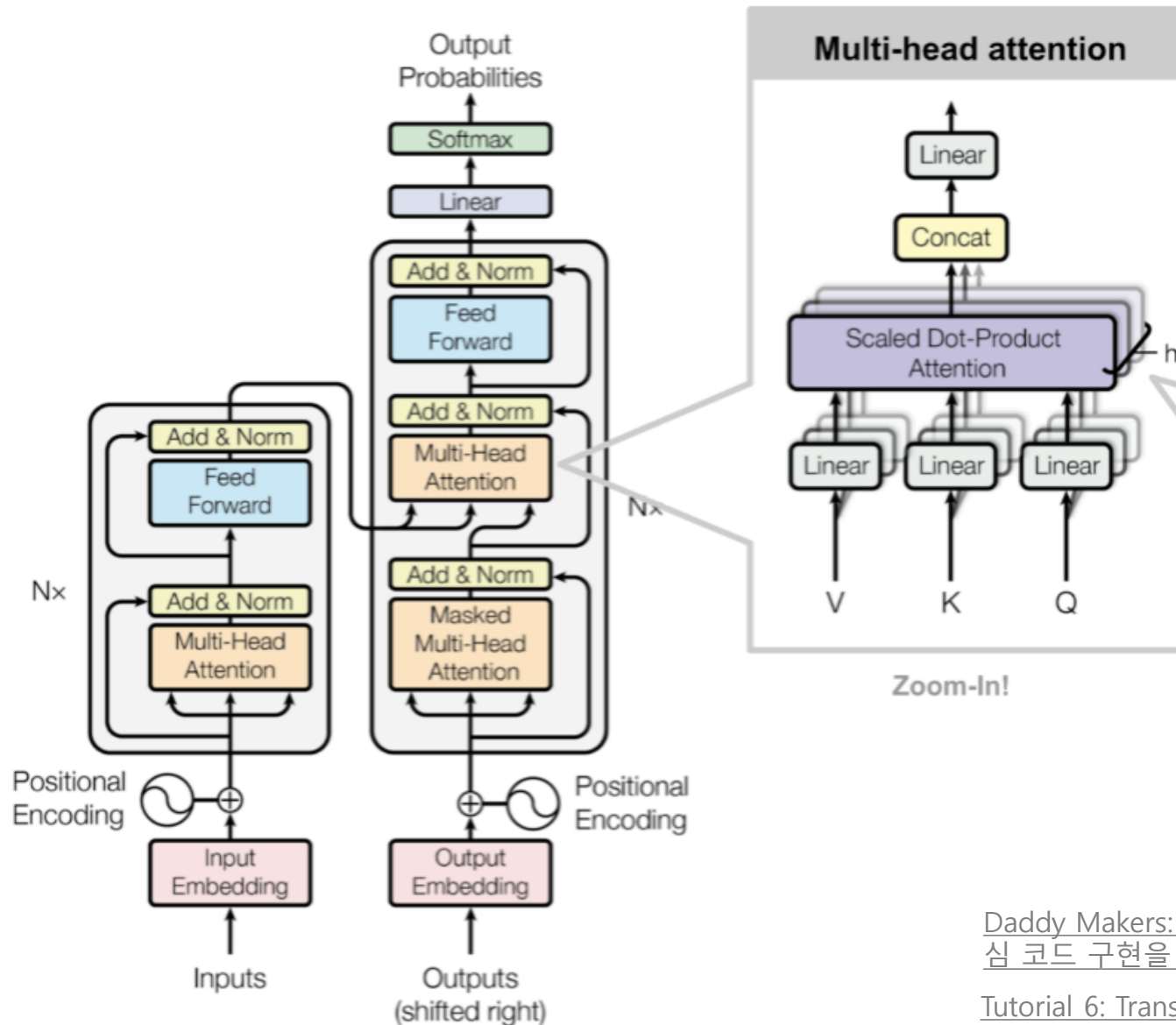
The	0	P00=sin(0) = 0	P01=cos(0) = 1	P02=sin(0) = 0	P03=cos(0) = 1
dog	1	P10=sin(1/1) = 0	P11=cos(1/1) = 0.54	P12=sin(1/10) = 0.10	P13=cos(1/10) = 1.0
ran	2	P20=sin(2/1) = 0.91	P21=cos(2/1) = -0.42	P22=sin(2/10) = 0.20	P23=cos(2/10) = 0.98
fast	3	P30=sin(3/1) = 0.14	P31=sin(3/1) = -0.99	P32=sin(3/10) = 0.30	P33=cos(3/10) = 0.96



[A Gentle Introduction to Positional Encoding in Transformer Models, Part 1 - MachineLearningMaster.com](#)

[Learning Position with Positional Encoding - Scaler Topics](#)

Transformer Architecture

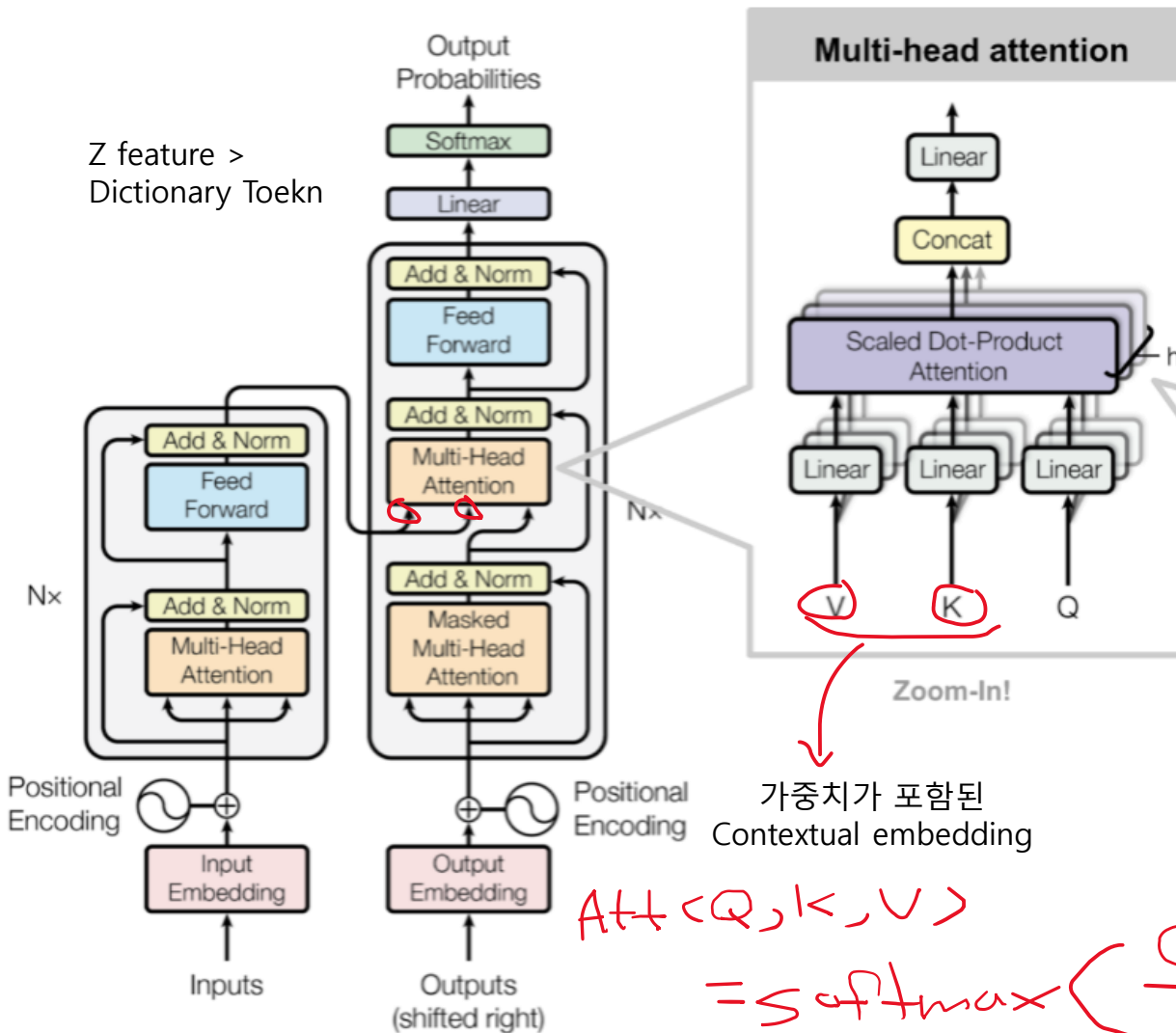


Daddy Makers: 딥러닝 모델 트랜스포머 인코더 핵심 코드 구현을 통한 동작 메커니즘 이해하기

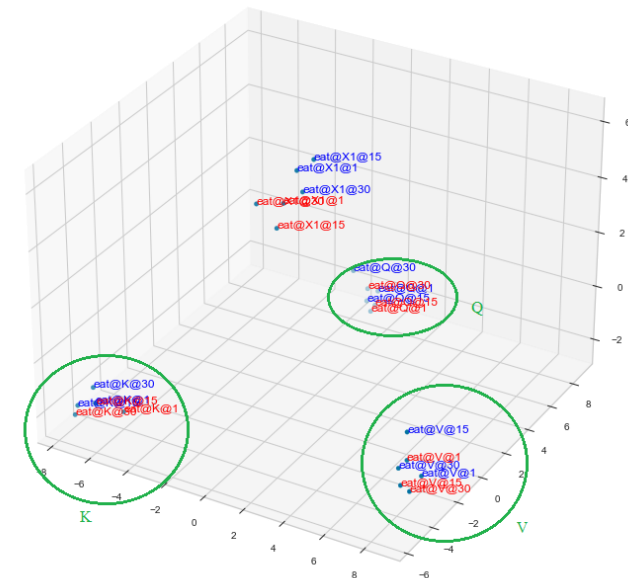
Tutorial 6: Transformers and Multi-Head Attention — UvA DL Notebooks v1.2 documentation

Transformer

Z feature >
Dictionary Toekn



Self attention
contextual embedding



Self Attention Space

가중치가 포함된
Contextual embedding

$$Att(Q, K, V)$$

$$= \text{softmax} \left(\frac{Q \cdot K^T}{\sqrt{d_k}} \right) V$$

ex) source code

Training transformer model

Contextual embedding

Self-Attention 을 통한 벡터값은 어떤 문맥(문장)에 놓이느냐에 따라 그 의미를 반영하는 고유한 벡터표현을 가지게 됨. 예를 들어, "과일 사과"의 임베딩과 "행위 사과"의 임베딩은 주변 단어의 영향을 받아 서로 다른 벡터 값을 가지게 됨

"나는 학생" → 어텐션 → out → FFN → logits (10000차원) → softmax → "입니다" (예측)

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V$$


out	문맥 반영된 벡터 (차원: <code>[seq_len, d_model]</code>)
Feed-Forward Network (FFN)	각 <code>out</code> 벡터를 비선형적으로 변환 (정보 강화)
디코더 스택 마지막 출력	최종 출력 hidden state
Linear + Softmax	각 hidden state에 대해 전체 vocabulary 크기만큼의 로짓(logits)을 만듦 → softmax를 적용해 확률로 변환
토큰 예측	확률이 가장 높은 단어가 다음 단어로 예측됨 (예: <code>"student"</code> 다음에 <code>"is"</code> 예측 등)

Training transformer model

1. 처음엔 QK^T 가 의미 없는 유사도를 계산함 → softmax 후 V를 평균해서 출력
2. 이 결과가 예측 라벨(예: 다음 단어)과 멀면 loss 증가
3. 역전파로 Q, K, V를 만드는 가중치 W_Q, W_K, W_V 가 업데이트됨
4. 배치 데이터셋에 대해 1-3을 반복하면서 각 QKV가 문맥에서 다른 역할을 하도록 학습

Scaled Dot Product Attention

이 모듈은 어텐션의 기본 동작을 수행하는 핵심 요소.

입력으로 주어진 Q(Query), K(Key), V(Value) 벡터를 이용하여 다음 연산을 수행

```
class ScaledDotProductAttention(nn.Module):
    def forward(self, Q, K, V):
        scores = torch.matmul(
            Q, K.transpose(-2, -1)) / math.sqrt(d_k)
        attn = F.softmax(scores, dim=-1)
        return torch.matmul(attn, V)
```

Multi-Head Attention

어텐션을 단일 벡터로 계산하면 정보 손실이 크기 때문에, 여러 개의 어텐션 "헤드"를 병렬로 실행한 후 결과를 concat하여 사용

```
class MultiHeadAttention(nn.Module):
    def forward(self, Q, K, V):
        # Q, K, V 선형 변환 및 split
        # 각 헤드별 attention 계산
        # concat 후 출력 선형 변환
        return output
```

Positional Encoding

트랜스포머는 순서를 고려하지 않기 때문에 각 단어의 위치 정보를 인코딩

이를 위해 사인(sin), 코사인(cos) 함수를 기반으로 위치 인코딩 벡터를 생성

```
class PositionalEncoding(nn.Module):
    def forward(self, x):
        # sin, cos 벡터 계산 후 입력에 더함
        return x + self.pe[:, :x.size(1)]
```

```
class EncoderLayer(nn.Module):
    def forward(self, x):
        x = x + self_attn(x, x, x) # Residual
        x = LayerNorm(x)
        x = x + FFN(x) # Residual
        x = LayerNorm(x)
        return x
```

Transformer scratch code

1. 입력 및 라벨 시퀀스

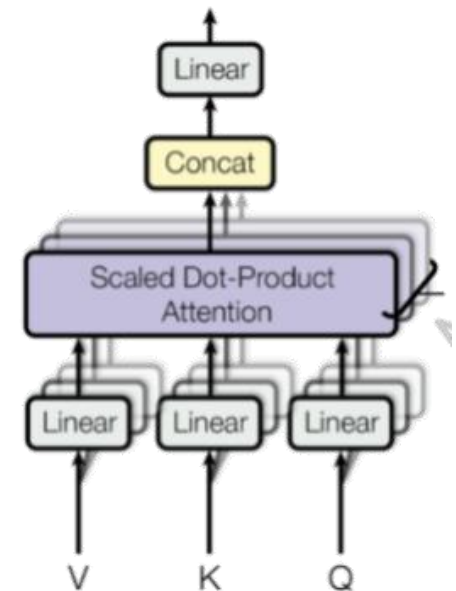
```
input_sentence = ["I", "am", "a", "student"]    # 영어 입력
target_sentence = ["<sos>", "저는", "학생", "입니다"] # 한국어 출력 입력 (디코더 입력)
target_labels = ["저는", "학생", "입니다", "<eos>"] # 예측할 실제 정답 (디코더 출력)
```

2. 인코더

```
X = embed(input_sentence)    # 입력 시퀀스 임베딩 → [x_l, x_am, x_a, x_student]
X = position_encoding(X)    # 위치 인코딩
Q_enc = linear_Q(X)         # Q from encoder input
K_enc = linear_K(X)         # K from encoder input
V_enc = linear_V(X)         # V from encoder input
```

인코더 출력: 입력 문맥이 반영된 벡터들

```
encoder_outputs = self_attention(Q_enc, K_enc, V_enc)
```



Transformer scratch code

```
Y = ["<sos>"]          # 디코더 초기 입력 (<sos> 부터 시작)
logits_sequence = []    # 출력 로짓을 저장
```

```
# 3. 디코더
for t in range(len(target_labels)):
```

```
    Y_embed = embed(Y)
```

현재까지 생성된 디코더 입력 임베딩

```
    # 디코더 Self-Attention (마스킹 포함)
```

```
    Q_dec = linear_Q_dec(Y_embed)
```

Q from decoder input

```
    K_dec = linear_K_dec(Y_embed)
```

K from decoder input

```
    V_dec = linear_V_dec(Y_embed)
```

V from decoder input

```
    dec_out = masked_self_attention(Q_dec, K_dec, V_dec) # 디코더 내부 문맥 계산(마스킹)
```

```
    # Cross-Attention: 디코더 → 인코더 입력 참조
```

```
    Q_cross = linear_Q_cross(dec_out)
```

Q from 디코더 문맥

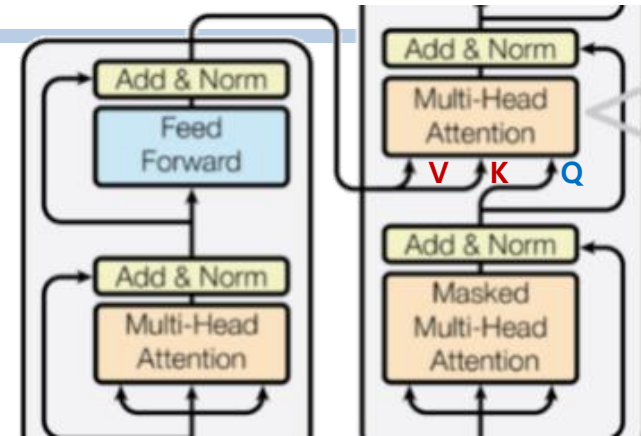
```
    K_cross = linear_K_cross(encoder_outputs)
```

K from 인코더 출력

```
    V_cross = linear_V_cross(encoder_outputs)
```

V from 인코더 출력

```
    cross_out = attention(Q_cross, K_cross, V_cross) # 입력 문장 정보에 기반한 번역 벡터
```



Transformer scratch code

```
for t in range(len(target_labels)):
```

```
...
```

```
# 출력 로짓 생성
```

```
logits = linear_output(cross_out)    # vocab 크기의 예측 로짓
```

```
logits_sequence.append(logits[-1])  # 토큰 예측
```

```
# 다음 디코더 입력을 위해 정답을 넣음 (teacher forcing)
```

```
next_token = target_sentence[t + 1]    # "저는" 다음은 "학생"
```

```
Y.append(next_token)                  # 다음 루프를 위한 디코더 입력에 추가
```

```
# 4. 손실(Loss) 계산
```

```
loss = 0
```

```
for logit, label_token in zip(logits_sequence, target_labels):
```

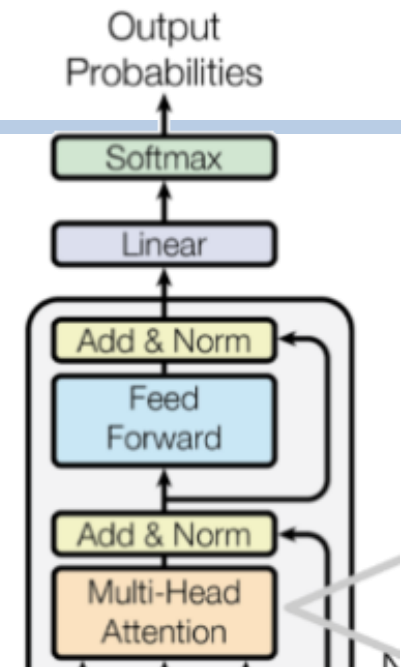
```
    label_id = vocab_id(label_token)    # 정답 토큰을 ID로 변환
```

```
    loss += cross_entropy_loss(logit, label_id)    # 예측 분포와 정답 ID 간의 CrossEntropy Loss
```

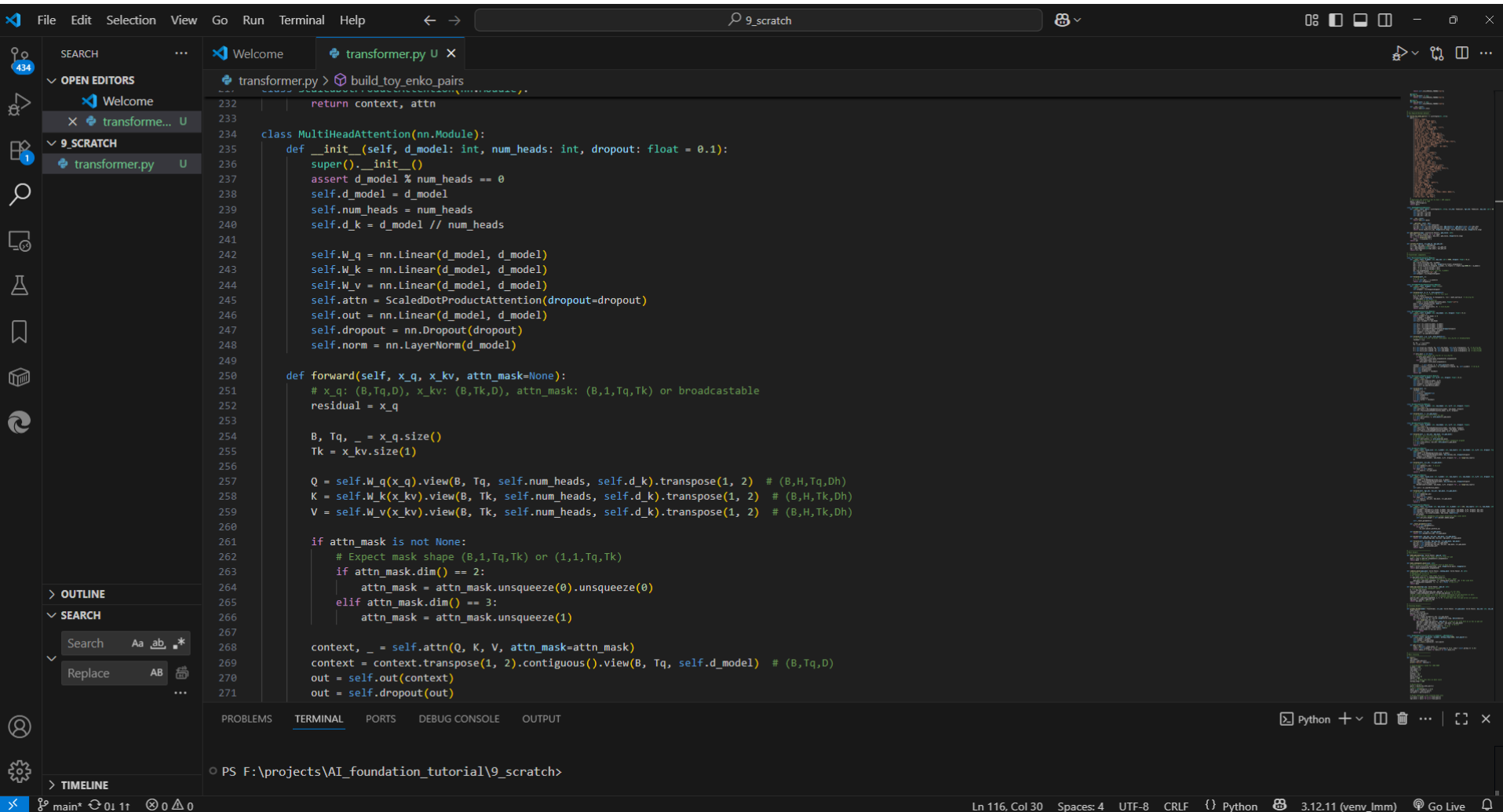
```
loss = loss / len(target_labels)
```

```
# 평균 손실값 (시퀀스 길이만큼 나눔)
```

```
# 5. 역전파 및 파라미터 업데이트 수행
```



Transformer scratch code



The image shows a Visual Studio Code editor window with a dark theme. The main editor area displays the code for a Transformer scratch implementation in a file named `transformer.py`. The code is written in Python and includes a `MultiHeadAttention` class. The class has an `__init__` method that initializes the model parameters and a `forward` method that implements the multi-head attention mechanism. The code is organized into several sections, with line numbers visible on the left side of the editor. The left sidebar shows the 'EXPLORER' view with the '9_SCRATCH' folder selected, and the 'transformer.py' file open. The bottom status bar shows the current file path as `F:\projects\AI_foundation_tutorial\9_scratch\` and the Python version as `3.12.11 (venv, Imm)`.

```
232     return context, attn
233
234 class MultiHeadAttention(nn.Module):
235     def __init__(self, d_model: int, num_heads: int, dropout: float = 0.1):
236         super().__init__()
237         assert d_model % num_heads == 0
238         self.d_model = d_model
239         self.num_heads = num_heads
240         self.d_k = d_model // num_heads
241
242         self.W_q = nn.Linear(d_model, d_model)
243         self.W_k = nn.Linear(d_model, d_model)
244         self.W_v = nn.Linear(d_model, d_model)
245         self.attn = ScaledDotProductAttention(dropout=dropout)
246         self.out = nn.Linear(d_model, d_model)
247         self.dropout = nn.Dropout(dropout)
248         self.norm = nn.LayerNorm(d_model)
249
250     def forward(self, x_q, x_kv, attn_mask=None):
251         # x_q: (B,Tq,D), x_kv: (B,Tk,D), attn_mask: (B,1,Tq,Tk) or broadcastable
252         residual = x_q
253
254         B, Tq, _ = x_q.size()
255         Tk = x_kv.size(1)
256
257         Q = self.W_q(x_q).view(B, Tq, self.num_heads, self.d_k).transpose(1, 2) # (B,H,Tq,Dh)
258         K = self.W_k(x_kv).view(B, Tk, self.num_heads, self.d_k).transpose(1, 2) # (B,H,Tk,Dh)
259         V = self.W_v(x_kv).view(B, Tk, self.num_heads, self.d_k).transpose(1, 2) # (B,H,Tk,Dh)
260
261         if attn_mask is not None:
262             # Expect mask shape (B,1,Tq,Tk) or (1,1,Tq,Tk)
263             if attn_mask.dim() == 2:
264                 attn_mask = attn_mask.unsqueeze(0).unsqueeze(0)
265             elif attn_mask.dim() == 3:
266                 attn_mask = attn_mask.unsqueeze(1)
267
268         context, _ = self.attn(Q, K, V, attn_mask=attn_mask)
269         context = context.transpose(1, 2).contiguous().view(B, Tq, self.d_model) # (B,Tq,D)
270         out = self.out(context)
271         out = self.dropout(out)
```

Train and Dataset

instruction string · lengths 	input string · lengths 	output string · lengths 
Find the most frequently used word in the given paragraph.	The cat is black, the dog is brown and the cow is white.	The: 3 times.
Generate a regex pattern to match the following string	String: "Hello123"	\w{5}[0-9]{3}
Write a code snippet to update the value of an element in an array.	Array: {3, 2, 5, 7, 8, 1} Index: 3 New value: 6	array[3] = 6;
Predict a label for the given sentence.	The phone is red .	Label: Color.
Print out the 5th element of an array.	arr = [55, 32, 10, 11, 8]	print(arr[4])
Classify the following text into two categories: 'Sports' and...	The critically acclaimed film 'Avatar' is about a paraplegic Marine who embark...	Entertainment
Create a formula in Excel to calculate the total cost given the price, tax,...	Price: 10 Tax: 0.08 Shipping: 6	=10+0.08*10+6
Classify the following given expression. Your response should be...	x = 0.5	float literal
Classify the following code as either a python function or code section.	x = "hello" z = x + " world"	Code section.
Generate a regex expression to match strings of length 8 with given...	Allowed letters: A-F and digits: 0-9	^[A-F0-9]{8}\$

Train and Dataset

Datasets:  junelee/ **sharegpt_deep1_ko**   like 53

 Dataset card  Viewer  Files and versions  Community

Dataset Preview

 API

 View in Dataset Viewer

Split (1)

train



► The full dataset viewer is not available (click to read why). Only showing a preview of the rows.

input string	instruction string	output string
	토니 로빈스가 설명하는 인간의 6가지 욕구를 요약해 보세요...	토니 로빈스는 인간의 행동과 동기를 이끄는 6가지 핵심 욕구를 설명합니다. 이 여섯 가지 욕구는 다음과 같습니다: 1. 확실성: 안전, 안정성, 예...
	고객 세그먼트가 잘 세분화되었는지 어떻게 알 수 있을까요? 세 가지로 요약할 수 있습니다.	1. 동질성: 세그먼트는 유사한 특성과 행동을 공유하는 고객으로 구성되어야 합니다. 2. 차별성: 세그먼트는 특성 및 행동 측면에서 다른 세그먼트...
	n Java에서 "이것은 {장소}에 있는 새로운 {객체}입니다"와 같은 문자열을 맵, {객체: "학생", "지점 3, 4"}로 바꾸고 "이 학생은 지점 3, 4...	을 사용하여 문자열의 자리 표시자를 맵의 값으로 바꿀 수 있습니다. 다음은 이를 수행하는 방법을 보여주는 코드 스니펫 예시입니다. ``java...
	전체 단락을 이와 같은 스타일로 작성하세요: 신들의 은총으로 은유적 언어의 신비롭고 수수께끼 같은 예술이 소환되어 우리 앞에 놓인 지침의 당혹...	보세요! 신성한 개입의 은총으로, 이해할 수 없고 불가사의한 은유적 언어의 예술이 우리 앞에 놓인 지침의 불가해한 전달 방식을 해명하기 위해 호...
	길이를 변경할 수 없습니다.	지식의 구도자 여러분, 잠시만요, 우리 앞에 제시된 지침의 복잡한 전달 방식을 설명하는 데 사용된 은유적 언어의 난해하고 수수께끼 같은 예술에 ...
	계속	은유적 언어의 사용은 단순히 보이는 이 지침에도 신비감과 초월성을 불어넣어 줍니다. 특히 '회상하다'라는 단어는 이 지침에 숨겨진 지식이나 비밀...
	다음과 같은 C++ 함수가 있습니다: void add_player(벡터& 플레이어)	주어진 C++ 함수의 'dummy' 변수는 'cin' 문을 사용하여

CLIP multimodal 2021

correct label: King Charles Spaniel

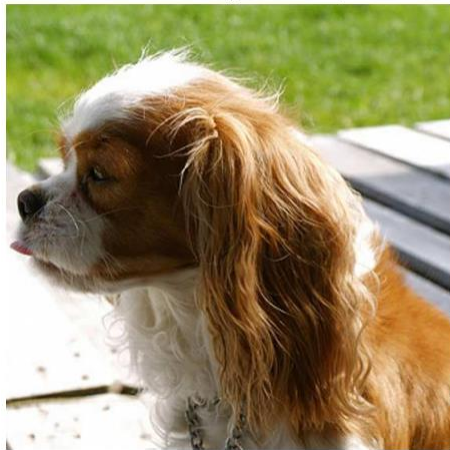


image
image

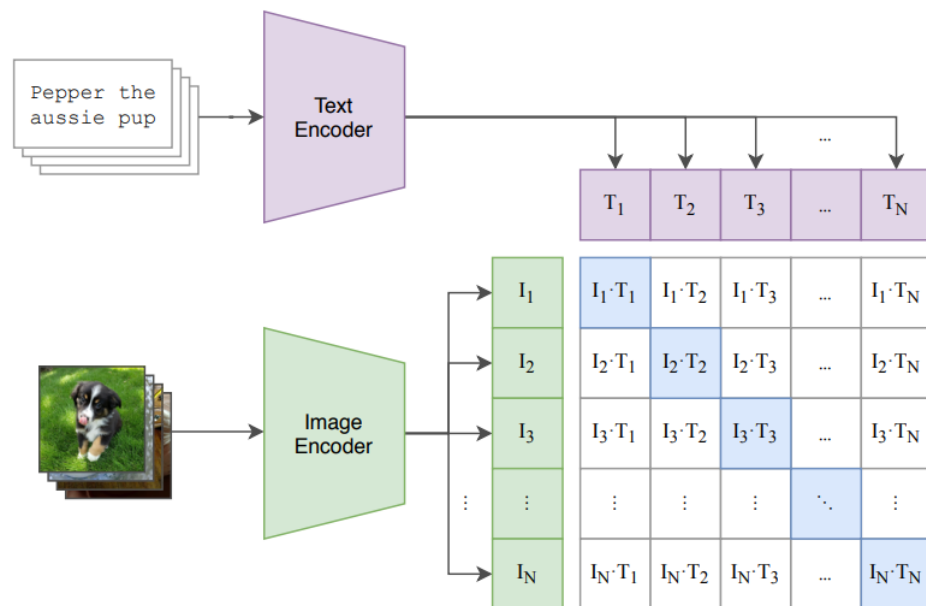


caption
list

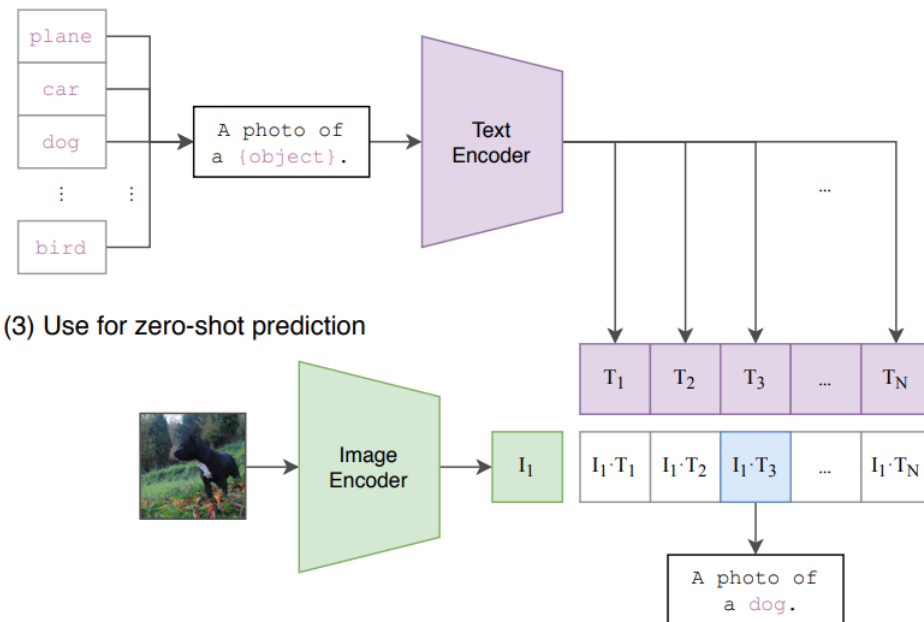
["Two young guys with shaggy hair look at their hands while hanging out in the yard.", "Two young, White males are outside near many bushes.", "Two men in green shirts are standing in a yard.", "A man in a blue shirt standing in a garden.", "Two friends enjoy time spent together."]

CLIP(Contrastive Language-Image Pre-Training. Open AI. 2021

(1) Contrastive pre-training

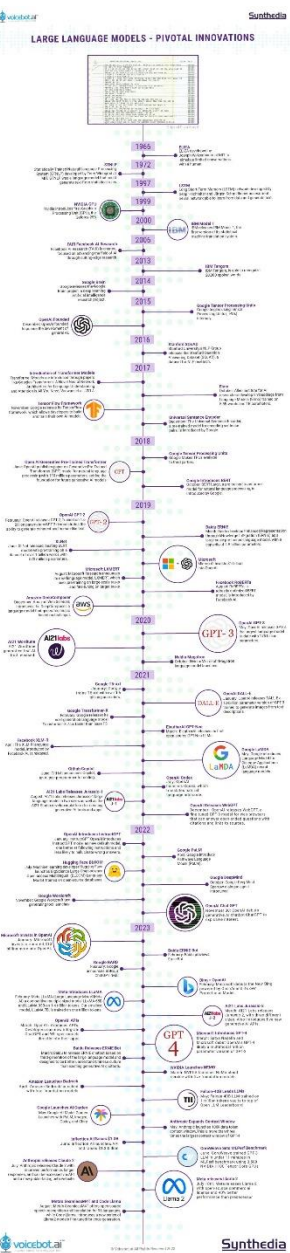


(2) Create dataset classifier from label text



(3) Use for zero-shot prediction

LLM large language model history



Introduction of Transformer Models
Transformer Models are introduced through papers like Google's Transformer: A Novel Neural Network Architecture for Language Understanding and Attention Is All You Need, Vaswani et al., 2017.

Tensor Flow Framework
November: Google releases its TensorFlow framework, which allows developers to build and train their own AI models.



2017

2018

Open AI Generative Pre-Trained Transformer
June: OpenAI publishes paper on Generative Pre-Trained Transformer (GPT) model for natural language processing with 110 million parameters, setting the foundation for future generative AI models



OpenAI Chat GPT
November 30: OpenAI debuts generative AI chatbot ChatGPT to explosive interest.



2023

Meta releases Llama 2
July 18th: Meta releases Llama 2 with open-source commercial license and 40% better performance than predecessor



$$y=f(w \cdot x+b)$$

Model	Parameters	Context Length
GPT-1	117M	512 tokens
GPT-4	175B - 1T (approx)	Up to 32,000 tokens
LLaMA 2	7B, 13B, 70B	4,096 tokens

$$\text{Memory (GB)} = \frac{\text{Parameters} \times \text{Bytes per Parameter}}{1024^3}$$

Substitute values:

$$\text{Memory (GB)} = \frac{175 \times 10^9 \times 2}{1024^3}$$

$$\text{Memory (GB)} = \frac{350 \times 10^9}{1,073,741,824} \approx 327 \text{ GB}$$

A Timeline of Large Language Model Innovation

Large Language Models as General Pattern Machines

Suvir Mirchandani Fei Xia Pete Florence Brian Ichter Danny Driess Montserrat Gonzalez Arenas
Kanishka Rao Dorsa Sadigh Andy Zeng



Sequence Transformation

Pattern transformations (symbolic)

Abstract Reasoning Corpus

Sequence Completion

Simple function classes (numeric)

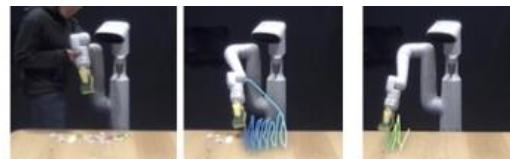
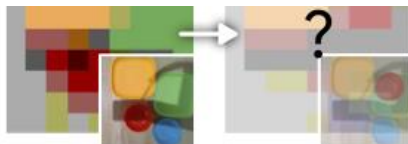
Sinusoid Extrapolation

Sequence Improvement

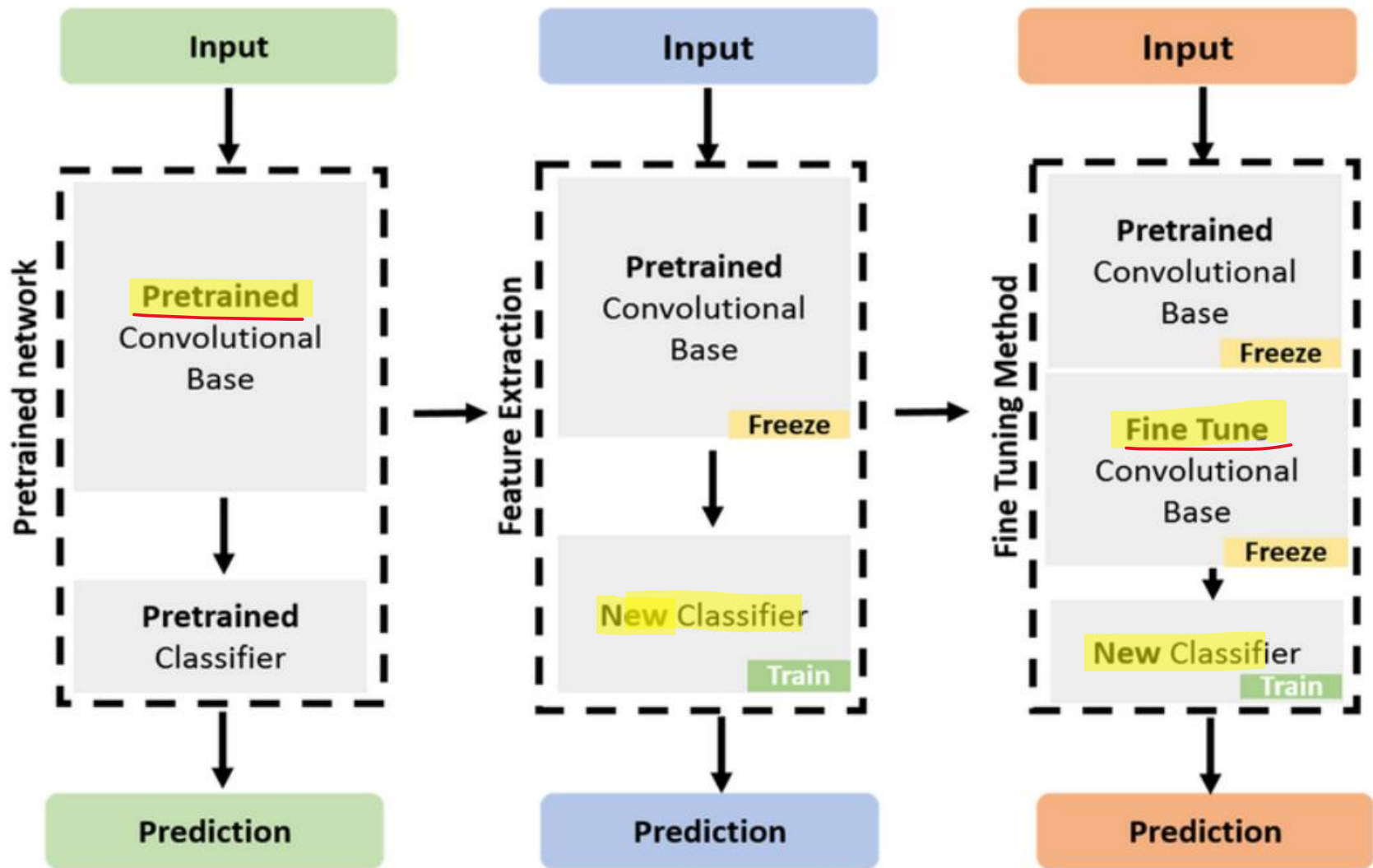
Online policies (numeric & symbolic)

CartPole

Limitations & Future Work. Today, the inference **costs** (and monetary costs) of using **LLMs** in the control loop are **quite high**. Predicting the next token for every sequence, e.g., every dimension of every time step in a trajectory, involves querying an LLM. State-action spaces which are higher dimensional and/or greater precision also result in longer representations, and thereby the extent to which they can be extrapolated or sequence optimized is bounded by the **context length** of models. **These limitations** may prevent deploying these models on more complex tasks in practice; however, they may be partially mitigated by incorporating mechanisms like external memory, and by current efforts to drive improvements in LLM **quantization** [85] and inference efficiency [86]. An additional limitation lies in the fact that, for best performance, some care must be taken to represent patterns with consistent tokenization (which requires knowledge of the model's tokenization scheme). Finally, as with any other language-only model, **LLM-based control** may (i) be **unpredictable**, and (ii) **lack visual/physical grounding**; thus, it is not currently suitable for application outside of constrained lab settings. We leave the exploration of these important topics for future work.



Fine turning



Fine turning

No fine tuning

```
[{'role': 'user', 'content': 'Hello doctor, I have bad acne. How do I get rid of it?'}]
```

Asking to truncate to max_length but no maximum length is provided and the model has no predefined maximum length. Default to no truncation.

The first thing you need to do is to keep your face clean. Wash your face twice a day with a mild soap. Do not use any oil base make up. Use only water base make up. Do not use any oil base cream. Use only water base cream. Do not use any oil base perfume. Use only water base perfume. Do not use any oil base toothpaste. Use only water base toothpaste. Do not use any oil base shaving cream. Use only water base shaving cream. Do not use any oil base hair oil. Use only water base hair oil. Do not use any oil base hair cream.

```
[{'role': 'user', 'content': 'What are the profile properties of IfcStructuralCurveMemberVarying described by?'}]
```

What are the profile properties of IfcStructuralCurveMemberVarying described by? So...

The profile properties of IfcStructuralCurveMemberVarying described by? So...

The profile properties of IfcStructuralCurveMemberVarying described by? So...

The profile properties of IfcStructuralCurveMemberVarying described by? So...

The profile properties of IfcStructuralCurveMemberVarying described by? So...

The profile properties of IfcStructuralCurveMemberVarying described by? So...

The profile properties of IfcStructuralCurveMemberVarying

```
[{'role': 'user', 'content': 'Hello doctor, I have bad acne. How do I get rid of it?'}]
```

Asking to truncate to max length but no maximum length is provided and the model has no predefined maximum length.

The first thing you need to do is to keep your face clean. Wash your face twice a day with a mild soap. Do not use m. Do not use any oil base perfume. Use only water base perfume. Do not use any oil base toothpaste. Use only water oil base hair oil. Use only water base hair oil. Do not use any oil base hair cream

```
[{'role': 'user', 'content': 'What are the profile properties of IfcStructuralCurveMemberVarying described by?'}]
```

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The profile properties of IfcStructuralCurveMemberVarying described by? So...

The profile properties of IfcStructuralCurveMemberVarying described by? So...

The profile properties of IfcStructuralCurveMemberVarying

Fine turning

After fine tuning

```
[{'role': 'user', 'content': 'Hello doctor, I have bad acne. How do I get rid of it?'}]
```

Hello. Acne is a universal issue in the age group of 15 to 25 years due to hormonal changes in adolescents. However, there are multiple effective treatment options depending on the grade and severity of the problem. You can visit a dermatologist for a clinical evaluation and further treatment. If you are looking for home remedies, you can apply tea tree oil. It has antibacterial properties that help in reducing acne. You can apply 3 to 4 teaspoons of tea tree oil in 8 oz of water twice a day for three weeks. Additionally, you can apply honey with lemon juice twice a day. Lemon juice

```
[{'role': 'user', 'content': 'What are the profile properties of IfcStructuralCurveMemberVarying described by?'}]
```

A start profile and an end profile (e.g. tapers) and the corresponding IfcStructuralCurveMemberConstants for linear structural elements. For non-linear structural elements, the profile properties are described by IfcStructuralCurveMemberHistory with an associated IfcStructuralCurveMemberConstants for linear structural elements only. For structural elements that represent reinforcements, the profile properties are described by IfcStructuralCurveMemberReactions with an associated IfcStructuralCurveMemberConstants for linear structural elements only. For structural elements that represent reinforcements, the profile properties are described by IfcStructuralCurveMemberReactions with

```
[{'role': 'user', 'content': 'Hello doctor, I have bad acne. How do I get rid of it?'}]
```

Hello. Acne is a universal issue in the age group of 15 to 25 years due to hormonal changes in adolescents. However, there are multiple effective treatment options depending on the grade and severity of the problem. You can visit a dermatologist for a clinical evaluation and further treatment. If you are looking for home remedies, you can apply tea tree oil. It has antibacterial properties that help in reducing acne. You can apply 3 to 4 teaspoons of tea tree oil in 8 oz of water twice a day for three weeks. Additionally, you can apply honey with lemon juice twice a day. Lemon juice

```
[{'role': 'user', 'content': 'What are the profile properties of IfcStructuralCurveMemberVarying described by?'}]
```

A start profile and an end profile (e.g. tapers) and the corresponding IfcStructuralCurveMemberConstants for linear structural elements. For non-linear structural elements, the profile properties are described by IfcStructuralCurveMemberHistory with an associated IfcStructuralCurveMemberConstants for linear structural elements only. For structural elements that represent reinforcements, the profile properties are described by IfcStructuralCurveMemberReactions with an associated IfcStructuralCurveMemberConstants for linear structural elements only. For structural elements that represent reinforcements, the profile properties are described by IfcStructuralCurveMemberReactions with

Fine turning

fine tuning. Epoch = 10 (over 5 hours)

```
[{'role': 'user', 'content': 'Hello doctor, I have bad acne. How do I get rid of it?'}]
```

Asking to truncate to max_length but no maximum length is provided and the model has no predefined maximum length.

Hello. Acne is a universal issue in the age group of 15 to 25 years due to hormonal changes in adolescents. However, the severity of the problem. Grade 1 to grade 3 can be treated conservatively, while grade 4 often requires oral antibiotics. Skin care products like Cetaphil and Clinique can be used at home. However, I suggest you consult a dermatologist. I don't know if I can assist you further. Regards, For more information consult a dermatologist online --> <https://bit.ly/3k8w8w8> Family Physician, Dermatology Specialist, Skin Care Specialist, Acne Specialist, Cosmetic Dermatology, Anti-Aging Specialist, and Cosmetic injection specialist. For more information consult a dermatologist online --> <https://bit.ly/3k8w8w8> provided me. Please do not hesitate to ask further clarifications. Regards, For more information consult a dermatologist online --> <https://bit.ly/3k8w8w8>

```
[{'role': 'user', 'content': 'What are the profile properties of IfcStructuralCurveMemberVarying described by?'}]
```

A start profile and an end profile (e.g. tapers) and the corresponding IfcStructuralCurveMemberEndProfile and IfcStructuralCurveMemberType. In addition, a sweep operation is described by IfcStructuralCurveMemberSweptAreaSolid. For IfcStructuralCurveMemberSweptAreaSolid and IfcStructuralCurveMemberVaryingSurface pressures have to be defined as well as the tolerance for curve. Finally, the local axis system has to be defined too. For more information, see Löwner, M.O.; Gröger, G. In: Computational BIM: 1313-0499. [CrossRef] [Web of Science (WoS)] [SCI] [e-EL] [Export to PDF] [Metadata] [BIMcert Handbuch] [BIMcert

Hallucination

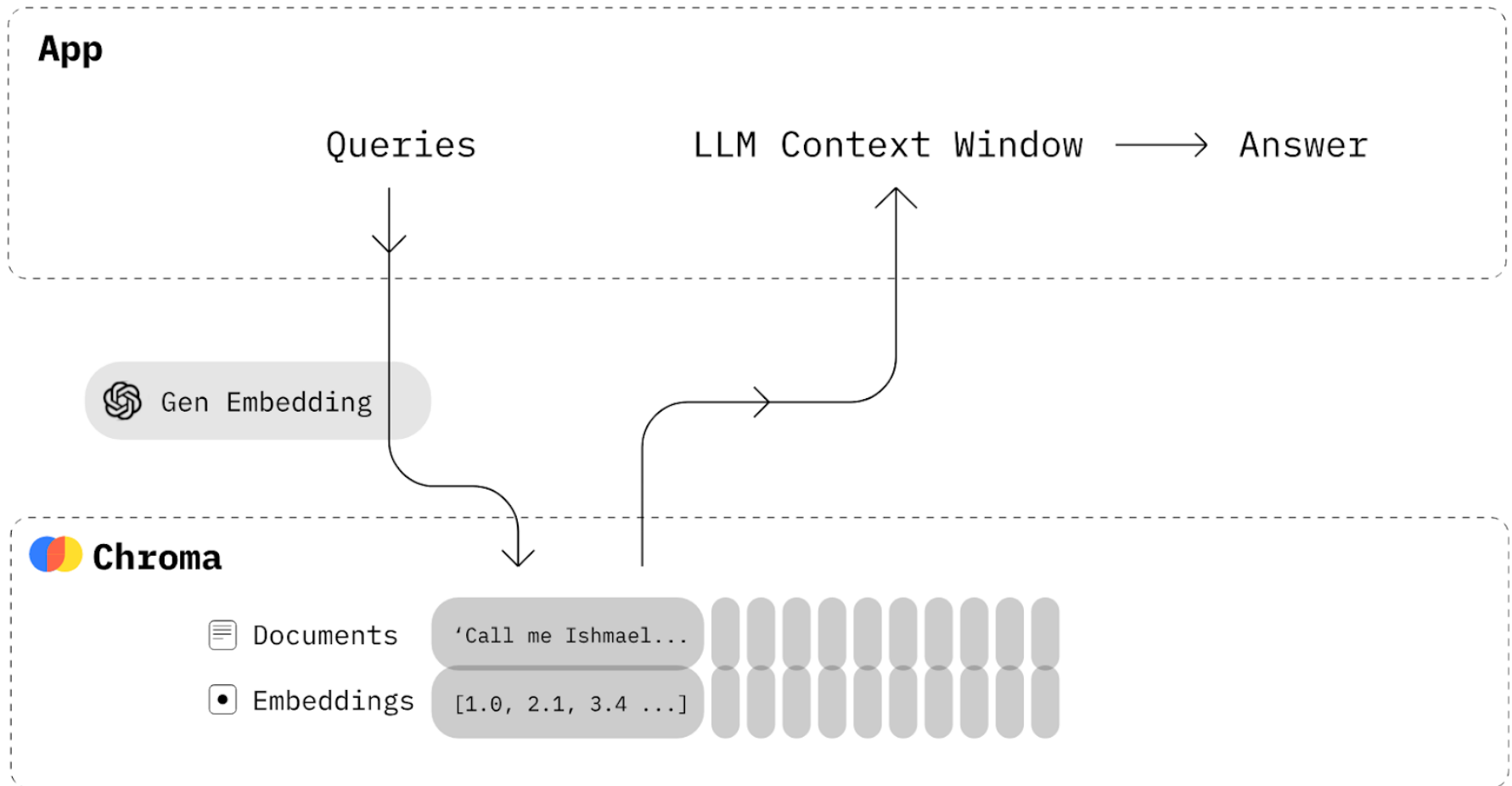


Why wasn't that answer 35 as you predicted?

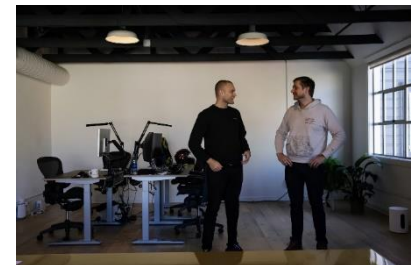


I apologize for my mistake earlier. The correct set of operations to arrive at 35 is to double the number 8, which is 16, subtract 3 from it to get 13, multiply it by 5 to get 65, subtract 30 to get 35, and then divide it by 2 to get the final answer of 17.

LLM & Vector database



Chroma CEO. [Jeff Huber](#), [Anton Troynikov](#)
18M\$. 2023



RAG Retrieval-Augmented Generation

ChatGPT ▾



Where can you access the NextGen-IFC project online?



The NextGen Integrated Facility for Computational Research (NextGen-IFC) project can be

Where can you access the NextGen-IFC project online?



The NextGen Integrated Facility for Computational Research (NextGen-IFC) project can be accessed online through several platforms and sources:

1. **Official Website:** The primary source of information and resources related to the NextGen-IFC project is its official website. This site will typically contain detailed information about the project's objectives, updates, publications, and contact details.

instance, the Na
the NextGen-IFC

5. **Conference Pro**
NextGen-IFC pro
computational r

Google

NextGen-IFC



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GitHub

<https://github.com/buildingSMART/NextGen-IFC>

buildingSMART/NextGen-IFC

Next Generation IFC · test new technologies to maintain IFC; · evaluate and explore new IFC schema changes; · test CI tools for quality assurance. Are you ...

Open Access Article

Development of a Conceptual Mapping Standard to Link Building and Geospatial Information

by Taewook Kang

Korea Institute of Civil Engineering and Building Technology, Ilsanseo-Gu Goyang-Si 10223, Korea

ISPRS Int. J. Geo-Inf. 2018, 7(5), 162; https://doi.org/10.3390/ijgi7050162

Submission received: 18 February 2018 / Revised: 7 April 2018 / Accepted: 19 April 2018 / Published: 24 April 2018

(This article belongs to the Special Issue Building Information Modeling and 3D GIS Integration: From the Theoretical to the Practical)

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Abstract

This study introduces the BIM (building information modeling)-GIS (geographic information system) conceptual mapping (B2GM) standard ISO N19166 and proposes a mapping mechanism. In addition, the major issues concerning BIM-GIS integration, and the considerations that it requires, are discussed. The B2GM is currently being standardized by the spatial-information international standardization organization TC211. Previous studies on BIM-GIS integration seem to focus on the integration of different types of model schemas and on the implementation of service interfaces. B2GM concerns the clear definition of the conditions and methods for mapping the object information required from the user's use-case viewpoint for city-scale mapping. The benefits of the B2GM approach are that the user is able directly control the BIM-GIS linkage and integration process in order to acquire the necessary objection information. This can reveal cases of possibly unclear BIM-GIS integration outside the black box in an explicit and standard way, so that the user can distinctively predict the final output obtainable from the BIM-GIS integration. This study examined B2GM in terms of its development background, components, and several utilization examples, as well as the levels and considerations of the integration of different BIM-GIS models.

Keywords: BIM; GIS; conceptual mapping; ISO 19166; BG-IL

Version: 551.61		CUDA Version: 12.4	
Bus-Id	Disp.A	Volatile	Uncorr. ECC
Memory-Usage		GPU-Util	Compute M.
			MIG M.
00000000:01:00.0	On		N/A
5074MiB /	8192MiB	86%	Default
			N/A

https://daddynkidsmakers.blogspot.com/2024/02/github-copilot-ai.html

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GeoBIM.pdf 3.6MB

Chat

what is BIM?

BIM (Building Information Modeling) is a method used in the construction industry to create and manage digital representations of physical and functional characteristics of places. It includes defining object information such as runtime, geometry, property, and relationship for building components like walls, doors, and rooms. The geometry information uses boundary representation with support for vertices, edges, loops, and faces. Property information categorizes attributes and defines name, value type, and initial value pairs. System properties, including the BIM element name and GUID, are automatically created when making a model. Relationships between elements are also defined, following UML relationships.

what is use cases of BIM?

BIM-GIS integration is used for scaling building information modeling (BIM) to city level use cases. This method includes a BIM-GIS element, geometry, and property set mapping with external datasets. The goal is to enable flexible information mapping and integration, focusing on user's specific requirements. Previous studies suggest various methods for BIM-GIS integration, but this study proposes using the B2GM standard under ISO/TC211.

tell me about BIM-GIS standard?

The BIM-GIS standard, also known as Building Information Modeling-Geographic Information System, is a method for

BLM expert agent using RAG

```
client = OpenAI(  
    base_url = 'http://localhost:11434/v1',  
    api_key='ollama',  
)  
  
template = {  
    "question": " ",  
    "answer": " "  
}
```

```
def generate_questions_answers(text_chunk):  
    messages = [  
        {'role': 'system', 'content': 'You are an API that converts bodies of text  
            into multiple questions and answers into a JSON format. Each JSON  
            should contain a single question with a single answer. Only respond  
            with the JSON and no additional text. \n.'},  
        {'role': 'user', 'content': 'Text: ' + text_chunk}  
    ]  
  
    response = client.chat.completions.create(  
        model='llama3',  
        messages=messages,  
        max_tokens=2048,  
        n=1,  
        stop=None,  
        temperature=0.7,  
    )
```

BIM expert agent using RAG

```
{
  "question": "What is the URL to access information on jetgeo?",
  "answer": ".com/jetgeo/IFC2GML"
},
{
  "question": "When was the ISO TC 211 introduction to UML accessed?",
  "answer": "20 March 2020"
},
{
  "question": "Who wrote Learning UML 2.0 and what is its edition?",
  "answer": "Miles, R.R.; Hamilton, K., 1st ed."
},
},
```

```
1555 {
1556   "question": "What problems were handled?",
1557   "answer": "Problems surrounding the integration of BIM and GIS heterogeneous m
1558 },
1559 {
1560   "question": "What is a method used for model integration?",
1561   "answer": "Schema integration, extension, linkage."
1562 },
1563 {
1564   "question": "What are limitations of model integration methods?",
1565   "answer": "Each method has its own respective limitation."
1566 },
1567 {
1568   "question": "What is BIM-GIS conceptual mapping?",
1569   "answer": "BIM-GIS conceptual mapping (B2GM) attempts to standardize the BIM-G
1570 },
1571 {
1572   "question": "How does B2GM divide heterogeneous processes?",
1573   "answer": "B2GM divides heterogeneous process into PD, EM, and LM, and links a
1574 },
1575 }
```

AI Agent with BIM



Available Tools

Select tools to use:

Choose an option

Upload BIM files

Drag and drop files here

Limit 200MB per file • IFC, PDF, JSON, CSV

Browse files

2D

3D

bar_mode

bar_camera

dxBar_options

bar_view

X

Y

Z

btn_storey_reset

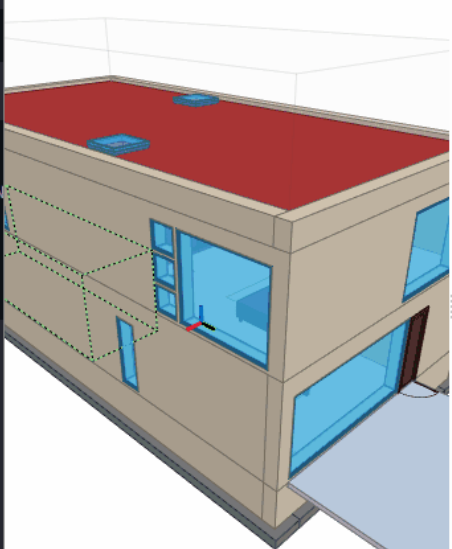
btn_allow_move_x

btn_allow_move_y

btn_allow_move_z

bar_offsets

bar_prog



col.Active

col.Type

col.Name

col.Description

☒	IfcSpace	A103	Kitchen
☒	DC_Meble		
☒	IfcSpace	A104	Bathroom 1
☒	IfcSpace	A101	Foyer
☒	IfcSpace	B102	Living Room

txt_properties

txt_location

txt_classification

txt_relations

col.Name

col.Value

txt_unit

Location		
Project	1.000	
Storey	Level 1	
Top Elevation	2.600	m
Bottom Elevation	0.019	m
Global Top Elevation	2.600	m
Global Bottom Elevation	0.019	m
Geometry		
Has Own Geometry	txt.yes	
Children Have Geometry	txt.no	
Global X	6.200	m
Global Y	-17.383	m
Global Z	0.019	m
Bounding Box Length	2.183	m
Bounding Box Width	9.308	m
Bounding Box Height	2.581	m
Membership		
Layer	A-AREA-IDEN	

BIMvision

0001 (Duplex_A_20110907.ifc) • Default • IfcBuilding • Level 1 • DC_Obszary • A101 • 10 txt_m 0.01 s 1.6 ★

GeoGPT

Google Calendar - 2024 | kiris.kict.re.kr | KICT웹메일 | IoT map simple dashboar | leaflet map function - G | Documentation - Leaflet

localhost:8000

Query ... I

Logout

Infra Connect

- Dashboard
- Scan view
- Flow view
- Devices
- Settings
- Accounts
- Link

Dashboard / My Dashboard

ISSUE

89

VIEW DETAILS >

OPEN ISSUE

11

VIEW DETAILS >

PROGRESS

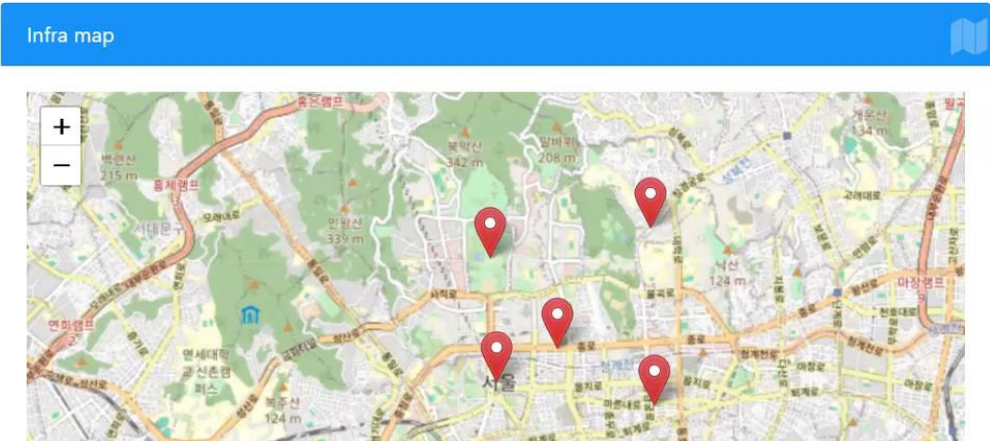
152

VIEW DETAILS >

CLOSE ISSUE

7

VIEW DETAILS >



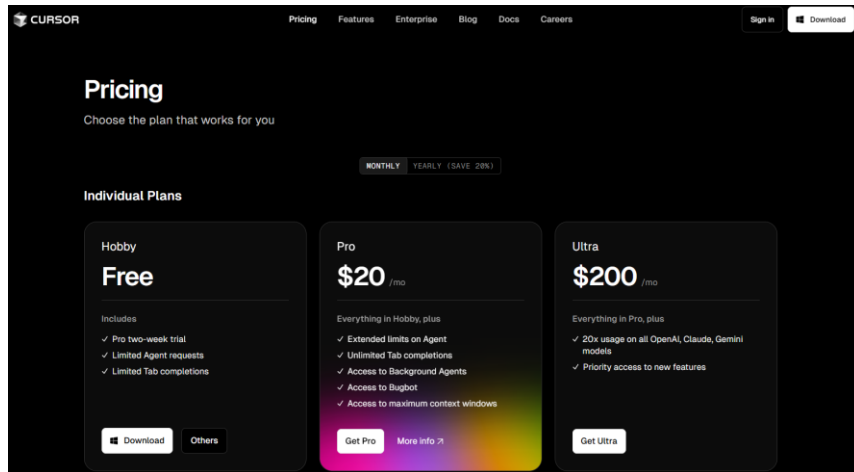
LLM & Transformer

[mac999/LLM-RAG-Agent-Tutorial: LLM-RAG-Agent-Tutorial](#)

Vibe coding
Hands-on

Github Copilot

Vibe coding



The image shows the Cursor Pricing page. At the top, there's a navigation bar with links for Pricing, Features, Enterprise, Blog, Docs, and Careers. On the right, there are buttons for Sign in and Download. The main heading is 'Pricing' with a subtext 'Choose the plan that works for you'. Below this, there are tabs for MONTHLY and YEARLY (SAVE 28%). Under the 'Individual Plans' section, there are three plans: Hobby (Free), Pro (\$20 /mo), and Ultra (\$200 /mo). Each plan lists its features and includes a 'Get Pro' or 'Get Ultra' button. The Pro plan also has a 'More info' link.

CURSOR Pricing Features Enterprise Blog Docs Careers Sign in Download

Pricing

Choose the plan that works for you

MONTHLY YEARLY (SAVE 28%)

Individual Plans

Hobby
Free

Includes

- ✓ Pro two-week trial
- ✓ Limited Agent requests
- ✓ Limited Tab completions

Download Others

Pro
\$20 /mo

Everything in Hobby, plus

- ✓ Extended limits on Agent
- ✓ Unlimited Tab completions
- ✓ Access to Background Agents
- ✓ Access to Bugbot
- ✓ Access to maximum context windows

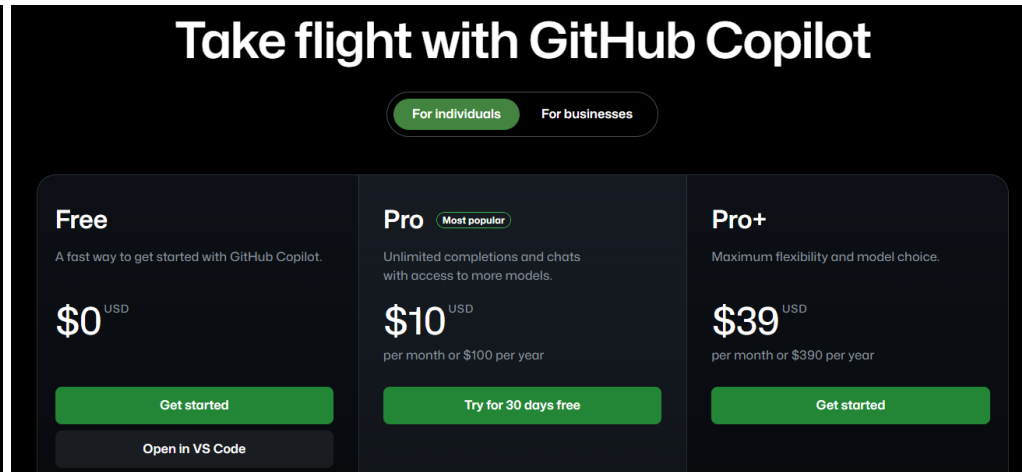
Get Pro More info ↗

Ultra
\$200 /mo

Everything in Pro, plus

- ✓ 20x usage on all OpenAI, Claude, Gemini models
- ✓ Priority access to new features

Get Ultra



The image shows the GitHub Copilot pricing page. The main heading is 'Take flight with GitHub Copilot'. Below this, there are tabs for 'For individuals' and 'For businesses'. There are three pricing tiers: Free (\$0 USD), Pro (\$10 USD), and Pro+ (\$39 USD). Each tier has a 'Get started' button. The Pro tier also has a 'Try for 30 days free' button. The Free tier has an 'Open in VS Code' button.

Take flight with GitHub Copilot

For individuals For businesses

Free

A fast way to get started with GitHub Copilot.

\$0 USD

Get started

Open in VS Code

Pro Most popular

Unlimited completions and chats with access to more models.

\$10 USD

per month or \$100 per year

Try for 30 days free

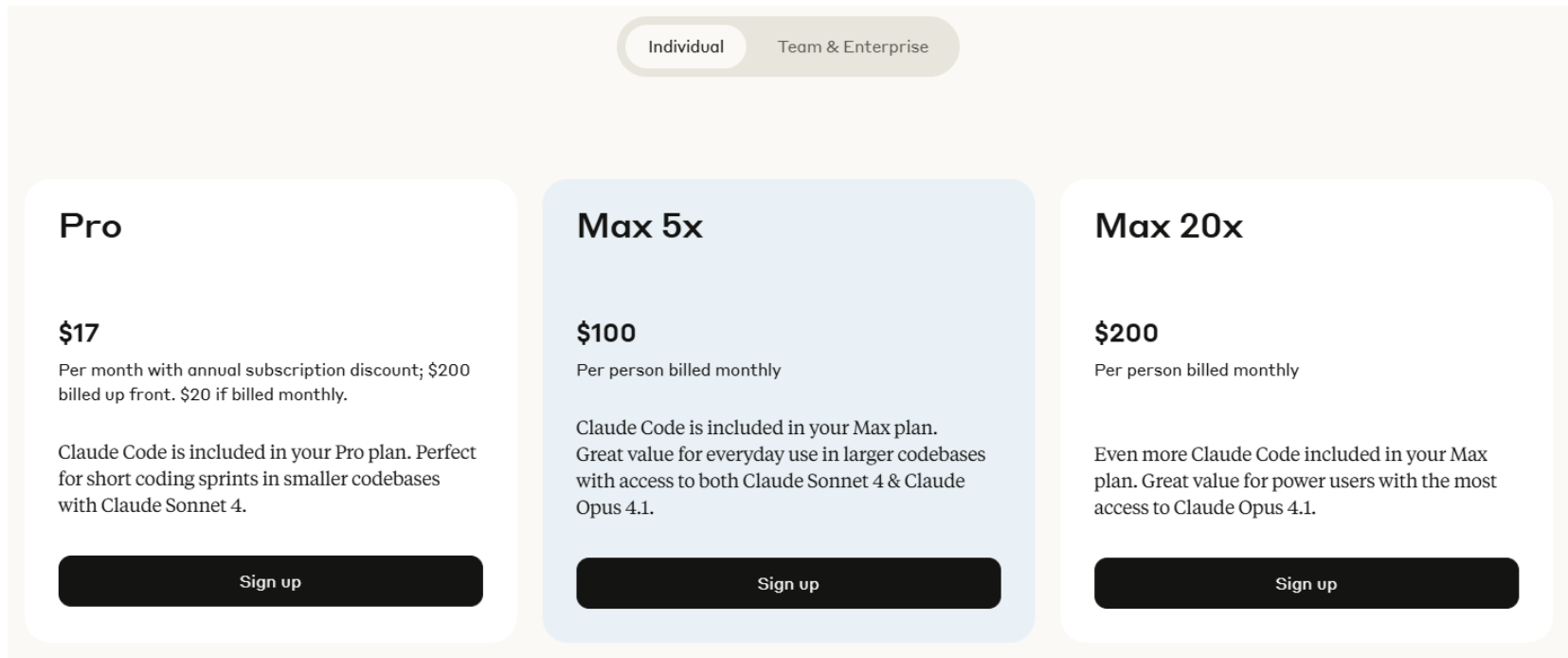
Pro+

Maximum flexibility and model choice.

\$39 USD

per month or \$390 per year

Get started



The image shows the Claude Code pricing page. At the top, there are tabs for 'Individual' and 'Team & Enterprise'. There are three pricing tiers: Pro (\$17), Max 5x (\$100), and Max 20x (\$200). Each tier has a 'Sign up' button. The Pro tier includes a description of the plan and its benefits. The Max 5x and Max 20x tiers also include descriptions of the plans and their benefits.

Individual Team & Enterprise

Pro

\$17

Per month with annual subscription discount; \$200 billed up front. \$20 if billed monthly.

Claude Code is included in your Pro plan. Perfect for short coding sprints in smaller codebases with Claude Sonnet 4.

Sign up

Max 5x

\$100

Per person billed monthly

Claude Code is included in your Max plan. Great value for everyday use in larger codebases with access to both Claude Sonnet 4 & Claude Opus 4.1.

Sign up

Max 20x

\$200

Per person billed monthly

Even more Claude Code included in your Max plan. Great value for power users with the most access to Claude Opus 4.1.

Sign up

Vibe coding

도구 (Tool)	가격 (월. 2025/8 시점)	장점	단점
Cursor	- Pro: \$20 - 무료: 기능 제한	- 코드 에디터 자체에 AI 기능이 완벽히 통합 - 프로젝트 전체 코드를 이해하고 답하는 @Codebase 기능이 강력함 - AI를 통한 코드 생성, 수정, 리팩토링이 직관적임 - VS Code 기반으로 기존 사용자에게 익숙함	- 일반 에디터보다 무겁고 시스템 자원을 많이 사용 - 최신 도구라 안정성이 상대적으로 낮을 수 있음 - 무료 플랜은 '빠른 모델' 사용 횟수가 제한적임
GitHub Copilot	- Individual: \$10 - Business: 사용자당 \$19 - 학생/교사/오픈소스 기여자 무료	- 업계 표준 수준의 정확하고 빠른 인라인 코드 완성 - VS Code, JetBrains 등 기존 IDE에 확장 프로그램으로 설치하여 사용 편의성이 높음 - Microsoft/GitHub의 지원으로 안정성과 지원이 우수함	- Cursor에 비해 프로젝트 전체의 복잡한 맥락 파악은 다소 부족 - 주로 코드 완성 기능에 집중되어 있어 고차원적인 작업 지시는 한계가 있을 수 있음
Claude-CLI	- API 기반 (사용량 과금) - 모델(Sonnet/Opus)과 토큰 사용량에 따라 비용 변동 - 개발자 평균 월 \$100 내외 (사용량에 따라 크게 달라짐)	- 복잡한 로직 생성, 버그 수정, 문서화 등 높은 수준의 추론이 필요한 작업에 매우 강력 - 터미널(CLI) 기반으로 스크립트 작성, 자동화 등 개발 워크플로우에 자유롭게 통합 가능 - 긴 컨텍스트 처리 능력이 뛰어나 전체 파일 또는 여러 파일의 리팩토링에 유리함	- IDE 통합이 아닌 별도 터미널 창에서 사용해야 해 번거로움 - 실시간 코드 완성과 같은 기능은 없음 - 사용량 기반 요금제로 비용이 예측하기 어렵고, 사용량이 많으면 비쌀 수 있음

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Vibe coding

What is GitHub Copilot? - GitHub Docs

What is GitHub Copilot?

Learn what GitHub Copilot is and what you can do with it.

GitHub Copilot is an AI coding assistant that helps you write code faster and with less effort, allowing you to focus more energy on problem solving and collaboration.

GitHub Copilot has been proven to increase developer productivity and accelerate the pace of software development. See [Research: quantifying GitHub Copilot's impact on developer productivity and happiness](#) in the GitHub blog.

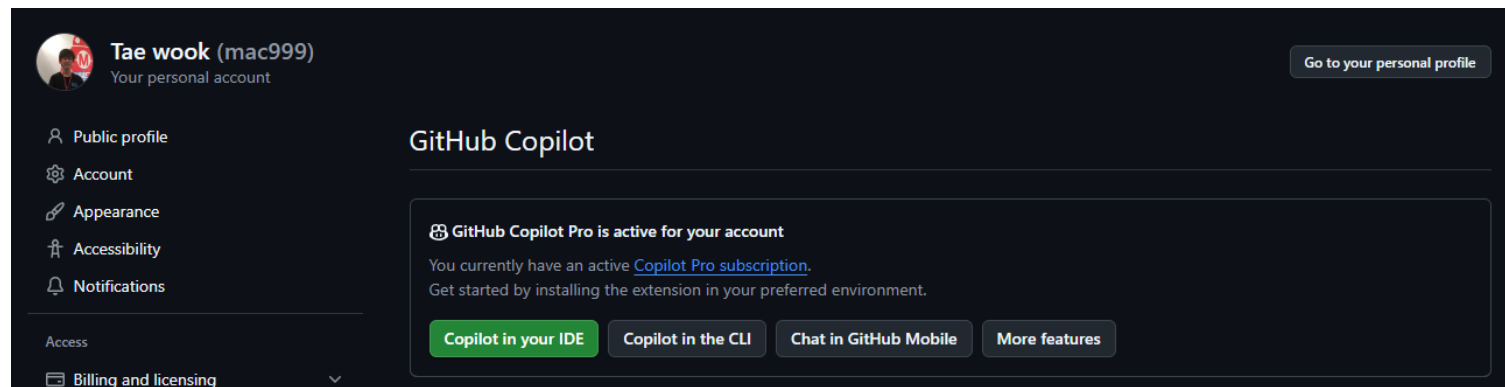
Copilot features

GitHub Copilot includes a suite of features. You can use Copilot to:

- Get code suggestions as you type in your IDE
- Chat with Copilot to ask for help with your code
- Ask Copilot for help using the command line
- Organize and share task-specific context with Copilot Spaces to get more relevant answers
- Generate a description of the changes in a pull request
- Create and manage collections of documentation, called knowledge bases, to use as a context for chatting with Copilot (*Copilot Enterprise only*)
- Work on code changes and create a pull request for you to review (*Copilot Pro+*, *Copilot Business*, and *Copilot Enterprise only*)

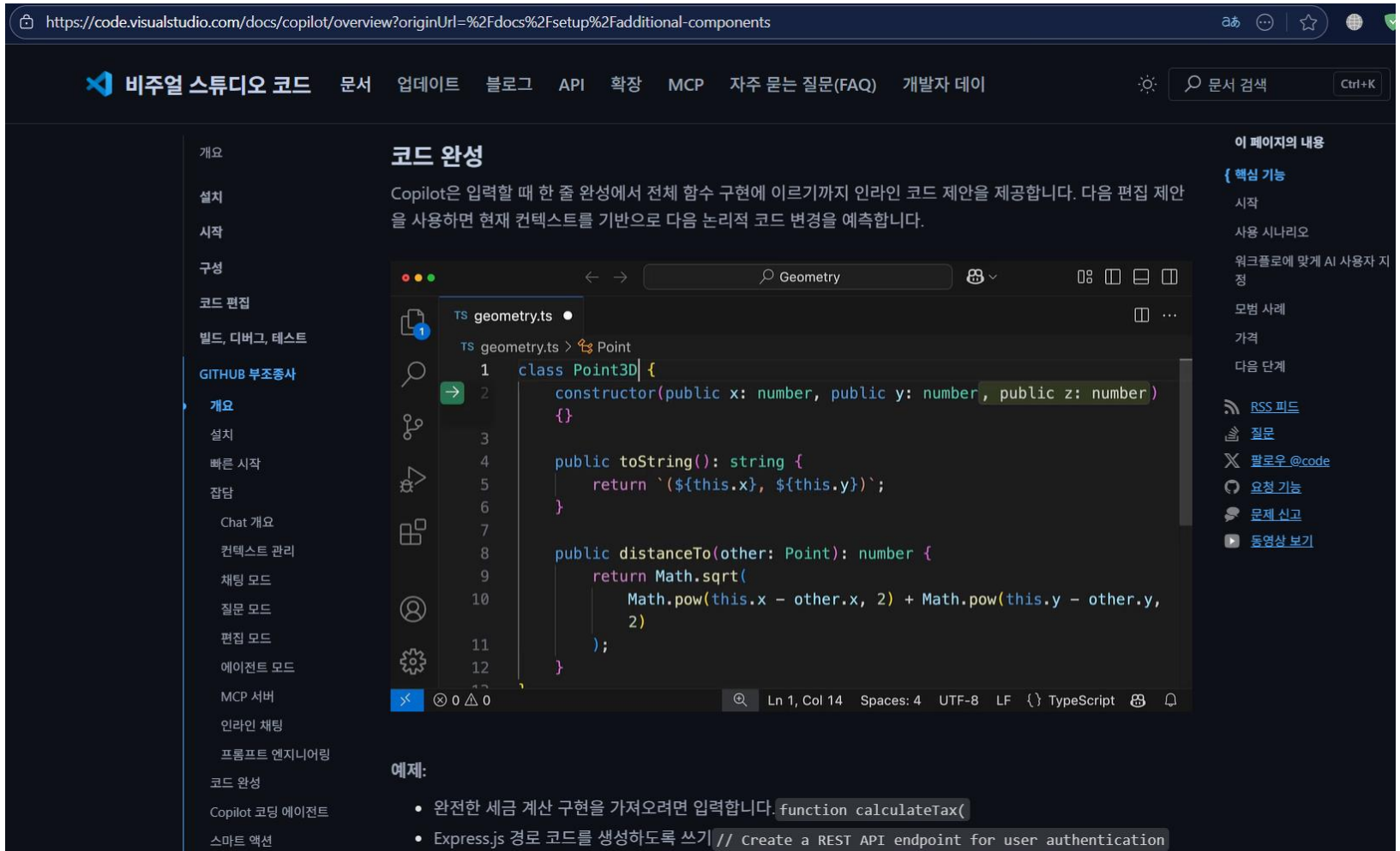
Vibe coding

- **Try Copilot for free:** Use Copilot Free to explore core Copilot features with no paid plan required.
- **Subscribe to a paid plan:** Upgrade to Copilot Pro or Copilot Pro+ for full access to premium features and more generous usage limits. You can [try Copilot Pro for free](#) with a one-time 30-day trial.
- **Eligible for free Copilot Pro access?** Students, teachers, and open source maintainers may qualify for Copilot Pro at no cost. See [Getting free access to GitHub Copilot Pro as a student, teacher, or maintainer](#).
- **Organization members:** If your organization or enterprise has a GitHub Copilot plan, you can request access to Copilot by going to <https://github.com/settings/copilot> and requesting access under "Get Copilot from an organization."
See [Getting started with a GitHub Copilot plan](#) for more information.



Vibe coding

VS Code의 GitHub Copilot



<https://code.visualstudio.com/docs/copilot/overview?originUrl=%2Fdocs%2Fsetup%2Fadditional-components>

비주얼 스튜디오 코드 문서 업데이트 블로그 API 확장 MCP 자주 묻는 질문(FAQ) 개발자 데이

문서 검색 Ctrl+K

코드 완성

Copilot은 입력할 때 한 줄 완성에서 전체 함수 구현에 이르기까지 인라인 코드 제안을 제공합니다. 다음 편집 제안을 사용하면 현재 컨텍스트를 기반으로 다음 논리적 코드 변경을 예측합니다.

```
TS geometry.ts > Point
1 class Point3D {
2   constructor(public x: number, public y: number, public z: number) {}
3
4   public toString(): string {
5     return `${this.x}, ${this.y}`;
6   }
7
8   public distanceTo(other: Point): number {
9     return Math.sqrt(
10      Math.pow(this.x - other.x, 2) + Math.pow(this.y - other.y,
11      2)
12    );
13  }
```

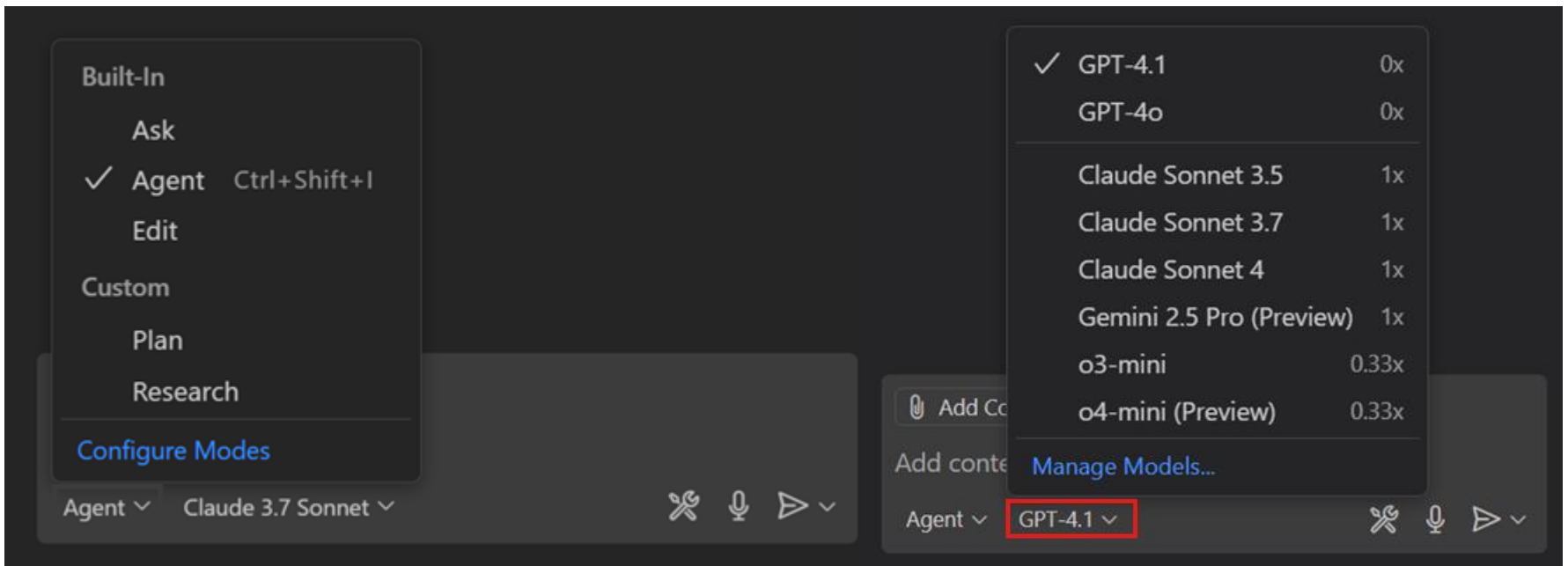
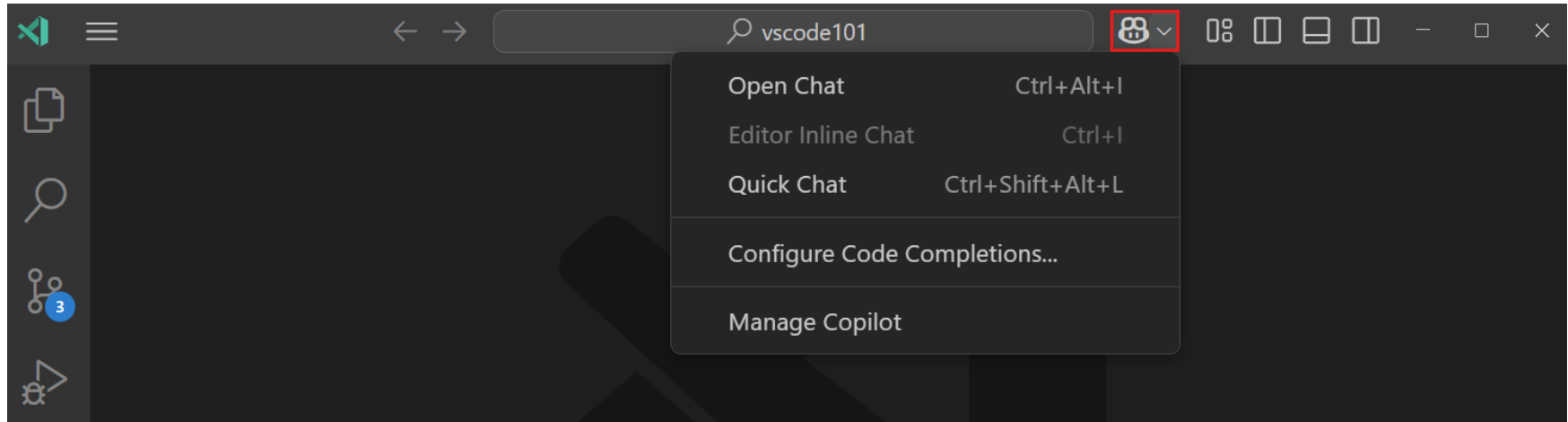
이 페이지의 내용

- { 핵심 기능
- 시작
- 사용 시나리오
- 워크플로에 맞게 AI 사용자 지정
- 모범 사례
- 가격
- 다음 단계
- RSS 피드
- 질문
- 팔로우 @code
- 요청 기능
- 문제 신고
- 동영상 보기

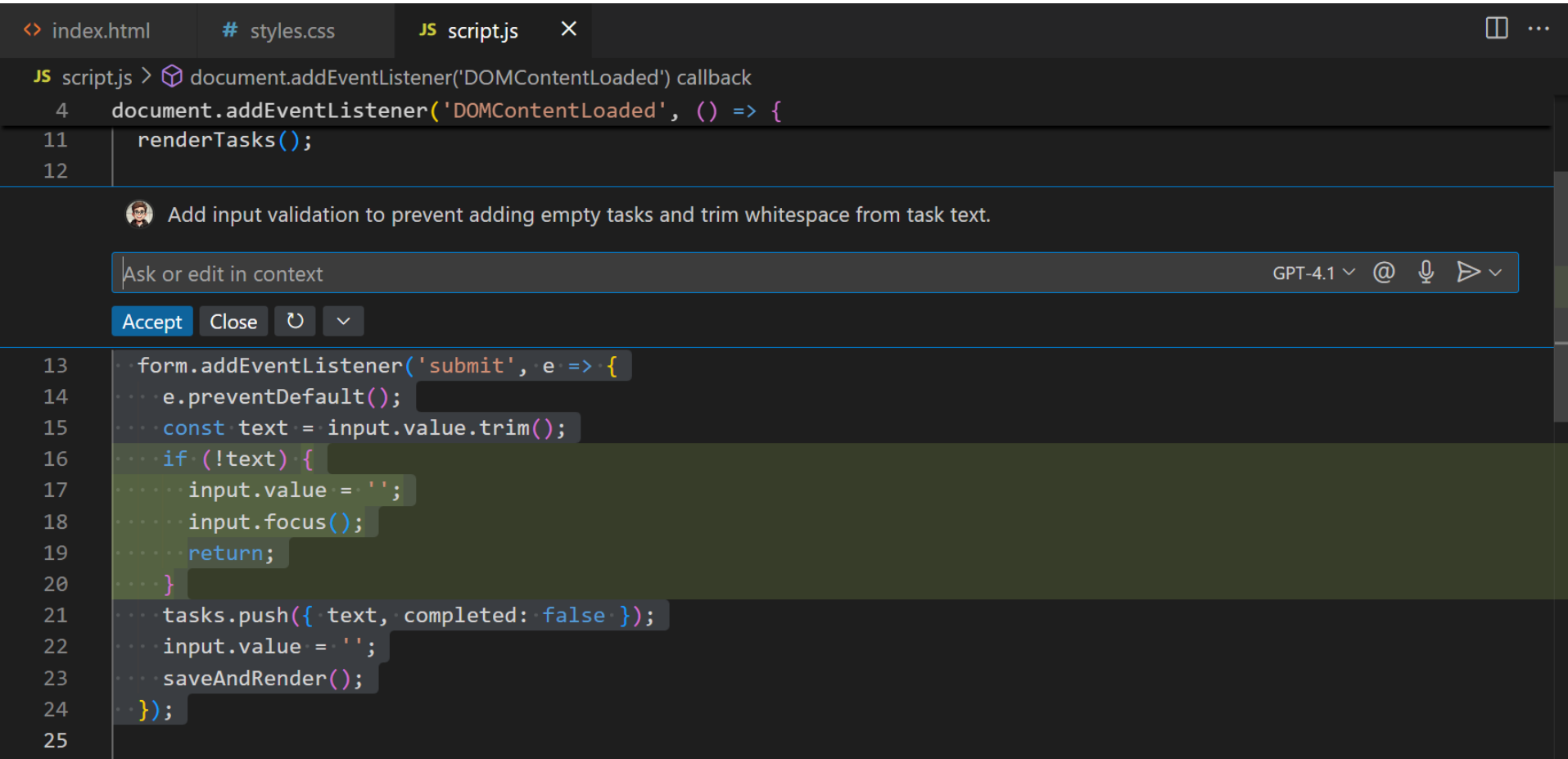
예제:

- 완전한 세금 계산 구현을 가져오려면 입력합니다. `function calculateTax()`
- Express.js 경로 코드를 생성하도록 쓰기 `// Create a REST API endpoint for user authentication`

Vibe coding



Vibe coding



```
index.html # styles.css JS script.js X
```

```
JS script.js > document.addEventListener('DOMContentLoaded') callback
4 document.addEventListener('DOMContentLoaded', () => {
11   renderTasks();
12
```

 Add input validation to prevent adding empty tasks and trim whitespace from task text.

Ask or edit in context GPT-4.1 @  

Accept Close  

```
13   form.addEventListener('submit', e => {
14     e.preventDefault();
15     const text = input.value.trim();
16     if (!text) {
17       input.value = '';
18       input.focus();
19       return;
20     }
21     tasks.push({ text, completed: false });
22     input.value = '';
23     saveAndRender();
24   });
25
```

Vibe coding

Discover and install MCP Servers in VS Code

 Visual Studio Code Docs Updates Blog API Extensions MCP FAQ Dev Days

☀️ 🔍 Search Docs Ctrl+K [Download](#)

Try [MCP servers](#) to extend agent mode in VS Code!

MCP Servers for agent mode

Extend agent mode in VS Code by installing MCP servers to bring extra tools for connecting to databases, invoking APIs, and performing specialized tasks. Below is a curated list of MCP servers, if you don't see an MCP server you're looking for, please [suggest it here](#). See the [docs](#) to learn how to use MCP tools in agent mode.

Developer Tools

**GitHub**
GitHub

Access GitHub repositories, issues, and pull requests through secure API integration.

[Install GitHub](#)

**Figma**
Figma

Extract UI content and generate code from Figma designs. Update to the latest desktop app version.

[Install Figma](#)

**Playwright**
Microsoft

Automate web browsers using accessibility trees for testing and data extraction.

[Install Playwright](#)

**Sentry**
Sentry

Retrieve and analyze application errors and performance issues from Sentry projects.

[Install Sentry](#)

**Hugging Face**
Hugging Face

Access models, datasets, and Spaces on the Hugging Face Hub.

[Install Hugging Face](#)

**DeepWiki**
Kevin Kern

Query and extract information from GitHub repositories indexed on DeepWiki.

[Install DeepWiki](#)

**MarkItDown**
Microsoft

Convert various file formats (PDF, Word, Excel, images, audio) to Markdown.

[Install MarkItDown](#)

**Microsoft Docs**
Microsoft

Search and retrieve content from Microsoft Learn, Azure documentation, and official Microsoft technical resources.

[Install Microsoft Docs](#)

Vibe coding

Discover and install MCP Servers in VS Code

Data & Analytics



DuckDB
Kentaro Tanaka

Query and analyze data in DuckDB databases locally and in the cloud.

[Install DuckDB](#)



PostHog
PostHog

Access PostHog analytics to create annotations and retrieve product usage insights.

[Install PostHog](#)



Neon
neondatabase labs

Manage and query Neon Postgres databases with natural language.

[Install Neon](#)



Apify
Apify

Extract data from websites and automate workflows through Apify's Actor ecosystem.

[Install Apify](#)



Microsoft Clarity
Microsoft

Access Microsoft Clarity analytics data including heatmaps, session recordings, and user behavior insights.

[Install Microsoft Clarity](#)



Firecrawl
Mendable

Extract data from websites with advanced web scraping, crawling, search, and structured data extraction.

[Install Firecrawl](#)



Prisma Postgres
Prisma

Database operations with Prisma ORM and PostgreSQL. Schema management, query execution, migrations, and data modeling.

[Install Prisma Postgres](#)



MongoDB
MongoDB

Database operations and management. Execute queries, manage collections, aggregation pipelines, and document operations.

[Install MongoDB](#)

Vibe coding

VS Code에서 MCP 서버 검색 및 설치

1 작업 영역에서 파일을 만듭니다. `.vscode/mcp.json`

2 **서버 추가** 단추를 선택하여 새 서버에 대한 템플릿을 추가합니다. VS Code는 MCP 서버 구성 파일에 대한 IntelliSense를 제공합니다.

다음 예제에서는 GitHub 원격 MCP 서버를 구성하는 방법을 보여줍니다. [VS Code의 MCP 구성 형식](#)에 대해 자세히 알아보세요.

JSON

```
{
  "servers": {
    "github-mcp": {
      "type": "http",
      "url": "https://api.githubcopilot.com/mcp"
    }
  }
}
```

3 또는 명령 팔레트에서 **MCP: 서버 추가** 명령을 실행하고 추가할 MCP 서버 유형을 선택하고 서버 정보를 제공합니다. 그런 다음 **작업 영역 설정**을 선택하여 작업 영역에 파일이 없는 경우 파일을 만듭니다. `.vscode/mcp.json`

Vibe coding

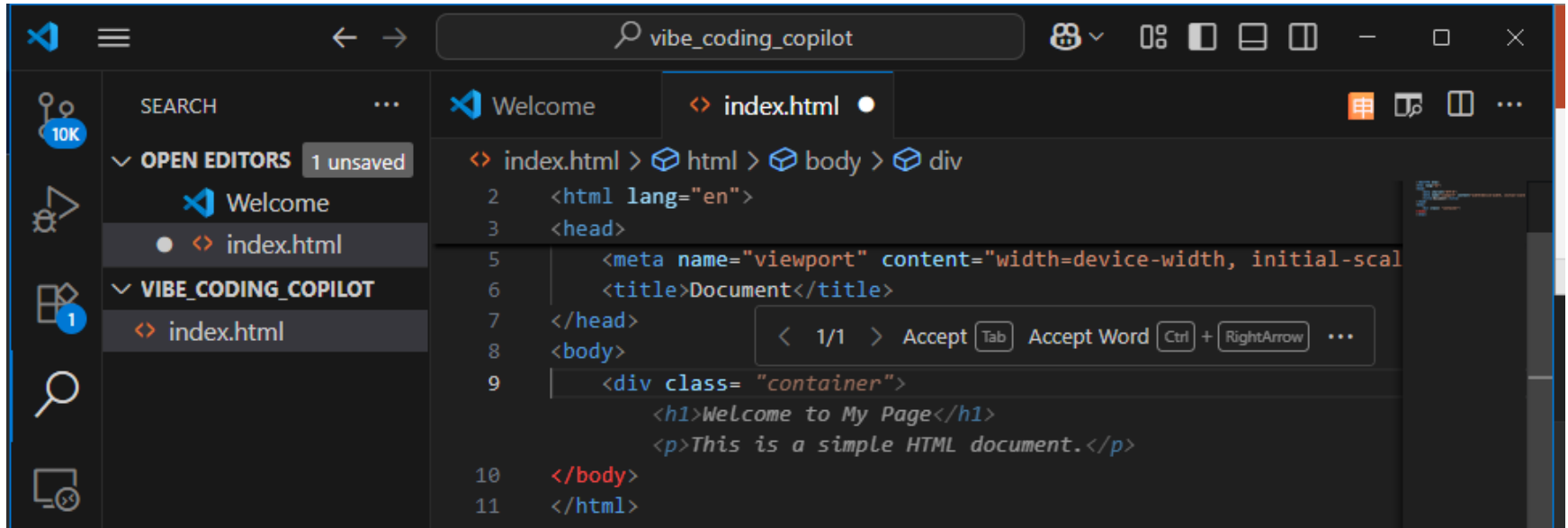
GitHub Copilot in VS Code cheat sheet

필수 키보드 단축키

- `Ctrl+Alt+I` - 채팅 보기 열기
- `Ctrl+I` - 채팅 보기에 음성 채팅 프롬프트 입력
- `Ctrl+N` - 채팅 보기에서 새 채팅 세션 시작
- `Ctrl+Shift+I` - 에이전트 모드로 전환
- `Ctrl+I` - 편집기 또는 터미널에서 인라인 채팅 시작
- `Ctrl+I` (길게 누르기) - 인라인 음성 채팅 시작
- 탭 - 인라인 제안을 수락하거나 다음 편집 제안으로 이동합니다.
- `Escape` - 인라인 제안 해제

Vibe coding

Get started with GitHub Copilot in VS Code



Vibe coding

Get started with GitHub Copilot in VS Code

The screenshot shows the Visual Studio Code editor with a file named `index.html` open. The code defines a simple web page with a toggle button and a list of areas for improvement. The toggle button is a dark gray rectangle with the text "Toggle Dark Mode" in white. The list of areas for improvement is a simple text area with a light gray background and a dark gray border.

```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <meta name="viewport" content="width=device-width, initial-scale=1.0">
6   <title>Document</title>
7   <style>
8     :root {
9       --background-light: #fff;
10      --background-dark: #181818;
11      --text-light: #181818;
12      --text-dark: #fff;
13    }
14    body {
15      background: var(--background-light);
16      color: var(--text-light);
17      transition: background 0.3s, color 0.3s;
18    }
19    body.dark-mode {
20      background: var(--background-dark);
21      color: var(--text-dark);
22    }
23    .dark-mode-toggle {
24      position: fixed;
25      top: 1rem;
26      right: 1rem;
27      padding: 0.5rem 1rem;
28      border: none;
29      border-radius: 4px;
30      background: #333;
31      color: #fff;
```

The right sidebar shows the "CHAT" panel with the following text:

toggle button is not found.

5. Code Readability

- The code is well-organized, with clear separation between structure, style, and behavior.
- Variable and class names are meaningful and self-explanatory.

Areas for Improvement

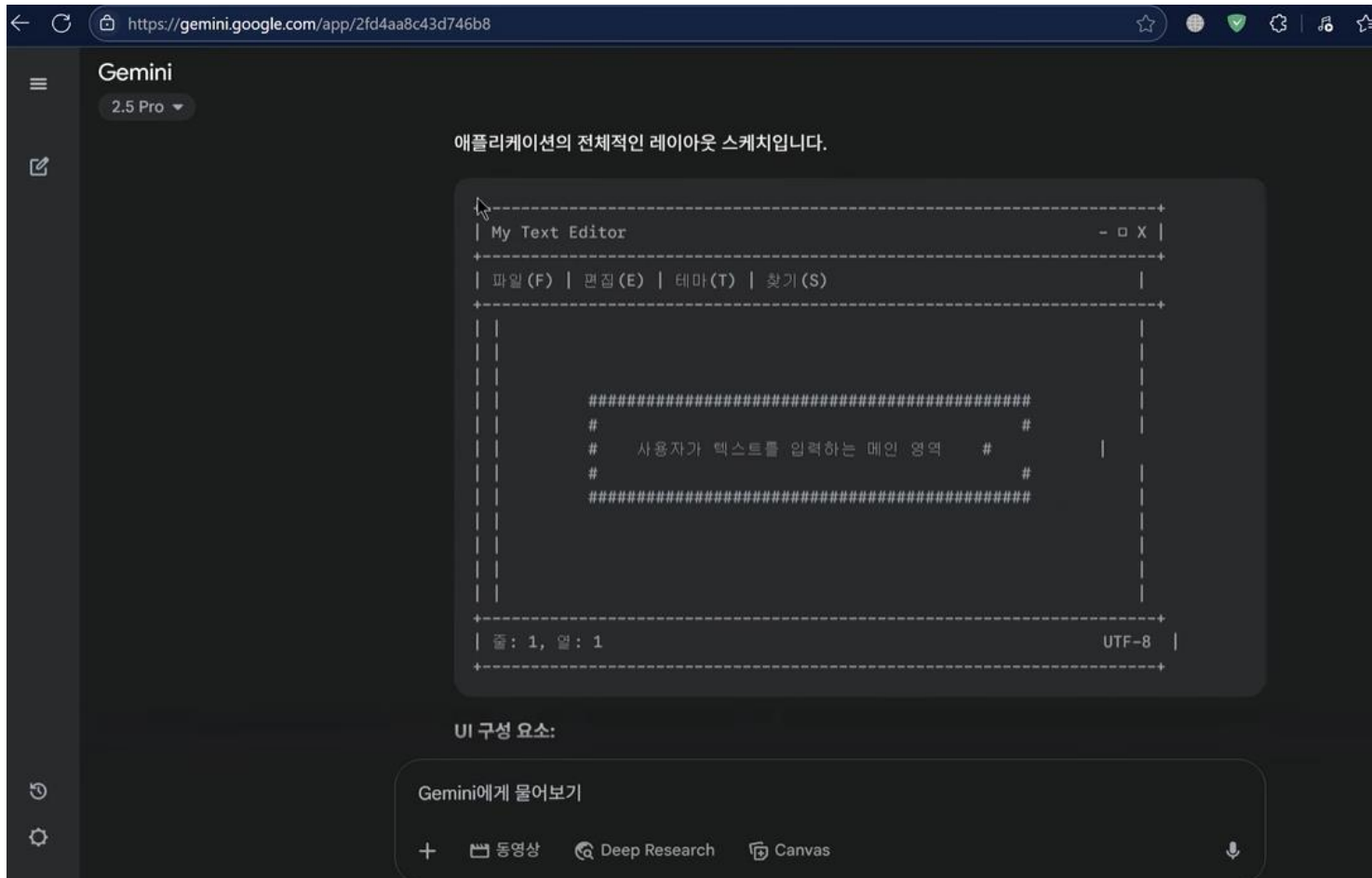
- Accessibility Enhancements
 - The toggle button could benefit from a more descriptive label or `aria-label` for users relying on assistive technology.
 - Consider using a `<button>` element with a clear label for toggling, which you have done, but ensure the label is descriptive enough for all users.
- User Preference Persistence
 - Currently, the dark mode state is not persisted. Users will lose their preference on page reload. Storing the mode in `localStorage` would improve user experience.
- Expandability and Maintainability
 - The inline `<style>` and `<script>` are fine for small projects, but as the project grows, consider moving them to separate CSS and JS files for better maintainability.
- HTML Title and Meta
 - The `<title>` is generic ("Document"). A more descriptive title would improve usability and SEO.
 - Consider adding a meta description for better search engine optimization and sharing.
- Placeholder Content
 - The `<main>` section contains only a comment. If this is a template, that's fine, but for a real project, ensure meaningful content is present.

The bottom status bar shows the file path: `Ln 9, Col 32 Spaces: 4 UTF-8 CRLF HTML Go Live`.

Vibe coding



GUI 기반 텍스트 편집기를 개발할꺼야. 파이썬과 Tkinter 라이브러리를 이용할꺼야. 메뉴는 파일 메뉴, 편집 메뉴, 테마 선택 메뉴(다크모드 하위 메뉴 포함), 찾기 메뉴로 구성됨. 이 프로그램 개발을 위한 PRD를 UI 스케치를 포함해 md 포맷으로 작성해.



Vibe coding



GUI 기반 텍스트 편집기를 개발할꺼야. 파이썬과 Tkinter 라이브러리를 이용할꺼야. 메뉴는 파일 메뉴, 편집 메뉴, 테마 선택 메뉴(다크모드 하위 메뉴 포함), 찾기 메뉴로 구성됨. 이 프로그램 개발을 위한 PRD를 UI 스케치를 포함해 md 포맷으로 작성해.

Python/Tkinter 기반 GUI 텍스트 편집기 제품 요구사항 명세서 (PRD)

1. 개요

본 문서는 파이썬(Python)과 표준 GUI 라이브러리인 티케이인터(Tkinter)를 사용하여 개발될 데스크톱 텍스트 편집기의 요구사항을 정의합니다. 이 애플리케이션은 기본적인 텍스트 편집 기능과 함께 사용자의 편의를 위한 다크 모드, 찾기 기능 등을 제공하는 것을 목표로 합니다.

2. 목표 사용자

- **초보 개발자 및 학생**: 간단한 코드나 스크립트, 메모, 과제 등을 작성하고 편집하려는 사용자.
- **일반 사용자**: Windows 메모장과 같은 가볍고 빠른 텍스트 편집기가 필요한 사용자.

```
+-----+
| My Text Editor                               - o X |
+-----+
| 파일(F) | 편집(E) | 테마(T) | 찾기(S)         |
+-----+

||                                           |
||                                           |
||                                           |
||                                           |
|| ===== |
|| #         # |
|| # 사용자 텍스트를 입력하는 메인 영역 # |
|| #         # |
|| ===== |
||                                           |
||                                           |
||                                           |
||                                           |
||                                           |
+-----+
| 줄: 1, 열: 1                                UTF-8 |
+-----+
...
```

UI 구성 요소

- **제목 표시줄**: 프로그램 이름과 창 조절 버튼이 표시됩니다.
- **메뉴 바**: 파일, 편집, 테마, 찾기 메뉴가 위치하며, 각 메뉴 클릭 시 하위 메뉴 항목들이 나타납니다.
- **텍스트 영역**: 스크롤 가능한 주 텍스트 편집 공간입니다.
- **상태 표시줄**: 하단에 커서의 현재 위치(줄, 열)나 인코딩 같은 간단한 상태 정보를 보여줍니다.

Daddy Makers: 확실하게 성공하는 바이브 코딩 도구 사용 방법

확실하게 성공하는 Vibe coding 방법

Vibe coding

The screenshot displays the Visual Studio Code editor interface. The main editor window shows a file named `PRD.md` with the following content:

```
1 # **Python/Tkinter 기반 GUI 텍스트 편집기 제품 요구사항 명세서 (PRD)**
2
3 ## **1. 개요**
4
5 본 문서는 파이썬(Python)과 표준 GUI 라이브러리인 티케이인터(Tkinter)를 사용하여
6 개발될 데스크톱 텍스트 편집기의 요구사항을 정의합니다. 이 애플리케이션은 기본적인
7 텍스트 편집 기능과 함께 사용자의 편의를 위한 다크 모드, 찾기 기능 등을 제공하는
8 것을 목표로 합니다.
9
10 ## **2. 목표 사용자**
11
12 * **초보 개발자 및 학생**: 간단한 코드나 스크립트, 메모, 과제 등을 작성하고
13   편집하려는 사용자.
14 * **일반 사용자**: Windows 메모장과 같은 가볍고 빠른 텍스트 편집기가 필요한
15   사용자.
16
17 ## **3. 핵심 기능**
18
19 ### **3.1. 파일 메뉴**
20
21 | 기능 | 단축키 | 설명 |
22 | :--- | :--- | :--- |
23 | **새 파일** | `Ctrl+N` | 새로운 빈 편집 창을 엽니다. |
24 | **열기** | `Ctrl+O` | 기존 텍스트 파일 (`.txt` 등)을 불러옵니다. |
```

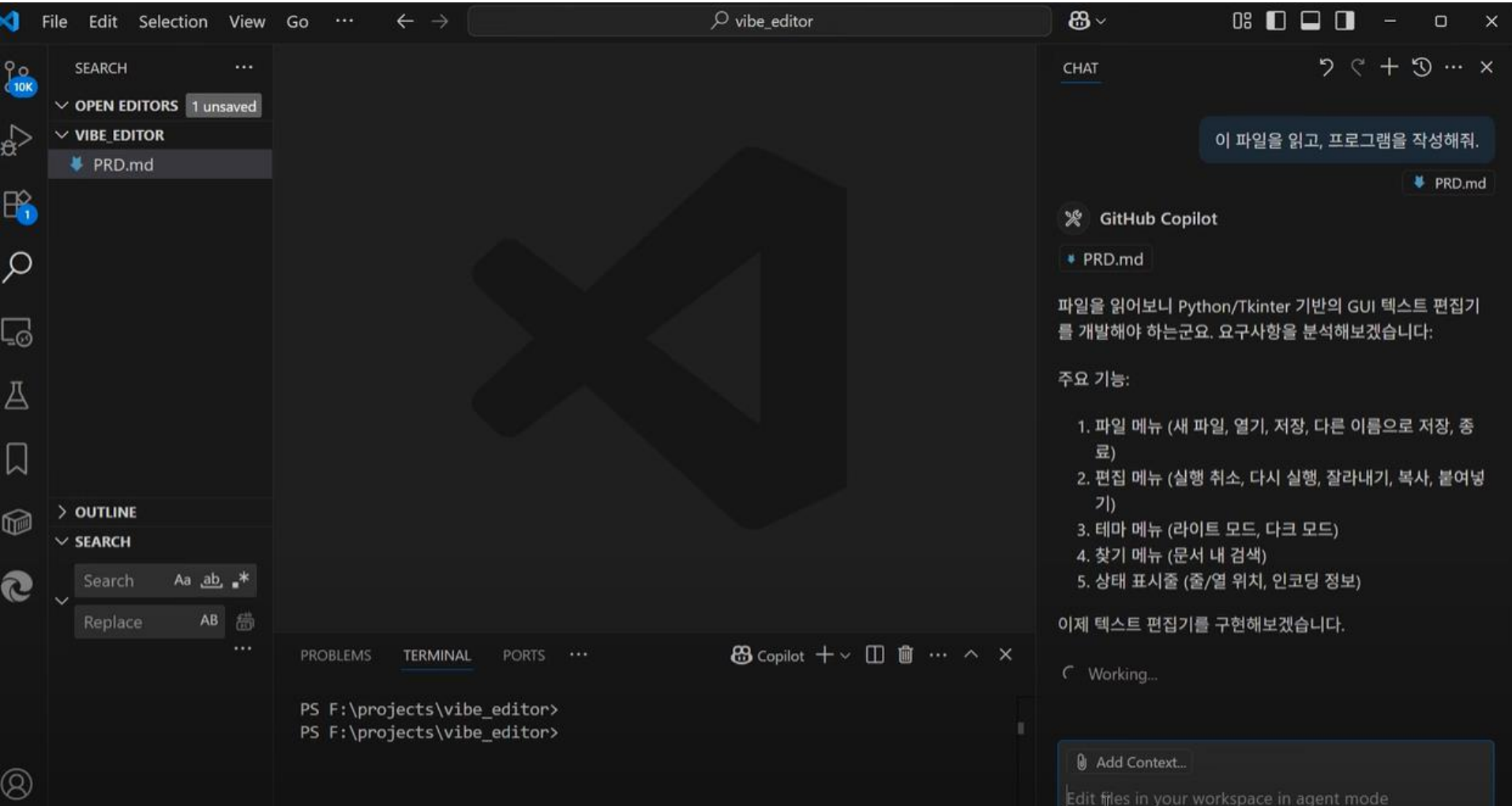
The left sidebar shows the `PRD.md` file in the `OPEN EDITORS` and `VIBE_EDITOR` panels. The `OUTLINE` panel is also visible. The bottom status bar shows the terminal output:

```
PS F:\projects\vibe_editor>
PS F:\projects\vibe_editor>
```

On the right side, the `CHAT` panel displays the `Edit with Copilot` interface in `Agent Mode`. It includes instructions on how to use Copilot to edit files and a warning message at the bottom:

The git repository at "f:\projects" has too many active changes, only a subset of Git features will be enabled.

Vibe coding



Vibe coding

The image shows a Visual Studio Code editor interface with a dark theme. The main editor window displays a Python file named `text_editor.py`. The code defines a `TextEditor` class using `tkinter` for the GUI. A blue Copilot suggestion box is visible over the `__init__` method, showing a checkmark and a question mark icon. The left sidebar contains the Explorer view with a search bar and a list of files: `text_editor.py`, `main.py`, and `README.md`. The bottom panel shows the TERMINAL view with a command prompt running `python main.py`, which has resulted in a `bad event type or keysym "Modified"` error. On the right, a CHAT window displays a conversation with an AI agent, where the agent suggests modifying the `on_modified` function to `on_text_modified` and improving file saving logic. Below the chat, a summary of changes shows three files modified: `main.py`, `README.md`, and `text_editor.py`.

```
1 import tkinter as tk
2 from tkinter import ttk, filedialog, messagebox, simpledialog
3 import os
4
5
6 class TextEditor:
7     def __init__(self):
8         self.root = tk.Tk()
9         self.root.title("Vibe Text Editor")
10        self.root.geometry("800x600")
11
12        # 현재 파일 정보
13        self.current_file = None
14        self.is_modified = False
15
16        # 테마 설정
17        self.is_dark_mode = False
18        self.light_theme = False
```

Keep Undo | 1 of 6

PS F:\projects\vibe_editor> cd f:\projects\vibe_editor && python main.py
애플리케이션 실행 중 오류가 발생했습니다: bad event type or keysym "Modified"
Traceback (most recent call last):
 File "F:\projects\vibe_editor\main.py", line 12, in main
 editor = TextEditor()
 ^^^^^^^^^^^
 File "F:\projects\vibe_editor\text_editor.py", line 34, in __init__
 self.setup_bindings()

CHAT

이제 `on_modified` 함수를 `on_text_modified`로 변경하고 로직을 수정하겠습니다.

또한 파일을 저장할 때와 새 파일을 만들 때 수정 상태를 정확히 초기화하도록 코드를 개선하겠습니다.

좀 더 구체적으로 수정하겠습니다.

Working...

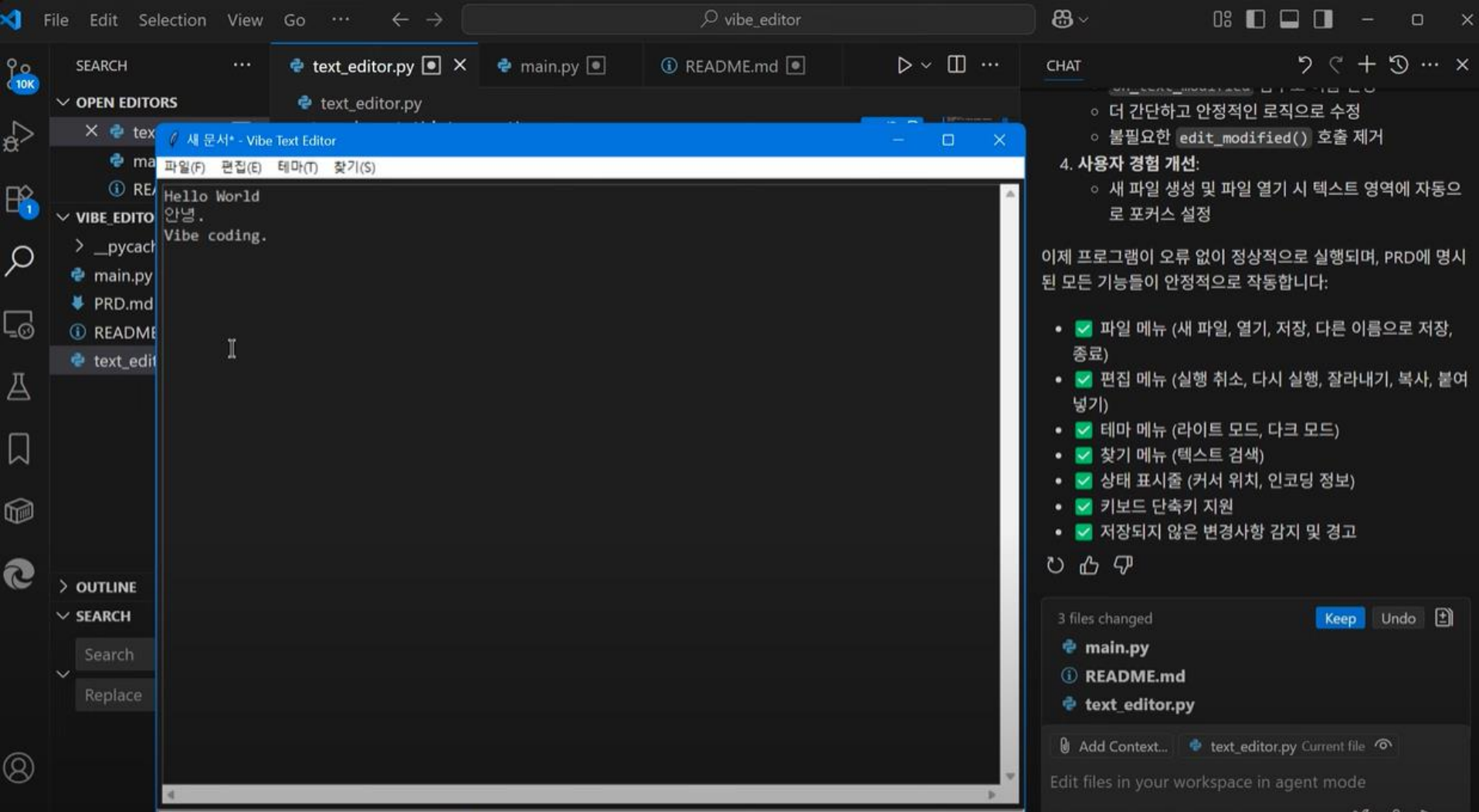
3 files changed

- main.py
- README.md
- text_editor.py

Add Context... text_editor.py Current file

Edit files in your workspace in agent mode

Vibe coding



Vibe coding

The screenshot displays the Visual Studio Code interface with a project named 'vibe_web_editor'. The left sidebar shows the 'OPEN EDITORS' list with 'index.html' and 'PRD.md' open. The 'VIBE_WEB_EDITOR' extension is also visible. The main editor area shows the 'index.html' file with the following code:

```
<!DOCTYPE html>
<html lang="ko" class="h-full">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <title>My Web Editor</title>
  <script src="https://cdn.tailwindcss.com"></script>
  <script>
    tailwind.config = {
      darkMode: 'class',
      theme: {
        extend: {
          fontFamily: {
            'mono': ['Consolas', 'Monaco', 'Courier New',
10         ]
11       }
12     }
13   }
14 }
15 }
16 }
17 }
18 }
19 }
20 }
21 }
22 }
23 }
24 }
25 }
26 }
27 }
28 }
29 }
30 }
31 }
32 }
33 }
34 }
35 }
36 }
37 }
38 }
39 }
40 }
```

The right sidebar shows the 'CHAT' panel with the following text:

이 파일을 읽고, 프로그램 개발해

PRD.md PRD.md:1-29

GitHub Copilot

이 PRD를 읽고 웹 기반 텍스트 편집기를 개발하겠습니다. PRD에 명시된 요구사항에 따라 Tailwind CSS를 사용한 웹 애플리케이션을 만들어보겠습니다.

먼저 프로젝트 구조를 파악하고 필요한 파일들을 생성하겠습니다.

✓ Read PRD.md vibe_web_editor

현재 PRD.md 파일만 있네요. 웹 기반 텍스트 편집기의 핵심 파일들을 생성하겠습니다.

✓ Created index.html

이제 JavaScript 파일을 생성하여

1 file changed

index.html

Add Context... index.html Current file

Edit files in your workspace in agent mode

Agent Claude Sonnet 4

Ln 141, Col 1 Spaces: 4 UTF-8 {} HTML Line 1 Count 0

Vibe coding

The screenshot shows a VS Code editor window with the following components:

- Left Sidebar:** Contains the 'SEARCH' panel with 'OPEN EDITORS' and 'VIBE...' sections. The 'VIBE...' section lists files: 'index.html', 'PRD.md', 'JS script.js', and 'JS sw.js'. The 'JS script.js' file is selected.
- Main Editor:** Displays the content of 'JS script.js'. The code defines a 'WebEditor' class with methods for initializing event listeners, theme, and status bar. The code is as follows:

```
1 // 웹 기반 텍스트 편집기 - 메인 JavaScript 파일
2
3 class WebEditor {
4   constructor() {
5     this.editor = document.getElementById('editor');
6     this.searchBar = document.getElementById('searchBar');
7     this.searchInput = document.getElementById('searchInput');
8     this.searchCount = document.getElementById('searchCount');
9     this.cursorPos = document.getElementById('cursorPos');
10    this.charCount = document.getElementById('charCount');
11    this.themeStatus = document.getElementById('themeStatus');
12    this.fileInput = document.getElementById('fileInput');
13
14    this.currentTheme = 'system'; // 'system', 'light', 'dark'
15    this.searchMatches = [];
16    this.currentSearchIndex = -1;
17    this.originalText = '';
18
19    this.initializeEventListeners();
20    this.initializeTheme();
21    this.updateStatusBar();
22  }
23
24  initializeEventListeners() {
25    // 파일 관련 버튼
26    document.getElementById('newBtn').addEventListener('click', () => {
27      document.getElementById('openBtn').addEventListener('click', () => {
28        document.getElementById('saveBtn').addEventListener('click', () => {
29
30      // 검색 관련
31      document.getElementById('searchBtn').addEventListener('click', () => {
32        document.getElementById('searchCloseBtn').addEventListener('click', () => {
33          document.getElementById('searchInput').addEventListener('input', () => {
34            document.getElementById('searchPrevBtn').addEventListener('click', () => {
35              document.getElementById('searchNextBtn').addEventListener('click', () => {
36
37      // 테마 관련
38      document.getElementById('themeBtn').addEventListener('click', () => {
39
40      // 에디터 관련
```
- Right Sidebar:** Contains the 'CHAT' panel. It displays a message from 'GitHub Copilot' and a list of files generated by the agent: 'PRD.md', 'index.html', 'script.js', and 'sw.js'. The chat interface includes a 'Keep' button and a 'Undo' button. The chat content is as follows:

이 파일을 읽고, 프로그램 개발해

이 PRD를 읽고 웹 기반 텍스트 편집기를 개발하겠습니다. PRD에 명시된 요구사항에 따라 Tailwind CSS를 사용한 웹 애플리케이션을 만들어보겠습니다.

먼저 프로젝트 구조를 파악하고 필요한 파일들을 생성하겠습니다.

현재 PRD.md 파일이 있네요. 웹 기반 텍스트 편집기의 핵심 파일들을 생성하겠습니다.

이제 JavaScript 파일을 생성하여 모든 기능을 구현하겠습니다.

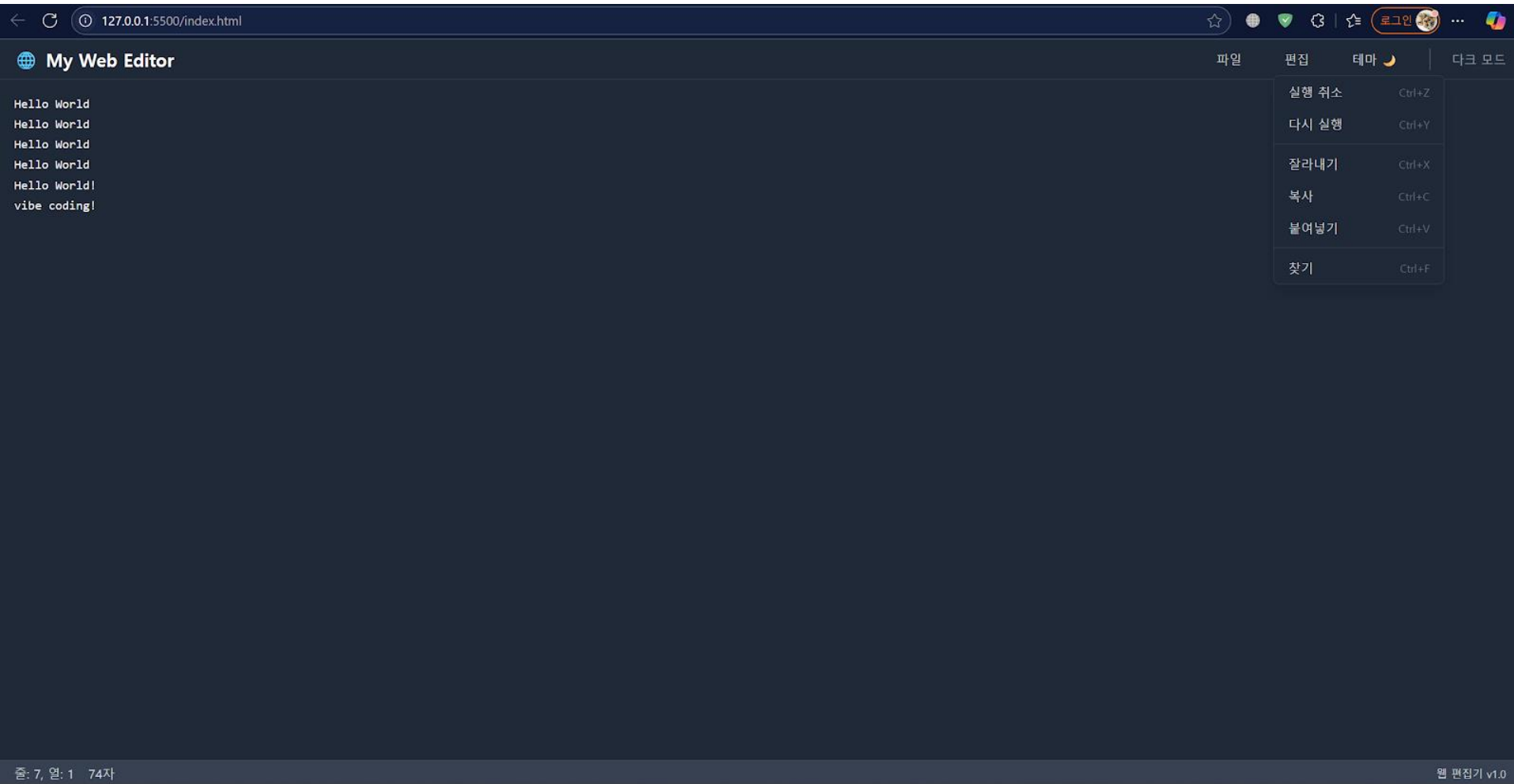
이제 PWA(Progressive Web App) 지원을 위한 서비스 워커와 매니페스트 파일을 생성하겠습니다.

Vibe coding

The image shows a screenshot of a VS Code editor interface. The left sidebar contains the 'EXPLORER' view with a search bar and a list of files: 'PRD.md', 'index.html', 'sw.js', 'script.js', and 'manifest.json'. The 'VIBE_WEB_EDITOR' view is also visible. The main editor area displays the 'index.html' file, which contains HTML code for a web editor. The code includes a header, a main content area, and a footer. The code is written in a dark theme with syntax highlighting. The right sidebar shows the 'CHAT' panel with a 'GitHub Copilot' chat window. The chat window displays a message in Korean: '현재 HTML 파일을 보니 편집 메뉴와 테마 메뉴가 없고, 단순한 버튼들만 있네요. PRD 요구 사항에 맞게 아름다운 메뉴 형태로 수정하겠습니다.' (Looking at the HTML file, I see there are no edit and theme menus, just simple buttons. I will modify it into a beautiful menu format according to the PRD requirements.) Below the chat window, there is a list of files changed: 'Index.html', 'manifest.json', 'script.js', and 'sw.js'. The bottom status bar shows the current file is 'index.html' at line 73, column 19, with a UTF-8 encoding and HTML file type.

```
2 <html lang="ko" class="h-full">
56 <body class="h-full bg-white dark:bg-gray-900 transition-colors duration-200">
57   <div class="h-full flex flex-col">
59     <header class="bg-white dark:bg-gray-800 border-b border-gray-200 dark:border-gray-700">
60       <div class="flex items-center justify-between">
65         <!-- Controls -->
66         <div class="flex items-center space-x-2">
67           <!-- File Controls -->
68           <div class="flex items-center space-x-1 mr-4">
69             <button id="newBtn" class="px-3 py-1 text-sm bg-blue-500 hover:bg-blue-600">
70               새 문서
71             <!-- Menu Bar -->
72             <div class="flex items-center space-x-6">
73               <!-- File Menu -->
74               <div class="relative dropdown">
75                 <button class="px-3 py-1 text-sm font-medium text-gray-700 dark:text-gray-300">
76                   파일
77                 <div class="dropdown-menu absolute top-full left-0 mt-1 w-48">
78                   <button id="openBtn" class="px-3 py-1 text-sm bg-green-500 hover:bg-green-600">
79                     열기
80                   <div class="dropdown-menu absolute top-full left-0 mt-1 w-48">
81                     <button id="newBtn" class="w-full text-left px-4 py-2">
82                       새 문서
83                     <span class="text-xs text-gray-500">Ctrl+N</span>
84                   </button>
85                   <button id="openBtn" class="w-full text-left px-4 py-2">
86                     파일 열기
87                   <span class="text-xs text-gray-500">Ctrl+O</span>
88                   </button>
89                   <div class="border-t border-gray-200 dark:border-gray-700">
90                     <button id="saveBtn" class="w-full text-left px-4 py-2">
91                       내보내기 (.txt)
92                     <span class="text-xs text-gray-500">Ctrl+S</span>
93                   </button>
94                 </div>
95               </div>
96             <!-- Edit Menu -->
```


Vibe coding



Vibe coding

127.0.0.1:7860



전자기학 방정식 시각화 도구

Maxwell 방정식과 관련된 전자기학 현상들을 시각화합니다.

전자기파 (Wave Equation) 가우스 법칙 ($\nabla \cdot \mathbf{E} = \rho/\epsilon_0$) 암페어 법칙 ($\nabla \times \mathbf{B} = \mu_0 \mathbf{J}$) 패러데이 법칙 ($\nabla \times \mathbf{E} = -\partial \mathbf{B} / \partial t$)

전류 주변의 자기장 회전

전류 (I)

1

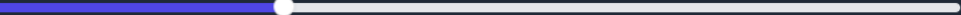


0.1  5

그리드 해상도

8

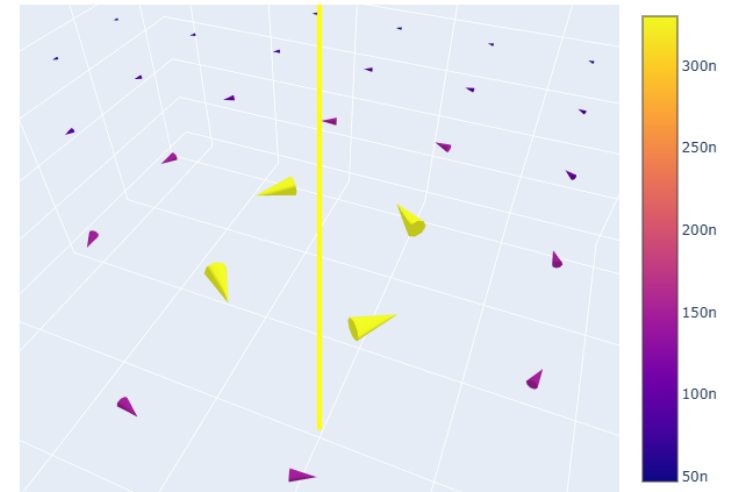


5  15

그래프 생성

자기장 회전

Magnetic Field around Current Wire ($\nabla \times \mathbf{B} = \mu_0 \mathbf{J}$)



Vibe coding
Hands-on

Gemini-CLI

Vibe coding

google-gemini/gemini-cli: An open-source AI agent that brings the power of Gemini directly into your terminal.



gemini cli vibe coding demo

Daddy Makers: 바이브 코딩을 위한 구글
Gemini CLI 도구 분석 및 사용

Vibe coding

<https://github.com/google-gemini/gemini-cli>

 **README**  Contributing  Apache-2.0 license  Security

Gemini CLI is an open-source AI agent that brings the power of Gemini directly into your terminal. It provides lightweight access to Gemini, giving you the most direct path from your prompt to our model.

Why Gemini CLI?

-  **Free tier:** 60 requests/min and 1,000 requests/day with personal Google account
-  **Powerful Gemini 2.5 Pro:** Access to 1M token context window
-  **Built-in tools:** Google Search grounding, file operations, shell commands, web fetching
-  **Extensible:** MCP (Model Context Protocol) support for custom integrations
-  **Terminal-first:** Designed for developers who live in the command line
-  **Open source:** Apache 2.0 licensed

Installation

Quick Install

Run instantly with npx

```
# Using npx (no installation required)
npx https://github.com/google-gemini/gemini-cli
```



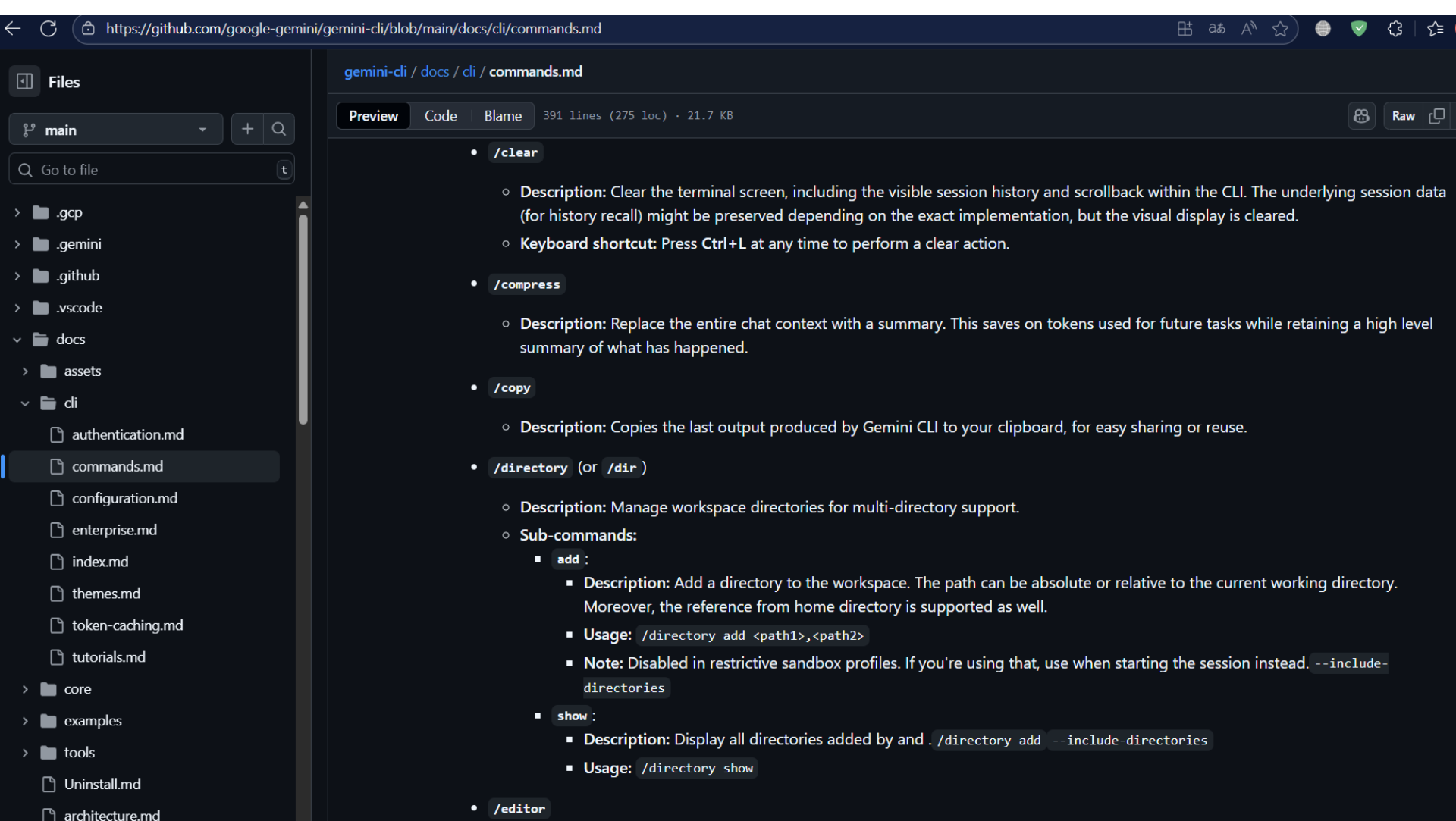
Install globally with npm

```
npm install -g @google/gemini-cli
```



Vibe coding

[gemini-cli/docs/cli/commands.md](https://github.com/google-gemini/gemini-cli/blob/main/docs/cli/commands.md) at main · google-gemini/gemini-cli

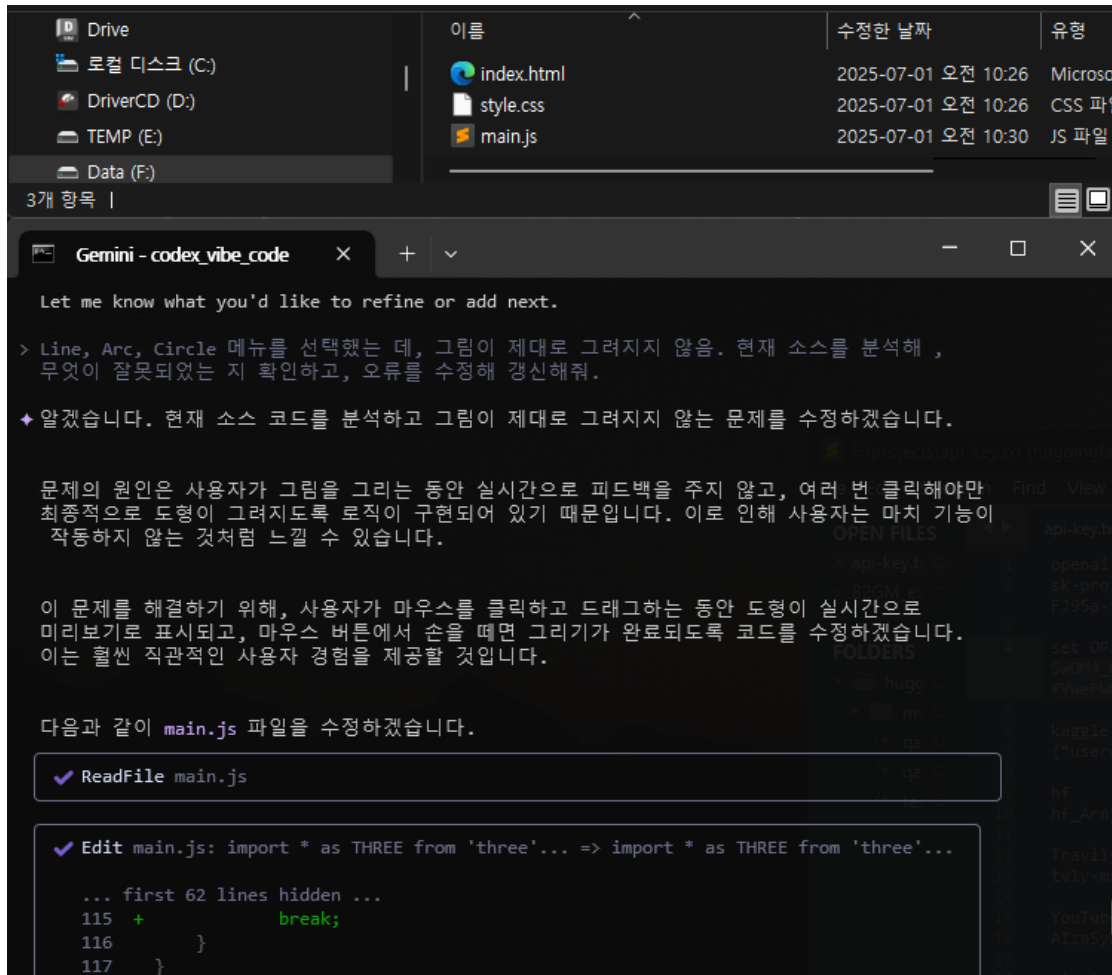


The screenshot shows the GitHub web interface for the file `gemini-cli / docs / cli / commands.md`. The file is 391 lines long, has 275 loc, and is 21.7 KB. The left sidebar shows the file explorer with the `cli` directory selected, containing files like `authentication.md`, `commands.md`, `configuration.md`, `enterprise.md`, `index.md`, `themes.md`, `token-caching.md`, and `tutorials.md`. The main content area displays the file's content, which is a list of commands and their descriptions.

- `/clear`
 - **Description:** Clear the terminal screen, including the visible session history and scrollbar within the CLI. The underlying session data (for history recall) might be preserved depending on the exact implementation, but the visual display is cleared.
 - **Keyboard shortcut:** Press `Ctrl+L` at any time to perform a clear action.
- `/compress`
 - **Description:** Replace the entire chat context with a summary. This saves on tokens used for future tasks while retaining a high level summary of what has happened.
- `/copy`
 - **Description:** Copies the last output produced by Gemini CLI to your clipboard, for easy sharing or reuse.
- `/directory` (or `/dir`)
 - **Description:** Manage workspace directories for multi-directory support.
 - **Sub-commands:**
 - **add :**
 - **Description:** Add a directory to the workspace. The path can be absolute or relative to the current working directory. Moreover, the reference from home directory is supported as well.
 - **Usage:** `/directory add <path1>,<path2>`
 - **Note:** Disabled in restrictive sandbox profiles. If you're using that, use when starting the session instead. `--include-directories`
 - **show :**
 - **Description:** Display all directories added by and `./directory add --include-directories`
 - **Usage:** `/directory show`
- `/editor`

Vibe coding

> make photoshop web app using three.js, bootstrap. Menu includes layer, line, arc, circle, fill color with transparent, border color, zoom in/out, pan, download file as JPG



Vibe coding

VS Code Interface:

- Left Panel:** SEARCH, OPEN EDITORS (JS main.js), GEMINI_VIBE_CODE (index.html, JS main.js, style.css), OUTLINE, SEARCH, REPLACE, TIMELINE.
- Main Editor:** JS main.js. Code includes:

```
document.addEventListener('DOMContentLoaded', () => {
  function onMouseMove(event) {
    // ...
    switch (drawingMode) {
      case 'line':
        updatePreviewLine(currentPoint);
        break;
      case 'circle':
        updatePreviewCircle(currentPoint);
        break;
    }
  }

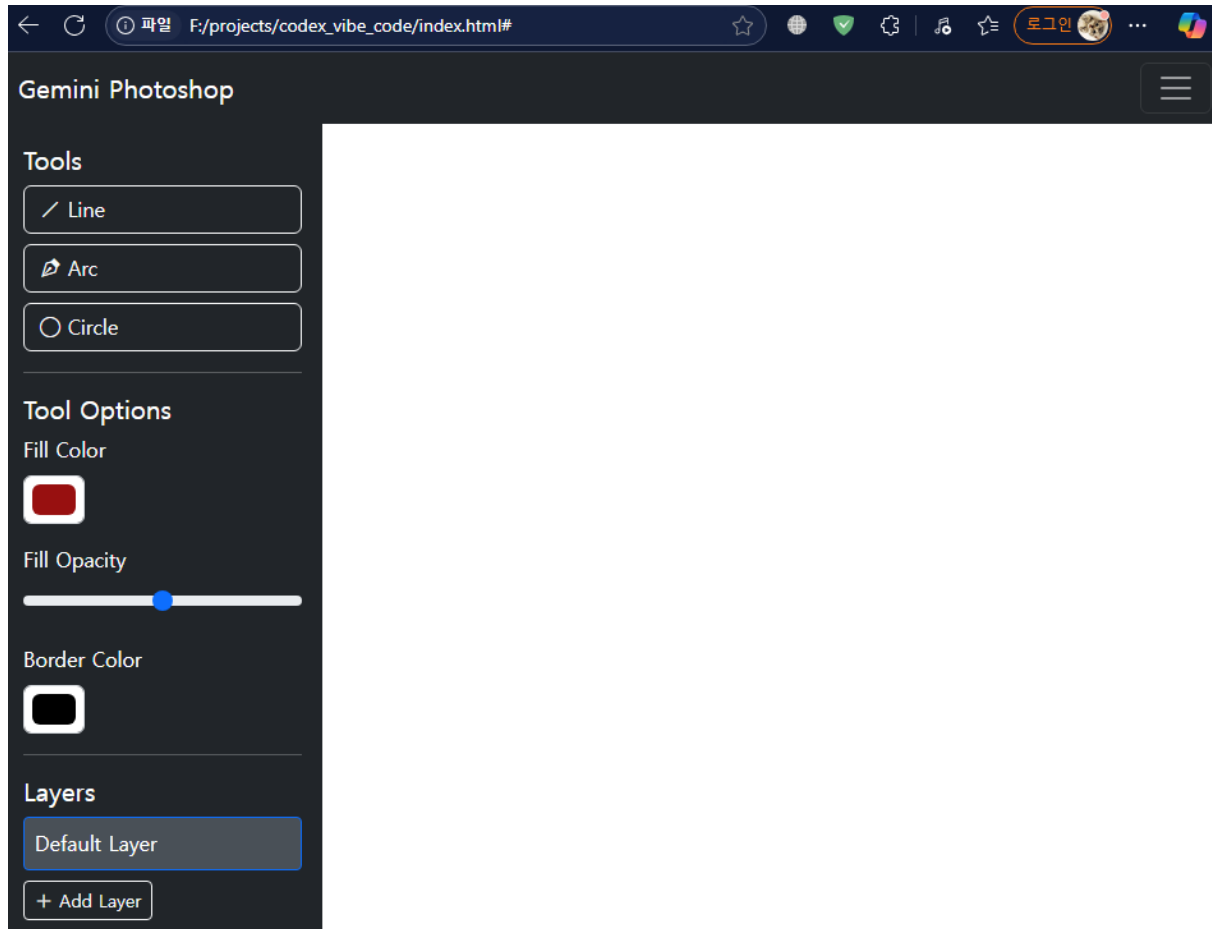
  function onMouseUp(event) {
    if (!isDrawing) return;
    isDrawing = false;
    previewObject = null; // The object is now part of the layer
    startPoint = null;
  }

  function createPreviewLine(p1) {
    const material = new THREE.LineBasicMaterial({ color: document... });
    const geometry = new THREE.BufferGeometry().setFromPoints([
    previewObject = new THREE.Line(geometry, material);
    activeLayer.add(previewObject);
  }

  function updatePreviewLine(p2) {
    previewObject.geometry.setFromPoints([startPoint, p2]);
    previewObject.geometry.verticesNeedUpdate = true;
    previewObject.geometry.attributes.position.needsUpdate = true;
  }

  function createPreviewCircle(center) {
    const geometry = new THREE.CircleGeometry(0.01, 64); // Start
  }
}
```
- Right Panel:** CHAT (GitHub Copilot).
 - Text: "코드를 검토해보니 몇 가지 문제점들이 있습니다. 주요 문제들을 수정하겠습니다."
 - List:
 1. `verticesNeedUpdate` 속성이 더 이상 사용되지 않음
 2. Circle geometry 업데이트 시 `needsUpdate` 플래그 누락
 3. Arc 기능이 단순한 선으로만 구현됨
 - Text: "이제 이러한 문제들을 수정하겠습니다."
 - Diff: JS main.js +1 -1, JS main.js +1 -0.
 - Text: "이제 HTML 파일과 CSS 파일을 확인해서 UI 요소들이 제대로 존재하는지 확인해보겠습니다."
 - Checkmarks:
 - ✓ Read index.html, lines 1 to 50
 - ✓ Read index.html, lines 51 to 87
 - Text: "HTML은 잘 설정되어 있습니다. 이제 JavaScript 코드에서 도구 선택 시 활성 상태를 표시하는 기능을 추가하고, 더 나은 시각적 피드백을 제공하도록 개선하겠습니다."
 - Diff: JS main.js +9 -3.
 - Status: Working...
 - Summary: 1 file changed. JS main.js.
 - Buttons: Add Context..., JS main.js Current file.
 - Text: "Edit files in your workspace in agent mode"
 - Footer: Agent Claude Sonnet 4 v.

Vibe coding



Vibe coding Hands-on

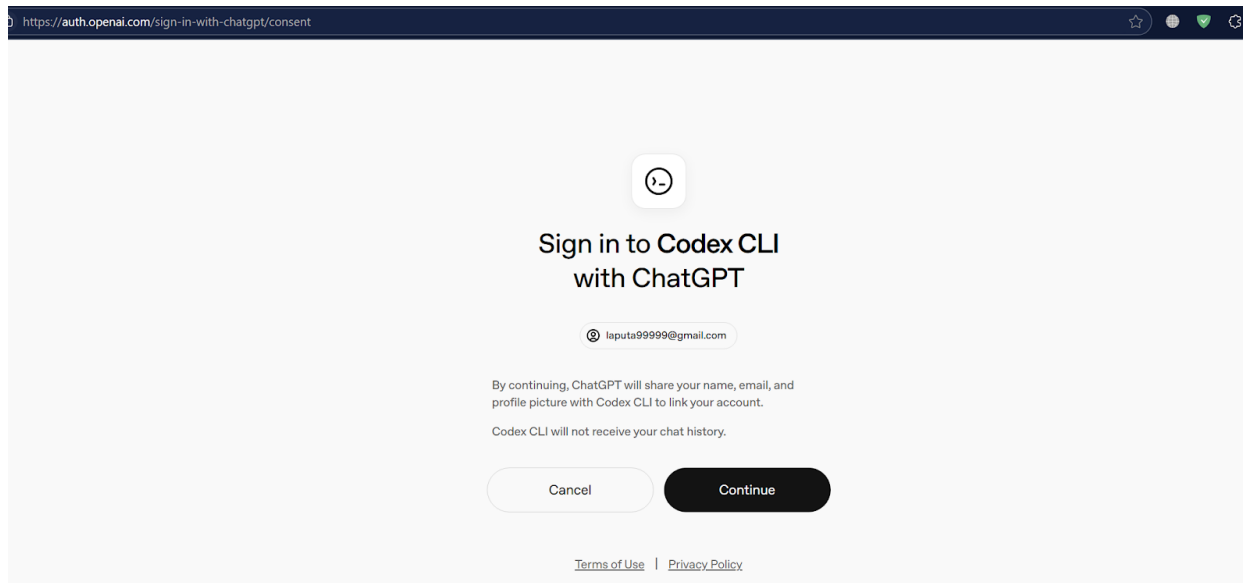
Codex

Vibe coding

openai/codex: Lightweight coding agent that runs in your terminal

What platforms are supported?

Codex officially supports macOS and Linux. Windows support is experimental and may require WSL.



Daddy Makers: OpenAI 바이브 코딩 지원 멀
티 에이전트 Codex 도구 사용법

Vibe coding

openai/codex: Lightweight coding agent that runs in your terminal

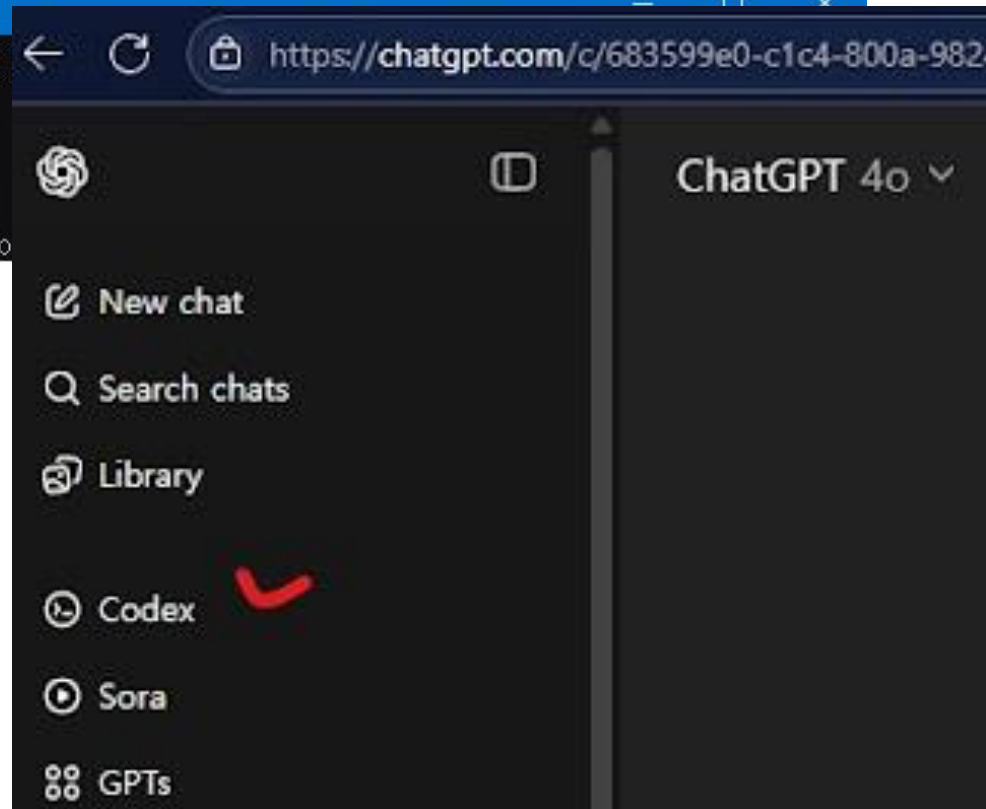
What platforms are supported?

Codex officially supports macOS and Linux. Windows support is experimental and may require WSL.

```
관리자: Windows PowerShell
(base) PS F:\projects\codex> codex "explain this codebase to
Sign in with ChatGPT to generate an API key or paste one you
[use arrows to move, enter to select]

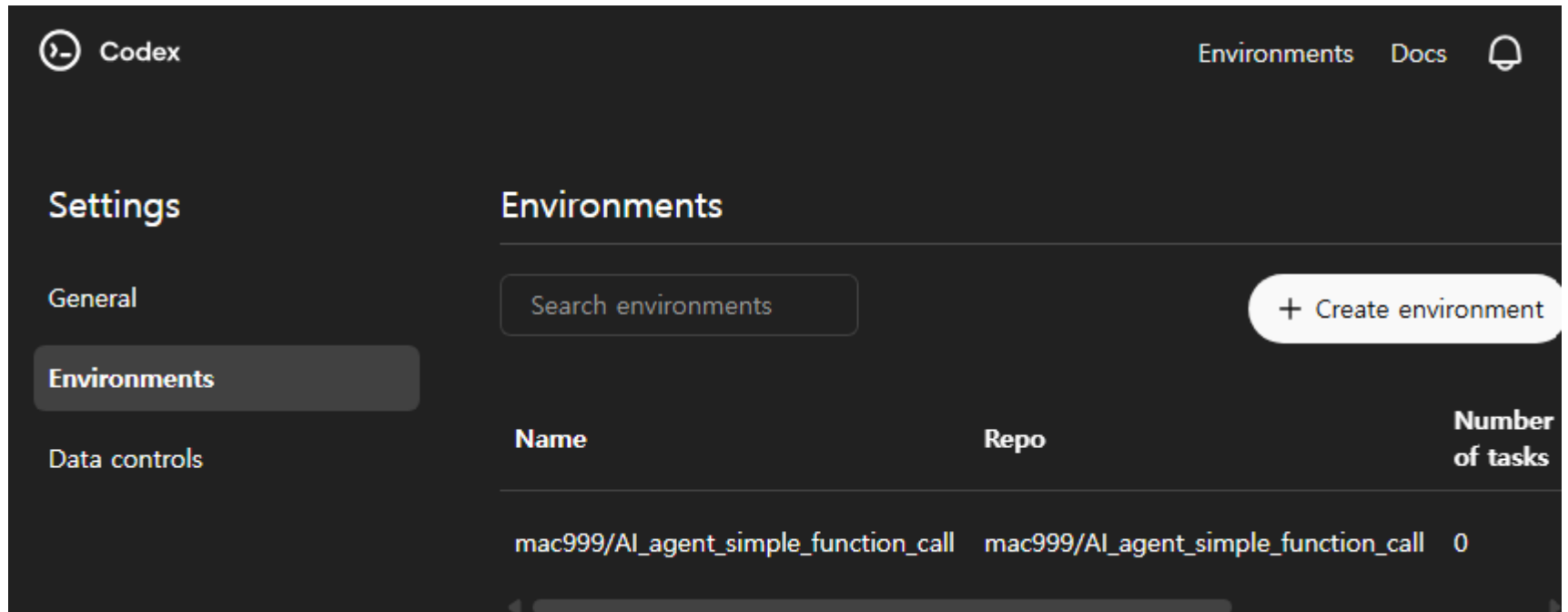
> Sign in with ChatGPT
  Paste an API key (or set as OPENAI_API_KEY)

Opening login page in your browser: https://auth.openai.com/o
```



Daddy Makers: OpenAI 바이브 코딩 지원 멀
티 에이전트 Codex 도구 사용법

Vibe coding



Vibe coding

What are we coding next?

Describe a task

mac999/AI_agent_simpl... ▾

main ▾



mac999/AI_agent_simple_function_call ✓

Manage environments

57 min ago · mac999/AI_agent_simple_function_call

+1 -1

Create AI agent with ReAct and OpenAI API

27 min ago · mac999/AI_agent_simple_function_call

Merged

+139 -0

Find and propose tasks for typo, bug, comment, and test improvements

58 min ago · mac999/AI_agent_simple_function_call

Vibe coding

 Codex Environments Docs

What are we coding next?

Describe a task

mac999/AI_agent_simpl... main

Tasks

Archive

Fix typo in comment

32 min ago · mac999/AI_agent_simple_function_call

+1 -1

Create AI agent with ReAct and OpenAI API

2 min ago · mac999/AI_agent_simple_function_call

Merged +139 -0

Find and propose tasks for typo, bug, comment, and test improvements

33 min ago · mac999/AI_agent_simple_function_call

Explain general structure of codebase

34 min ago · mac999/AI_agent_simple_function_call

Find and fix a bug in codebase

26 min ago · mac999/AI_agent_simple_function_call

Merged +78 -47

Vibe coding

2분 8초 동안 일했습니다. >

확인된 문제

1. 댓글에 오타

- Python 스크립트는 "prototype"의 철자를 "protype"로 잘못 씁니다. ⓘ

제안된 작업
주석의 오타 수정

작업 보기

2. query_llm 메모리 로깅의 버그

- 이 함수는 유효한 속성이 아닌 `response.response` 를 사용하여 응답을 저장하려고 시도합니다. (또는 LLM 응답의 적절한 속성)을 사용해야 합니다. `response.response` `response.content` ⓘ

제안된 작업
query_llm의 올바른 응답 속성

작업 시작 ▶

3. 주석/주석 불일치

- `query_llm` 반환 유형을 로 선언하지만 전체 응답 개체를 반환합니다. 이 주석은 오해의 소지가 있습니다. `str` ⓘ

제안된 작업
query_llm에 대한 반환 유형 주석 업데이트

작업 시작 ▶

4. 누락된 테스트

- 저장소에 테스트가 없습니다. 와 같은 기능에는 자동 검사가 없습니다. `get_function_prototype`

제안된 작업
get_function_prototype에 대한 단위 테스트 추가

작업 시작 ▶

요약

- 댓글에서 오타("protype")를 식별했습니다.
- 잘못된 응답 속성을 저장하는 버그를 발견했습니다. `query_llm`
- 에서 오해의 소지가 있는 반환 유형 주석을 발견했습니다. `query_llm`
- 에 대한 기본 테스트를 추가하는 것이 좋습니다. `get_function_prototype`

이러한 문제를 해결하면 코드의 품질과 안정성이 향상될 것입니다. 더 많은 문제를 발견하려면 코드를 더 자세히 검토하십시오.

7 Request changes or ask a question

Vibe coding

← ↻ https://github.com/mac999/AI_agent_simple_function_call

mac999 / AI_agent_simple_function_call

<> Code Issues Pull requests Actions Projects Wiki Security Insights Settings

 **AI_agent_simple_function_call** Public Pin Unwatch 1

 main  2 Branches  0 Tags t Add file <> Code

 **mac999** Merge pull request #1 from mac999/codex/find-and-fix-a-bug-in-codebase 2b22c5c · now 6 Commits

 tests	Add safe code block handling and tests	1 minute ago
 LICENSE	Initial commit	5 months ago
 README.md	Update README.md	5 months ago
 llm_func_call_for_ai_agent.py	Add safe code block handling and tests	1 minute ago

 **README**  MIT license ✎ ☰

Simplifying LLM Function Calls for AI Agents

Vibe coding

What are we coding next?

Describe a task

Tasks

Archive

Fix typo in comment

5 min ago · mac999/AI_agent_simple_function_call

+1 -1

• Create AI agent with ReAct and OpenAI API

8 min ago · mac999/AI_agent_simple_function_call

+139 -0

Find and propose tasks for typo, bug, comment, and test improvements

7 min ago · mac999/AI_agent_simple_function_call

Explain general structure of codebase

7 min ago · mac999/AI_agent_simple_function_call

Find and fix a bug in codebase

3 min ago · mac999/AI_agent_simple_function_call



Merged

+78 -47

Vibe coding

https://github.com/mac999/Al_agent_simple_function_call/pull/1

Open mac999 wants to merge 1 commit into main from codex/find-and-fix-a-bug-in-codebase

Conversation 0 Commits 1 Checks 0 Files changed 2 +78 -47

mac999 commented 1 minute ago

Owner

Summary

- handle missing Python code blocks in `preprocess_code`
- wrap runnable demo code in `if __name__ == '__main__':`
- add unit tests for `preprocess_code`

Testing

- `python3 -m unittest tests/test_preprocess_code.py -v`

https://chatgpt.com/codex/tasks/task_e_68428c6b5174832a9e67c77e31810860

Add safe code block handling and tests

mac999 added the **codex** label 1 minute ago — with ChatGPT Connector

No conflicts with base branch

Merging can be performed automatically.

Merge pull request

You can also merge this with the command line. [View command line instructions.](#)

Add a comment

Reviewers

Suggestions

Copilot

Still in progress? [Convert to draft](#)

Assignees

No one—[assign yourself](#)

Labels

codex

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

None yet

Notifications

Customize

Unsubscribe

You're receiving notifications because you're watching this repository.

1 participant

Vibe coding

← Create AI agent with ReAct and OpenAI API
Jun 6 mac999/AI_agent_simple_function_call +139 -0

make AI agent simple source code using python to implement ReAct, with openai api. it supports CLI for singline query mode, chatbot mode.

Worked for 1m 34s >

Notes

- Added `react_agent.py` containing a minimal ReAct-style loop that uses OpenAI function calling for searching and calculations.
- Updated `README.md` with instructions on how to run the new CLI tool.

Summary

- Documented the usage of the new ReAct CLI in the README, including single-query and chatbot modes ⁽ⁱ⁾
- Implemented `react_agent.py` to provide search and calculator tools and handle interactive or one-shot prompts ⁽ⁱ⁾

Testing

- ✖ `pytest -q` (no tests were found) >

Files (2)

README.md +19 -0

react_agent.py New

👍 🗨

Request changes or ask a question

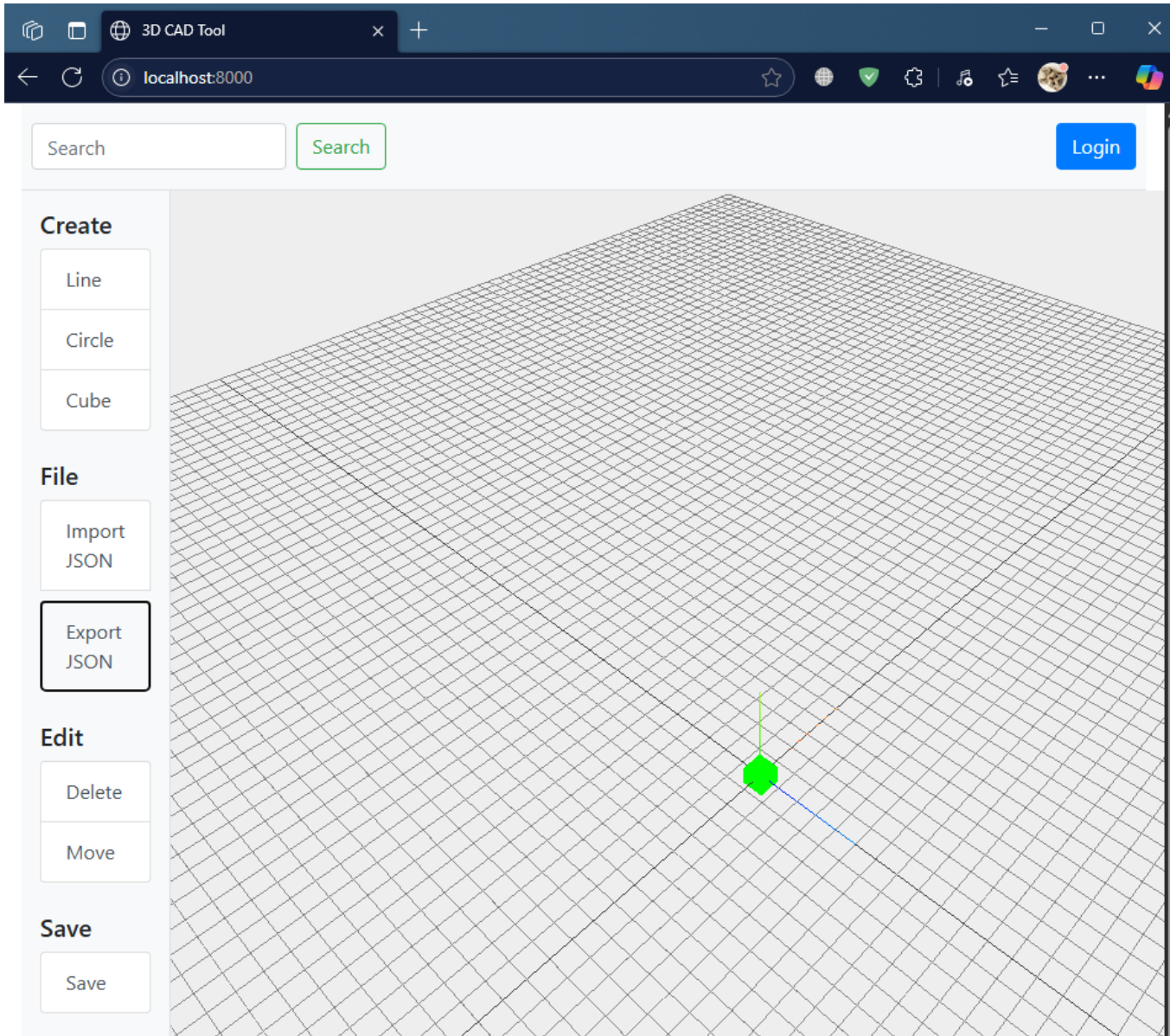
🎤

Diff Logs

▼ react_agent.py New

```
54 + }
55 +
56 +
57 + def react_loop(messages):
58 +     """Run the ReAct reasoning loop until a final answer is produced."""
59 +     while True:
60 +         response = openai.ChatCompletion.create(
61 +             model="gpt-4o",
62 +             messages=messages,
63 +             tools=tools,
64 +             tool_choice="auto",
65 +         )
66 +         msg = response["choices"][0]["message"]
67 +         if msg.get("tool_calls"):
68 +             messages.append(msg)
69 +             for call in msg["tool_calls"]:
70 +                 name = call["function"]["name"]
71 +                 args = json.loads(call["function"]["arguments"])
72 +                 result = FUNCTION_MAP[name](**args)
73 +                 messages.append({
74 +                     "role": "tool",
75 +                     "tool_call_id": call["id"],
76 +                     "content": result,
77 +                 })
78 +                 continue
79 +             messages.append(msg)
80 +             return msg["content"]
81 +
82 +
83 + def single_query(prompt: str) -> str:
84 +     messages = [
85 +         {"role": "system", "content": "You are a helpful assistant using ReAct."},
86 +         {"role": "user", "content": prompt},
87 +     ]
88 +     return react_loop(messages)
89 +
90 +
91 + def chat_mode():
92 +     messages = [{"role": "system", "content": "You are a helpful assistant using ReAct."}]
93 +     print("Starting chat. Type 'exit' to quit.")
94 +     while True:
95 +         user = input("> ")
96 +         if user.lower() in {"exit", "quit"}:
```

Vibe coding



Vibe coding Challenge



Vibe Coding

Build me an epic 3D game with cool features that'll make me \$\$\$!



Why is it not working?!



Vibe Engineering

Let's build a flying car game in Three.js, backend in Go, multiplayer with WebSocket...



That was easy!



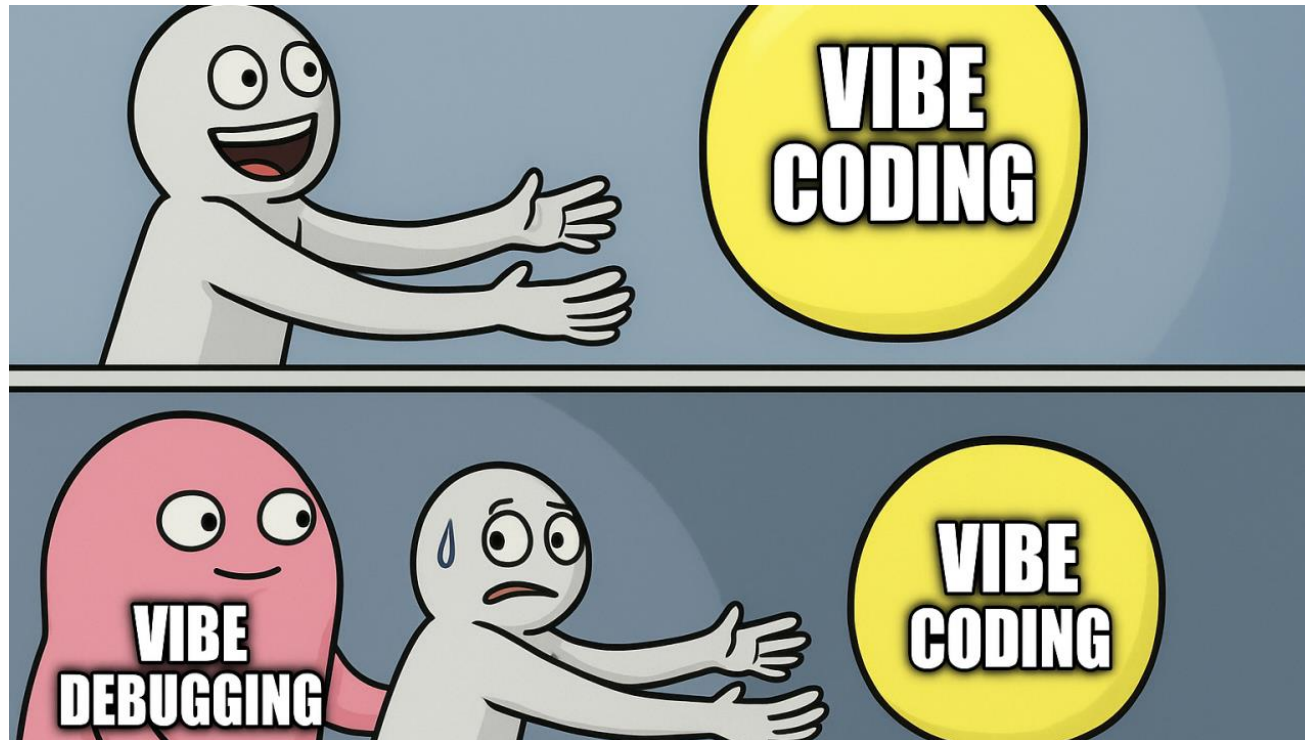
Vibe coding Challenge



Vibe coding Challenge



Vibe coding Challenge



Vibe coding Challenge



Vibe coding Challenge

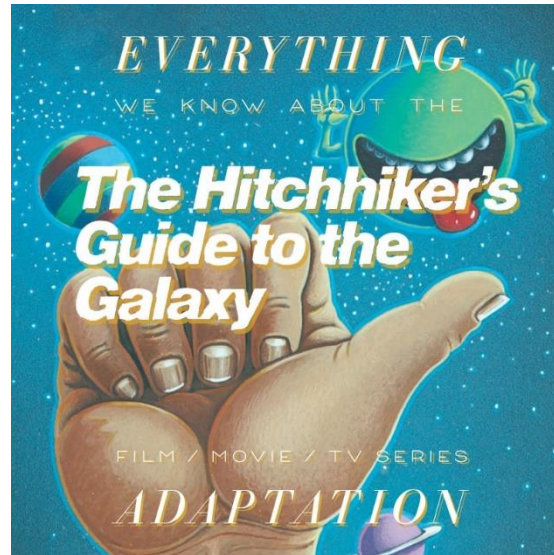
잘하는 것

생산성 향상
진입장벽 완화
창의성 집중
신속한 학습 및 적용

못하는 것

코드 신뢰성 및 품질 개선
라이브러리/알고리즘 과도한 의존성
디버깅과 유지보수 어려움
복잡한 맥락 이해

Join AI world

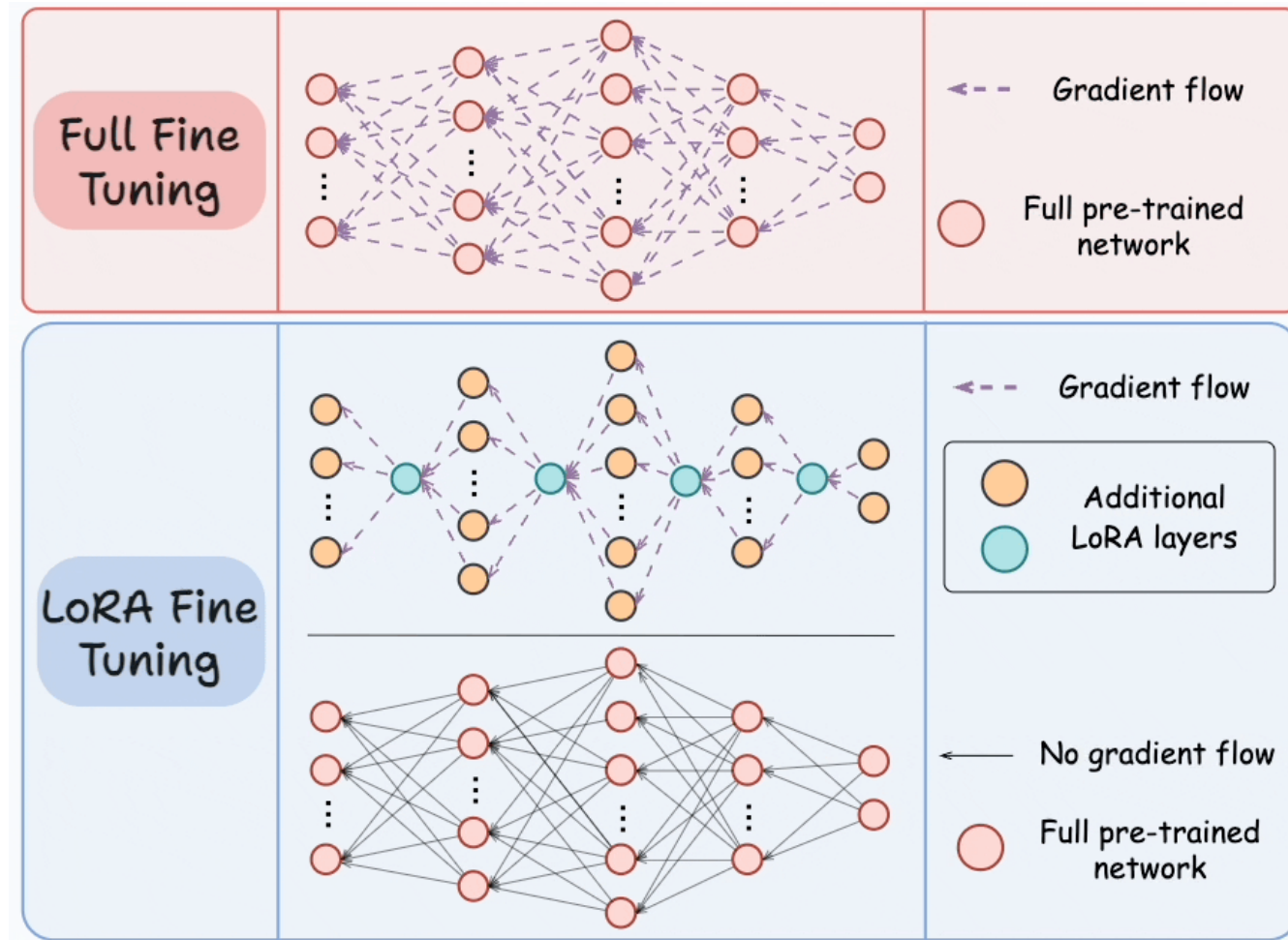


Daddy Makers

BIM digest podcast
Computer graphics digest
Software engineering digest

laputa99999@gmail.com

Fine turning & LoRA Low-rank adaptation

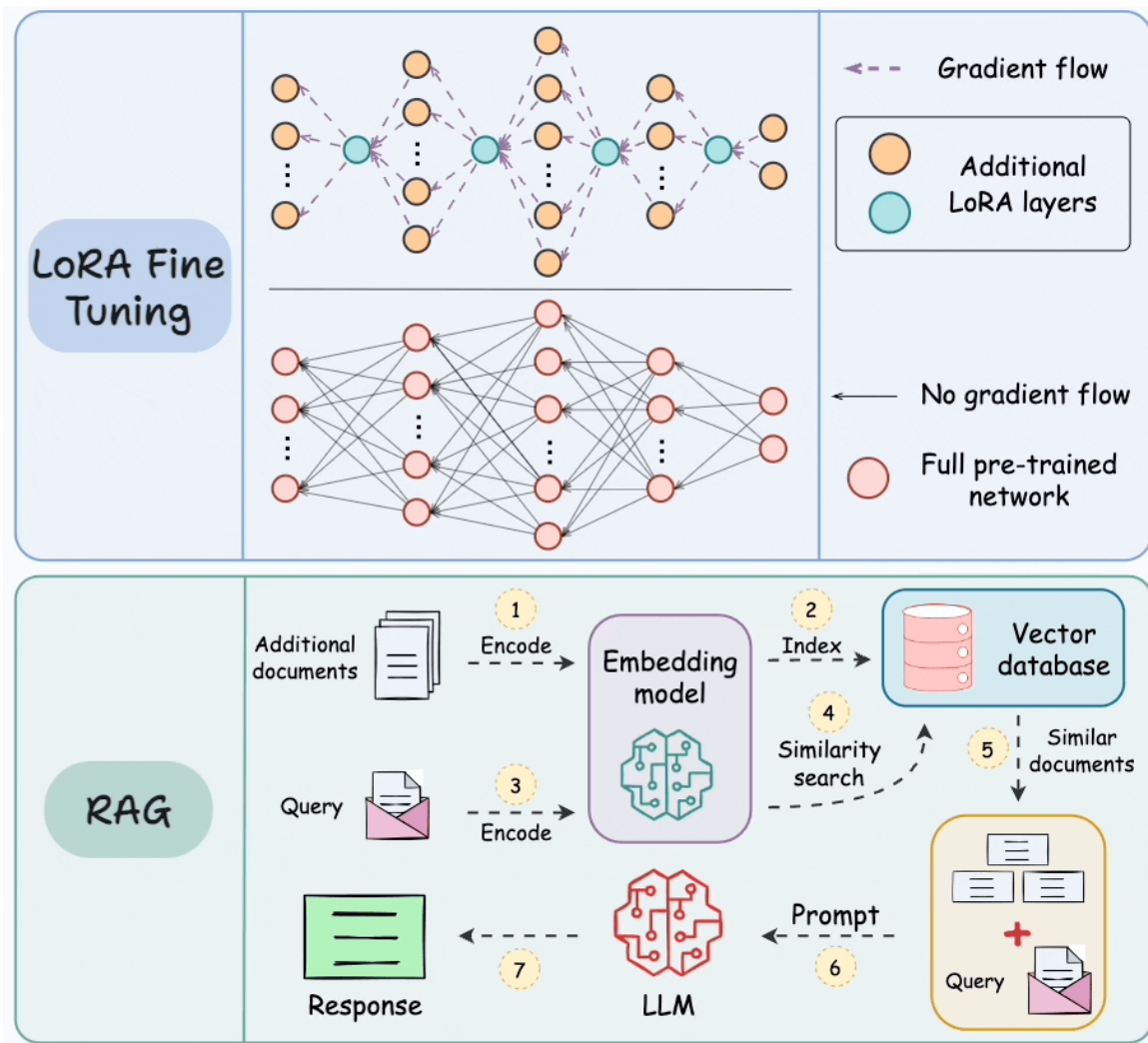


$$z=f(W \cdot x+b)$$

$$W'=W+\Delta W$$

$$\Delta W=A \cdot B$$

LoRA



Retrieval-
Augmented
Generation